

The background of the slide features a stylized image of the Earth from space, with a blue and white color scheme. Overlaid on the Earth is a complex network of white lines and dots, resembling a global communication or data network. The network lines are more prominent in the foreground and fade into the background.

SKYRD033 / GaN PAM Development

November 11, 2022 L&L

Jun “Bong” Jadormio, Ray Waugh, Eric Marsan, Michele Lewis

Connecting Everyone and Everything, All the Time.



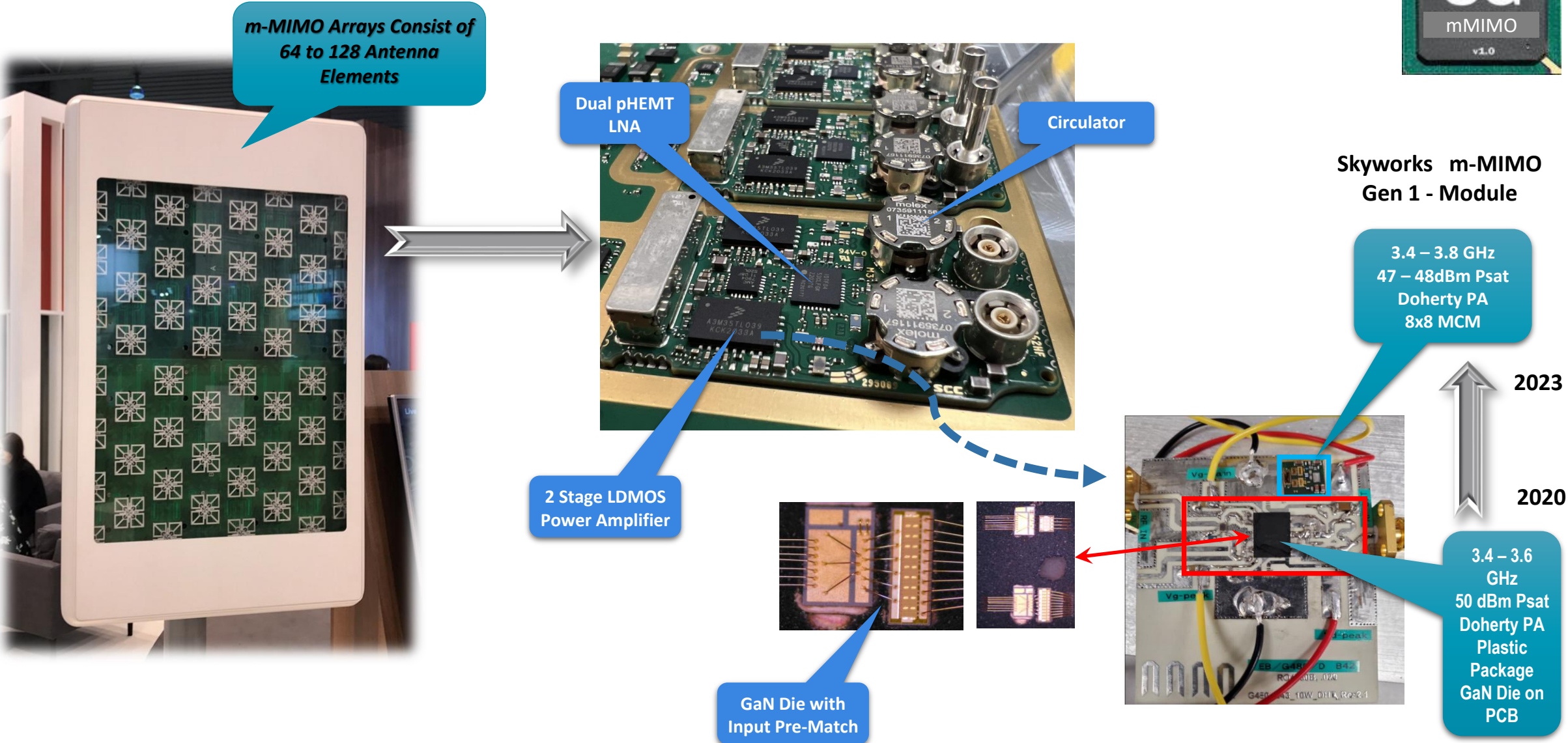
Outline

- Overview / Background
- Output Stage (GaN1 6.3W Doherty Power Amplifier)
 - History / Theory of Operation
 - Simulation / Design
 - Layout
 - Results
- Thermal Challenges
- Cascaded Performance Summary

Thank you for your support.....

- Michele Lewis
- Gokul Somasundaram
- Yu Zhu
- Alexy Klimashov
- Kai Kwok
- Woburn RPL

Massive MIMO Radio Example



Opportunity to Offer Total Solution for Both Tx & Rx

Summary / Overview of SKYRD033

Objective:

To have necessary IP developed & in place when Internal GaN is available.

- GaN PA for use in Tx Chain.
- High Power MCM / Packaging
- Control Circuitry

Deliverables:

**An integrated chip and wire MCM GaN Tx Chain.
Defined Tx architecture**

This is from the SKYRD033 P0. Over time, portions of other IP development needs were split off as separate SKYRD programs.

The Analog Controller became SKYRD046

The MCM & Assembly become SKYRD056

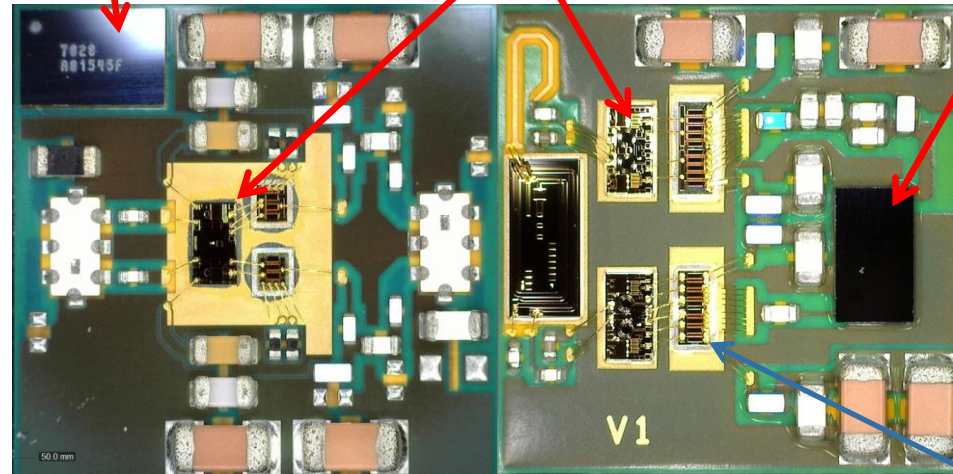
Internal HVIPD process became SKYRD520

Analog Controller

- Gate Bias Control (negative)
- Temperature Compensation
- PA enable / disable
- Supply sequencing

HVIPD Matching

Bumped Si-IPD Transformer



GaN Tx Chain

- 2 Stage GaN

MCM

- Chip and Wire Assembly
- Thermal Heat Slug under GaN

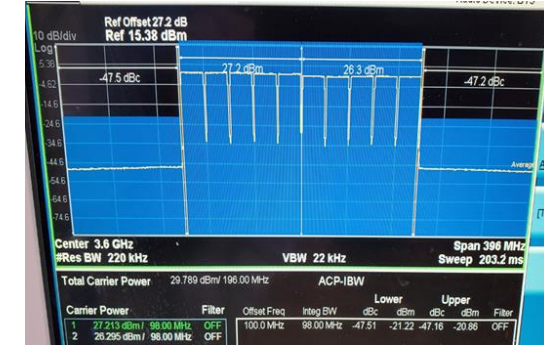
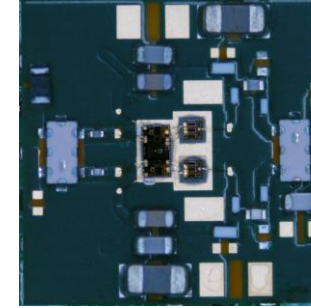
GaN PA Development / SKYRD033

Executive Summary

- Integrated 2 Stage GaN PA with Internal Bias Control for Small Cell & mMIMO Tx.
- Doherty-Lite Driver
 - Strategy: More efficient than Class AB PA, more linear than a traditional high PAE Doherty.
 - Promising Initial Results w/DPD
 - “Pipe Cleaner” for assembly and early look ahead quals.
- Final / Output Stage Amplifier
 - Asymmetric Doherty optimized for max PAE at 10dB back off from Psat.
 - Implemented in GaN1 internal process.
- Cascade of Doherty-Lite & Output Stage Balances PAE with “Ease” of Linearization (minimize DPD complexity).

Driver Amplifier

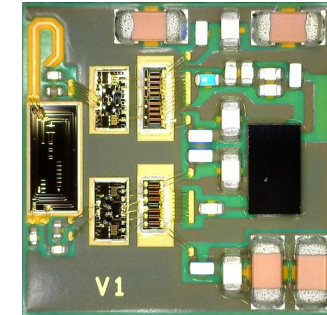
- Total Periphery = 1.6mm
- Avg Pout = +27.5dBm
- Psat = + 40dBm
- Gain = 11.5dB
- Drain Eff >25%



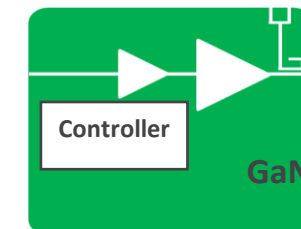
+

Final Amplifier

- Total Periphery = 7.7mm
- Avg Pout = +38dBm
- Psat = + 47dBm
- Gain = 10.5dB
- Drain Eff = 46%



Integrated PAM+



38 dBm Pout Avg
47 dBm Psat (50W)
Integrated
Controller
PAE > 40%

Architecture and Lineup Designed For Optimal System Level Performance



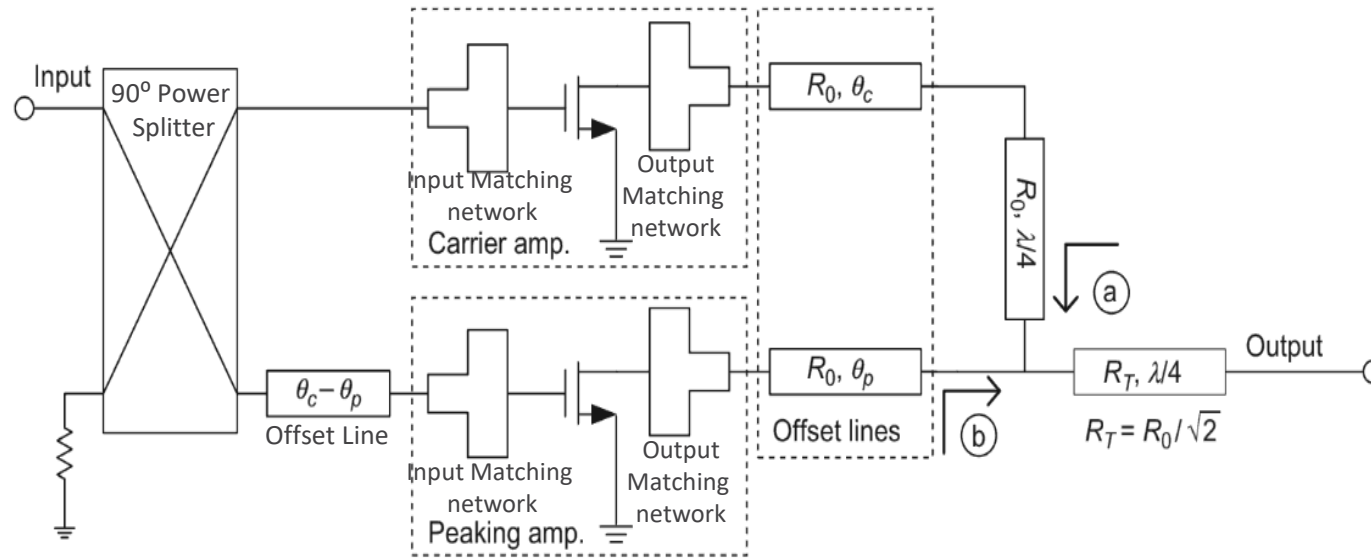
Doherty Power Amplifiers

Fundamental Principles

Introduction to Doherty Power Amplifier (Brief history)

- William H. Doherty – born in Cambridge, MA in 1907.
- He received bachelor's degree in communication engineering (1927) and master's degree in engineering (1928) at Harvard University.
- Joined in Bell Labs in 1929
- He invented unique amplifier approach in 1936 using vacuum tube and was first used in 50kW transmitter.
- 1940 Western Electric incorporate Doherty amplifier in commercial radio station worldwide.
- 1970 IRE recognized Doherty's important contribution in the development of RF power amplifier.

The Doherty Power Amplifier (DPA)

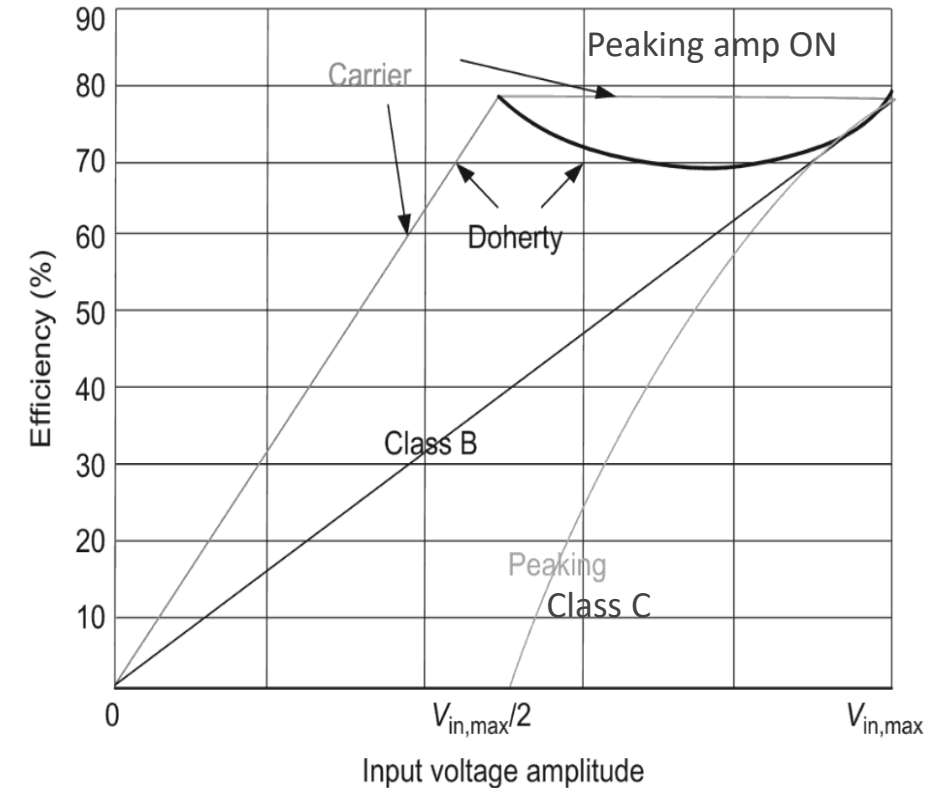


Types of DPA

- ❑ Symmetric – improves PAE @ 6dB OPBO
- ❑ Asymmetric – improves PAE @ 10dB OPBO

DPA Design Figure of Merits

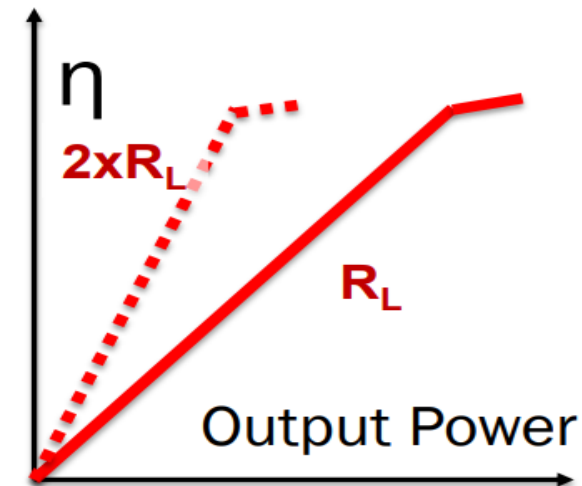
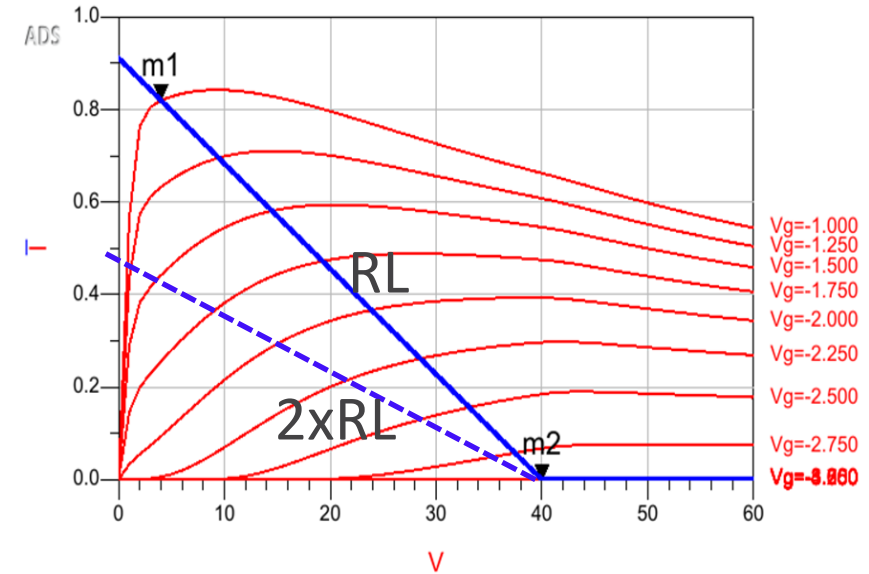
- Carrier amp, biased class B or class AB, is always on and handles signal at average power
- Peaking amp, biased class C, only handles signal peaks
- C and P's operation is dependent on each other
- Peak Power Ratio
- Z0 of $\lambda/4$ Impedance Inverter (not necessary always 90° inversion)
- Input Power Dividing ratio
- Parasitic due to packaging and matching circuitry all have phase shift effect
- Input/output offset lines to achieve in-phase output



The Doherty Power Amplifier

- Carrier Mode (Main Amplifier)
 - Carrier is only ON, load line is $2xR_L$
 - $\frac{1}{2}$ Max Pout
- Peaking Mode
 - Carrier and Peaking amp are both ON, Load line is R_L
 - $2x$ MaxPout

IV Curves & Class B Load lines



Basic Operation and Principle of DPA

- Theory of Load Modulation

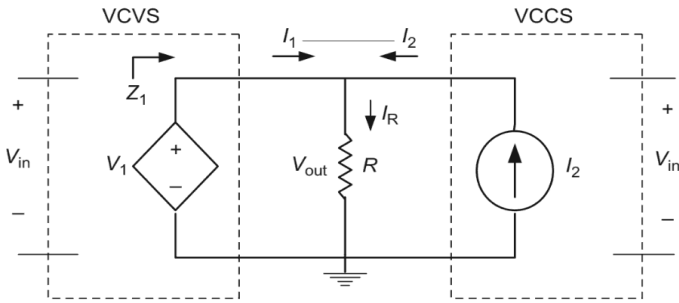


Figure1: Load Modulation Circuit driven by voltage and current source

$$Z_1 = \frac{V_1}{I_1} = \frac{V_1}{I_R - I_2} \quad (1)$$

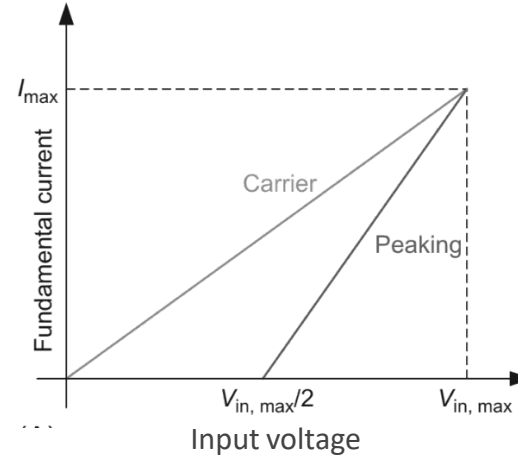


Figure3: Current profile

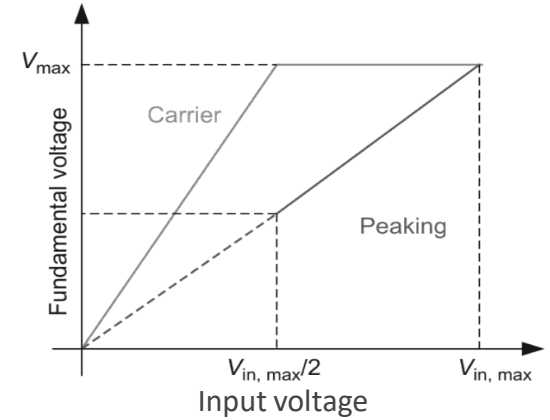


Figure4: Voltage profile

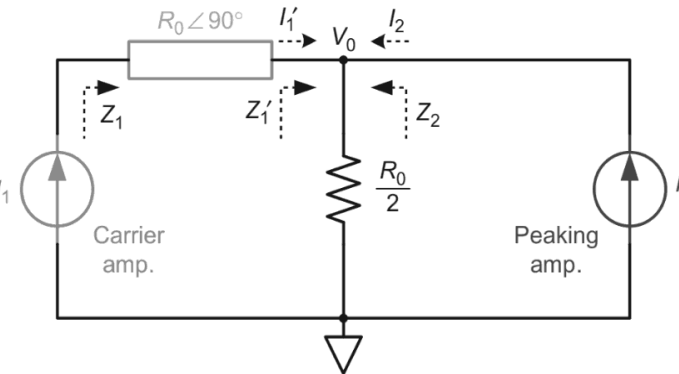


Figure2: Operational Diagram of DPA

$$Z_1' = \frac{V_0}{I_1'} = \frac{R_0}{2} \cdot \left[\frac{I_1' + I_2}{I_1'} \right] \quad (2)$$

$$Z_1 = \frac{R_0^2}{Z_1'} = \frac{2R_0}{1 + \frac{I_2}{I_1}} \quad (3)$$

$$Z_2 = \frac{V_0}{I_2} = \frac{R_0}{2} \cdot \left[\frac{I_1' + I_2}{I_2} \right] \quad (4)$$

Both PA is ON

$$\begin{aligned} I_1' &= I_2 \\ Z_1 &= R_0 \\ Z_2 &= R_0 \end{aligned}$$

Peaking PA is OFF

$$\begin{aligned} I_2 &= 0 \\ Z_1 &= 2R_0 \\ Z_2 &= \infty \end{aligned}$$

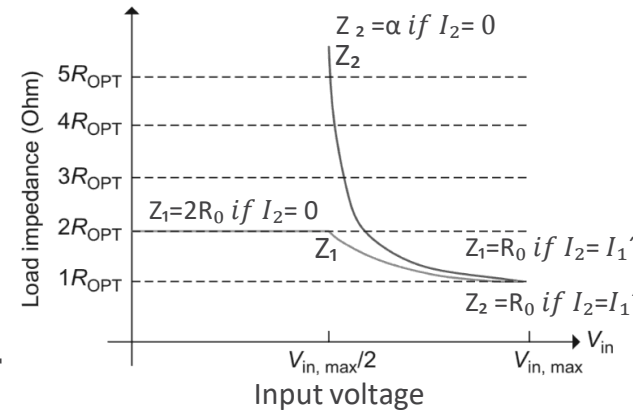


Figure5: Load Impedance profile

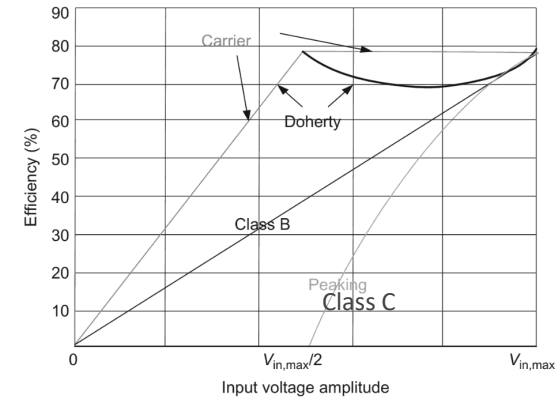
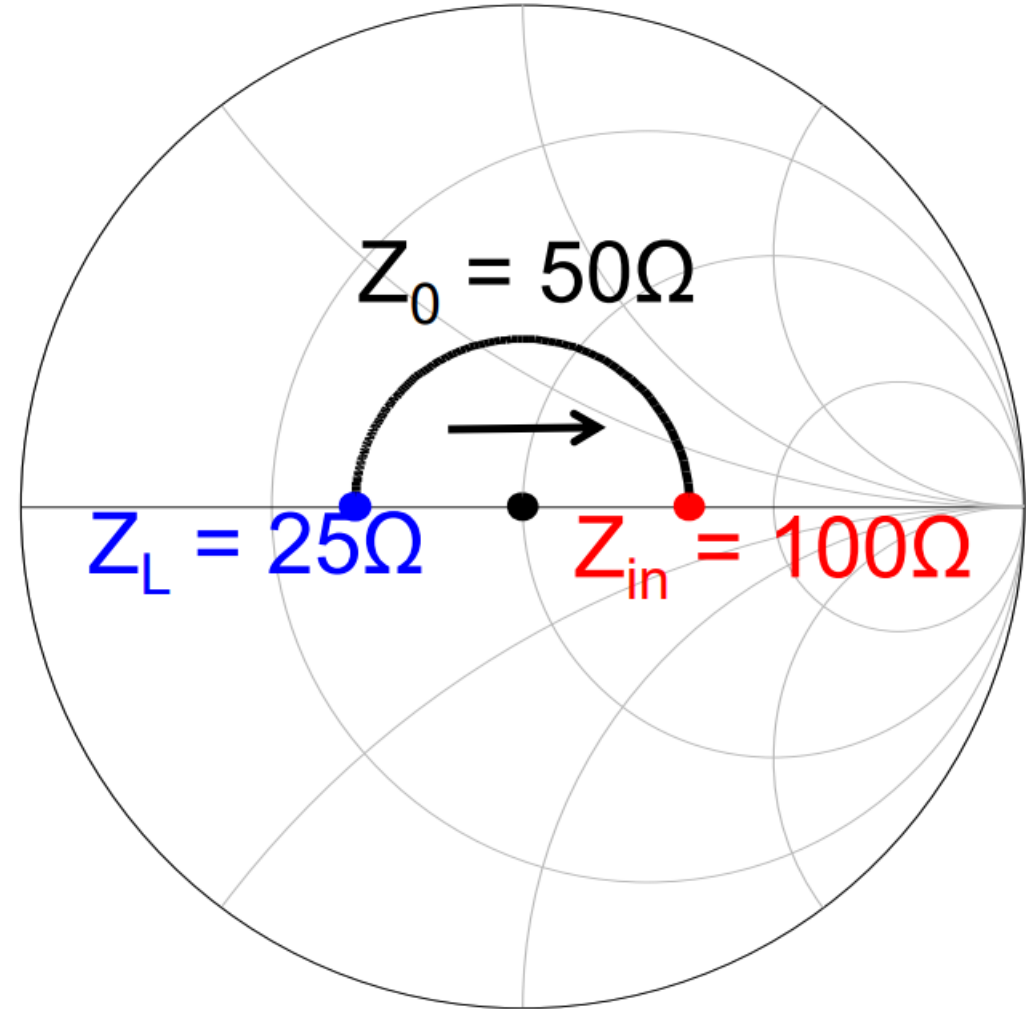
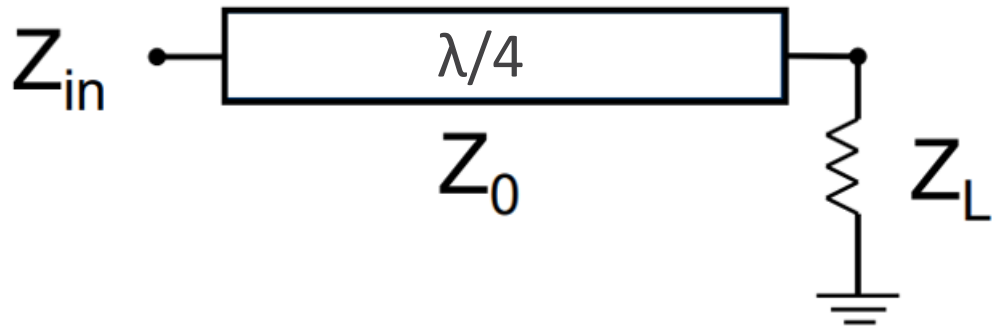


Figure6: PA's Efficiencies

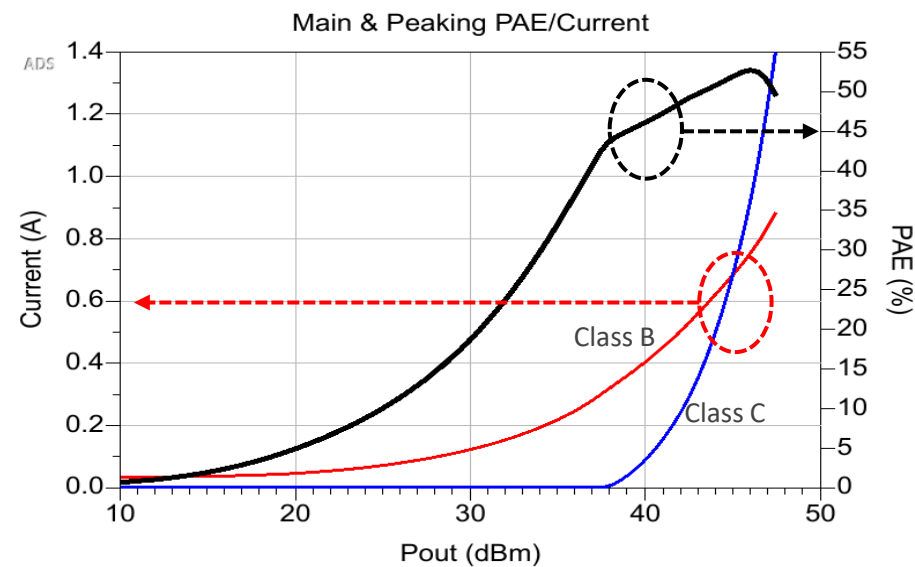
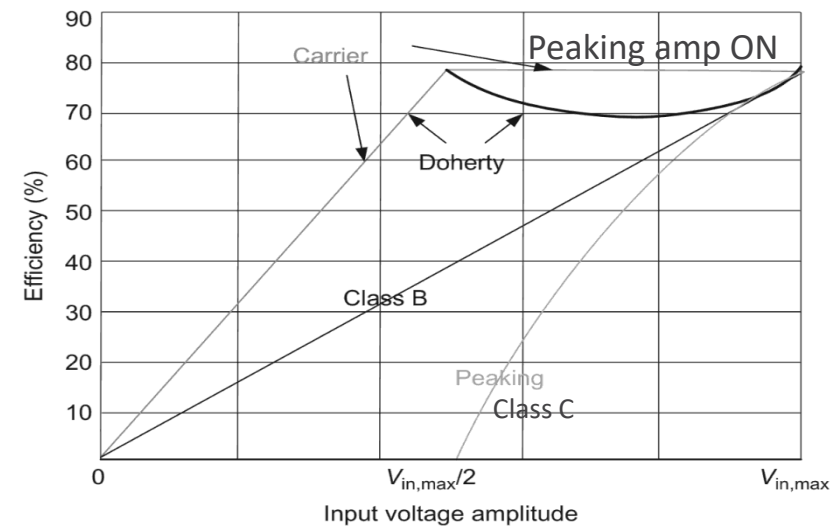
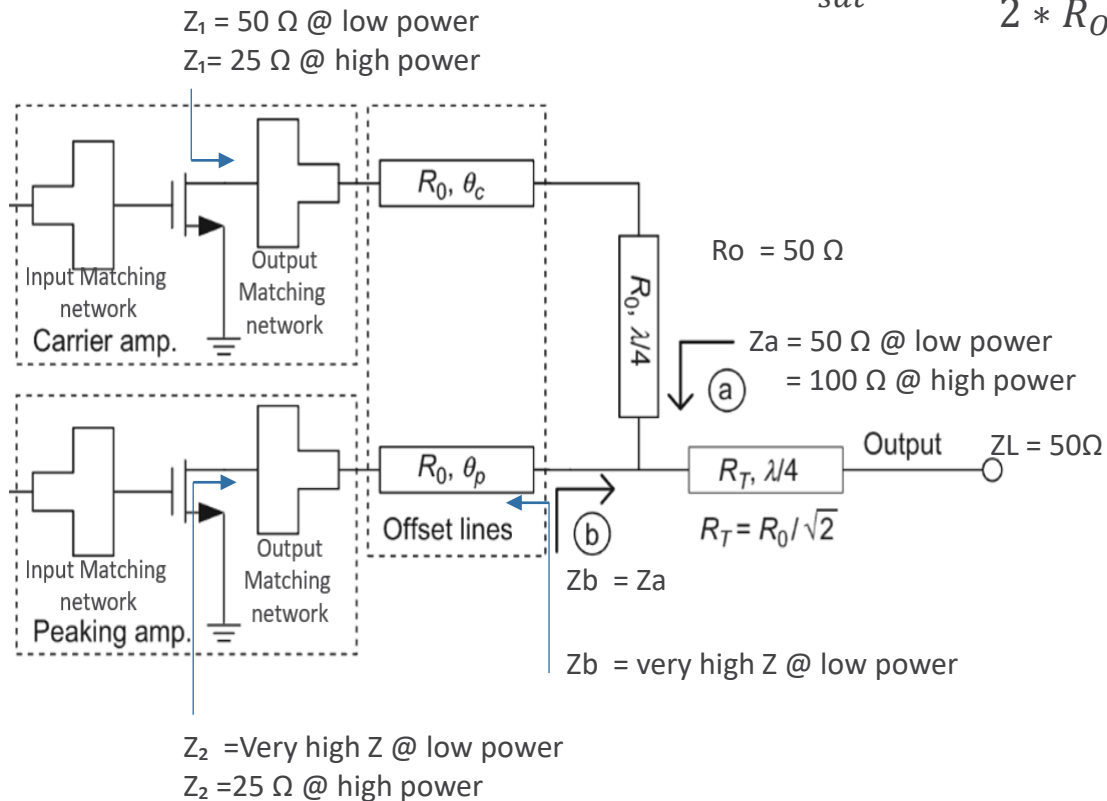
Function Quarter Wave Impedance Inverter in Load Modulation

$$Z_{in} = Z_0^2 / Z_L$$



DPA Load Lines Condition

$$P_{sat} = \frac{(V_{dd} - V_{knee})^2}{2 * R_{OPT}}$$



- At low input power, Carrier PA is ON while Peaking PA is OFF. High impedance is seen looking back to Peak PA.
- At high input power, Peaking PA turns ON, both PAs are connected at combining network R_T .
- $R_o, \lambda/4$ is not always 90° , offset lines is added to get peak signal in phase with carrier in the combining node.



Output Stage Design

GaN1 6.3W Doherty Power Amplifier

DPA Design Process

- Specifications
- Output impedance inverter design
- Output combining network analysis
 - Output Matching
- Input Matching Network
- Input power splitter design
- Combining the carrier and peaking amplifier
- Full module EM analysis
- Simulation and Test Data
- Thermal Analysis

SKYRD033A_WB DPA Final Stage – Performance Summary

Specified at Room Temp

Specification	Spec				Measurement @ 25C		
	Units	Min	Target	Max	Min	Max	Comment
Frequency Band	GHz	3.4	3.6	3.8	3.6	3.8	
Small Signal Gain	dB	12.0	12.5		10.2	10.8	
Input Return Loss	dB		-10	-8		-9	
Large Signal Gain	dB	11.0	11.5		9.5	10	
Gain Flatness (200 MHz BW)	dB		0.5	0.7	0.13	0.24	
Gain Flatness (400 MHz BW)	dB		1.0	1.2		0.7	
PAE at +38 dBm	%	48	50		48	49.9	
P3dB	dBm	45	46		46.7	47.0	
VBW	MHz	400	500				
Corrected ACPR (+38dBm, 100 MHz, 8dB PAR)	dBc	-50					
Corrected ACPR (+38dBm, 200 MHz, 8dB PAR)	dBc	-46			-46	-45	
AM-PM (Pavg to Psat)	°		10	12	6	18	Worst @ LBE
Stability (K factor)		1	1.5		1.3		
Turn on time	uS		0.75	1.0			

SKYRD033A_WB Final Stage MCM Layout

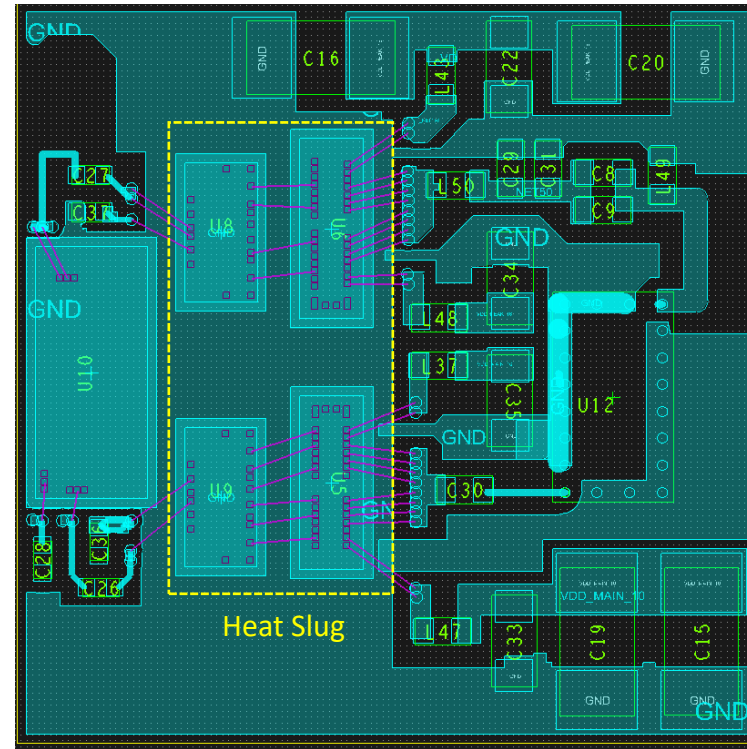
Primary die components

- U5, U6 = GaN1 SWKS
- U8, U9 = HIPD1 input Match SWKS
- U10 = LIPD power splitter SWKS
- U12 = RLC06E ST-IPD Z-inverter

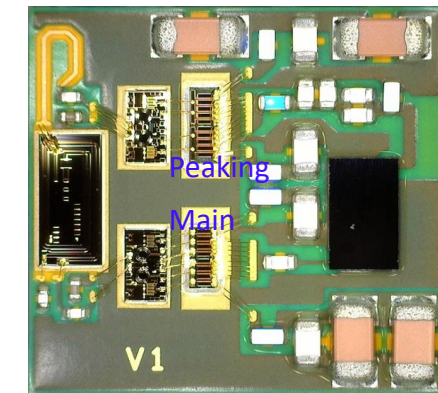
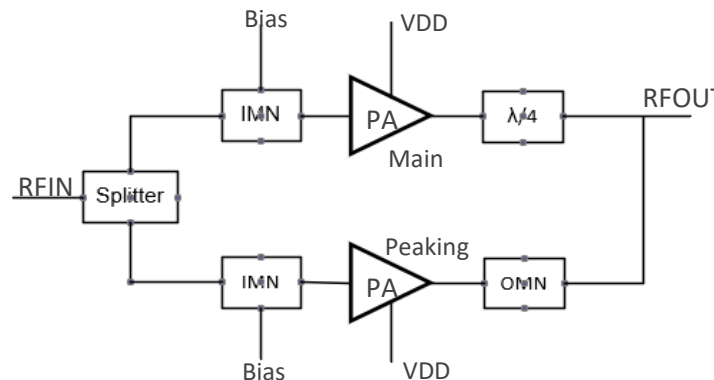
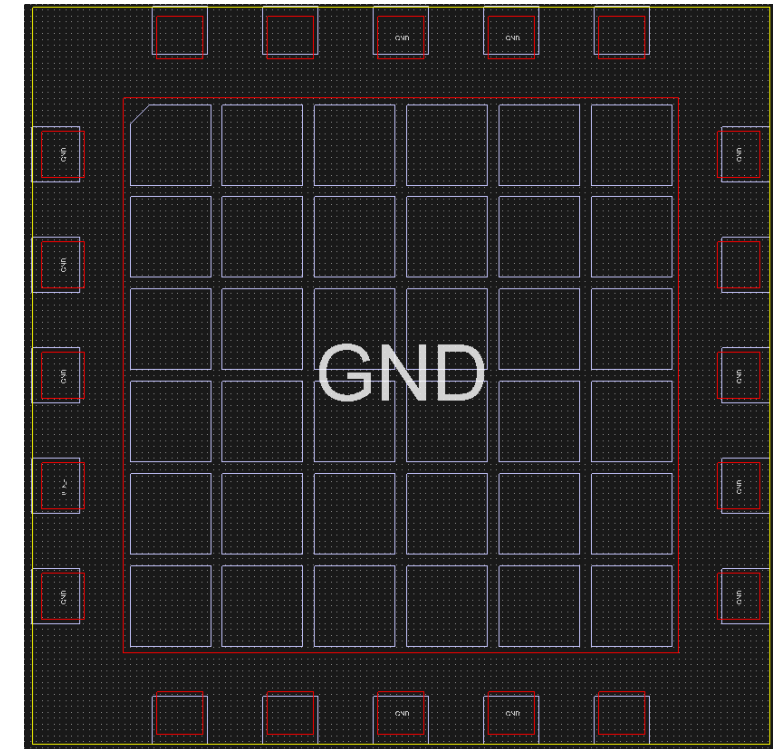
NOTES:

- MCM has 4 layers
- Heat Slug under U5, U6, U8, U9
- SMDs sizes are 0603, 0402, 0201, 01005
- 25um gold wire

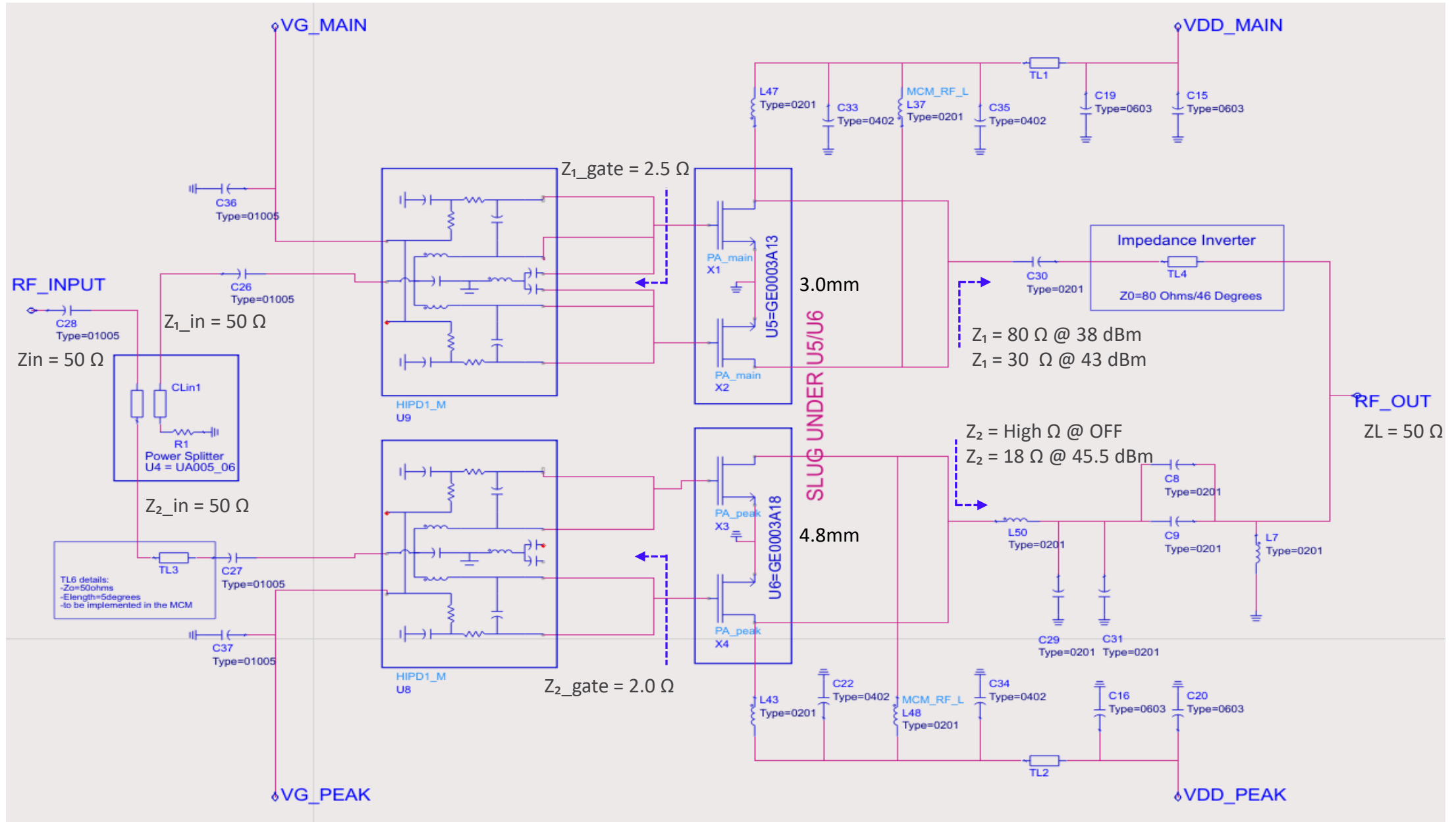
Top View



Bottom View

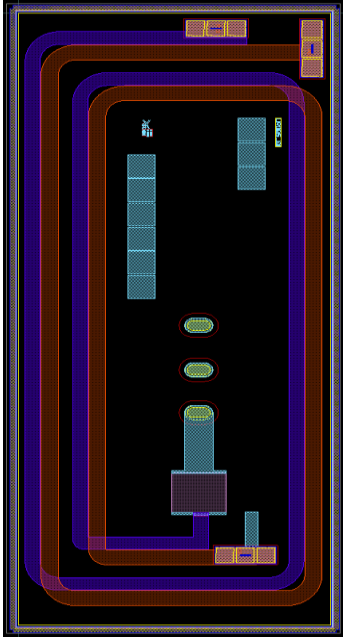


SKYRD033A_WB Final Stage Schematic Diagram



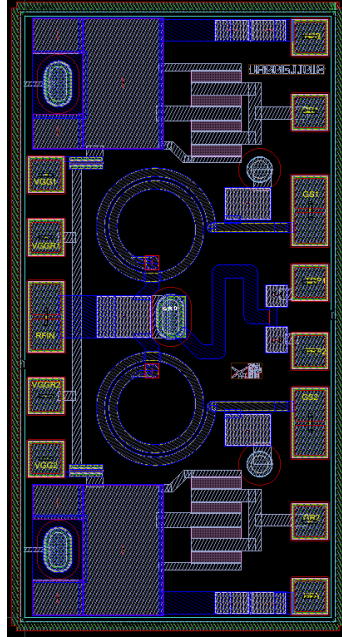
Primary die sizes and layout (1st Generation)

U10



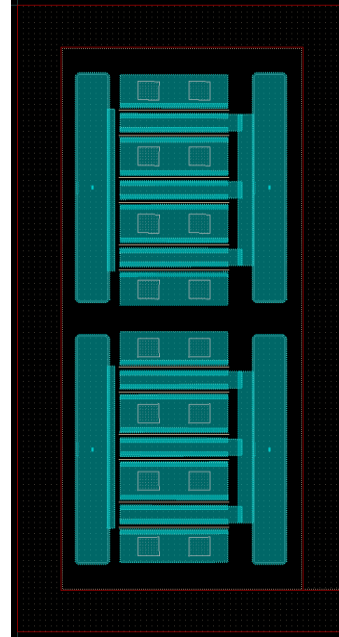
1192x2700um
LIPD
Power Splitter

U8/U9



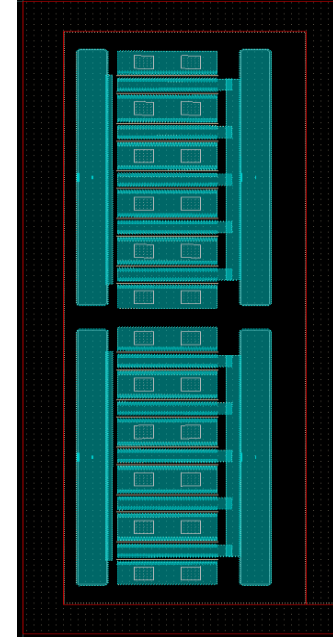
785x1500um
HIPD1
Input Match

U5



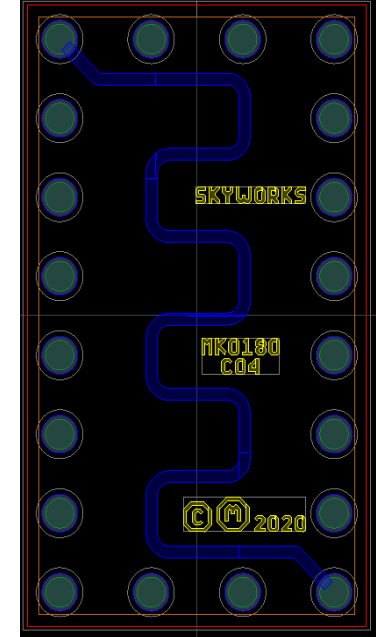
760x1495um
GaN1
Carrier Amp

U6



810x2140um
GaN1
Peaking Amp

U12



1265x2235um
ST-IPD
Impedance Inverter

R.F. GaN on SiC Spec Targets (GaN1)

Goal: Develop a 6 Inch R.F. GaN on SiC Process in the Woburn fab.

Device Features

- 0.35um gate
 - Field plate
- 2 Thick metals
 - M1 2um
 - M2 3um
- M1 field plate
- 50 Ohm/sq TFR
- Source TWV grounding

Device Performance Goals

- R.F. power amplifier to 8GHz
- I_{max} 800mA/mm*
- V_p -3V
- V_b 100->200V
- $F_t \sim 20$ GHz $F_{max} \sim 40$ GHz
- P_{out} 4 -> 6.5 W/mm***
- Operating voltage 28 now 40V

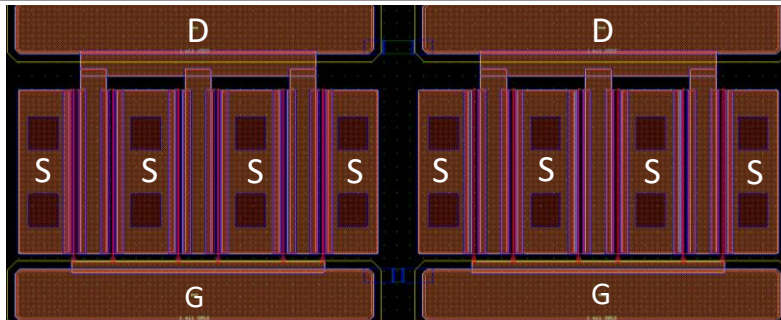
Device Spec. Goals

- I_{max}^{**} mA/mm min 780 typ. 900 max. 1100⁺
- I_{dss} mA/mm min 530 typ. 650 max. 930⁺
- V_p V min -3.8 typ. -2.9 max. -2.4⁺
- G_m mS/mm min 200 typ. 269 max. 330⁺
- $I_{doff_{30V}}$ uA/mm typ. 250 max 500
- Small signal⁺⁺ typ. 21dB
- P_{out} / PAE^{++} typ. 6.5 W/mm / 56 %

⁺⁺4x250 Power match load pull 3.5GHz, 10% I_{dss} 40V V_{ds}

PA FET cells

Infrastructure



Wire bond drain

TWV source

Wire bond gate

⁺Device spec. windows derived from 3 benchmarks with available parametric data

* measured at +1V V_{gs}

** measured at +2V V_{gs}

*** saturated power class AB

SKYRD033A_WB 6.3W

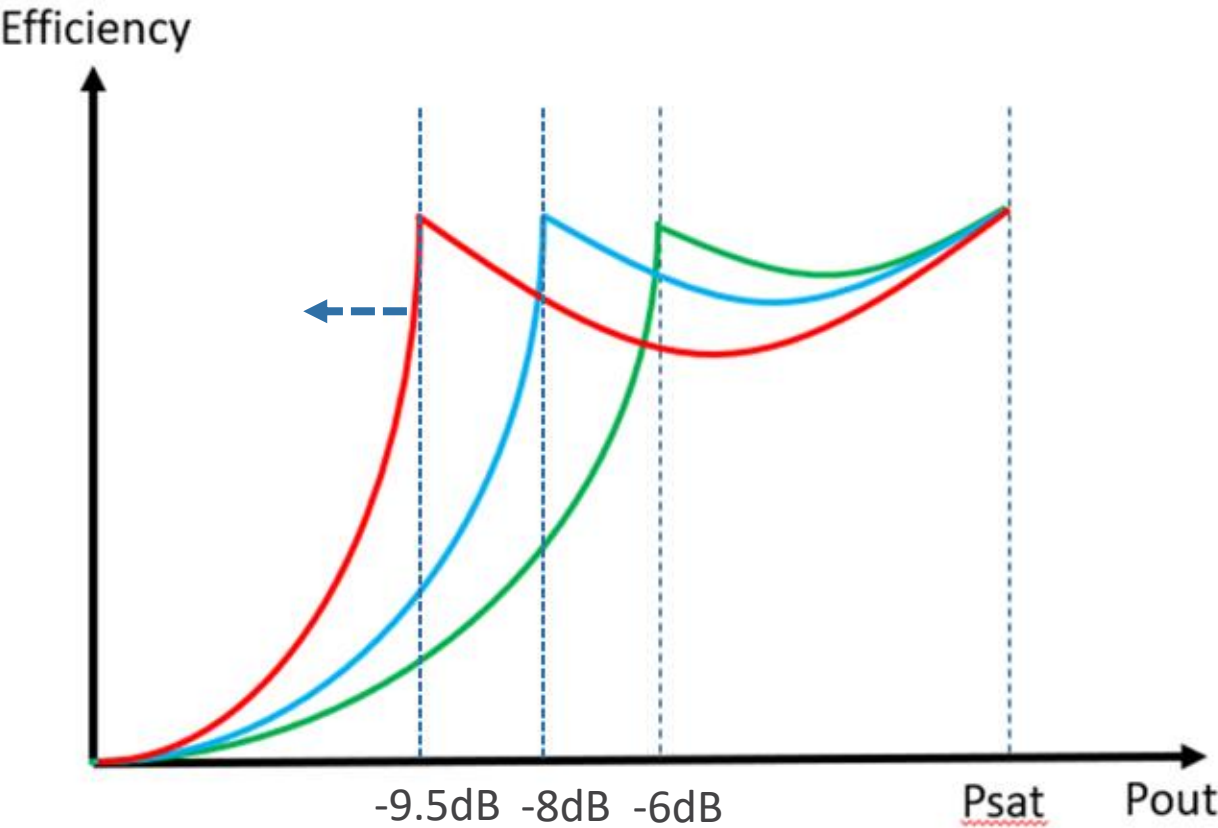
Doherty Power Amplifier Design

Final Stage Transistor Device Size and Loadlines

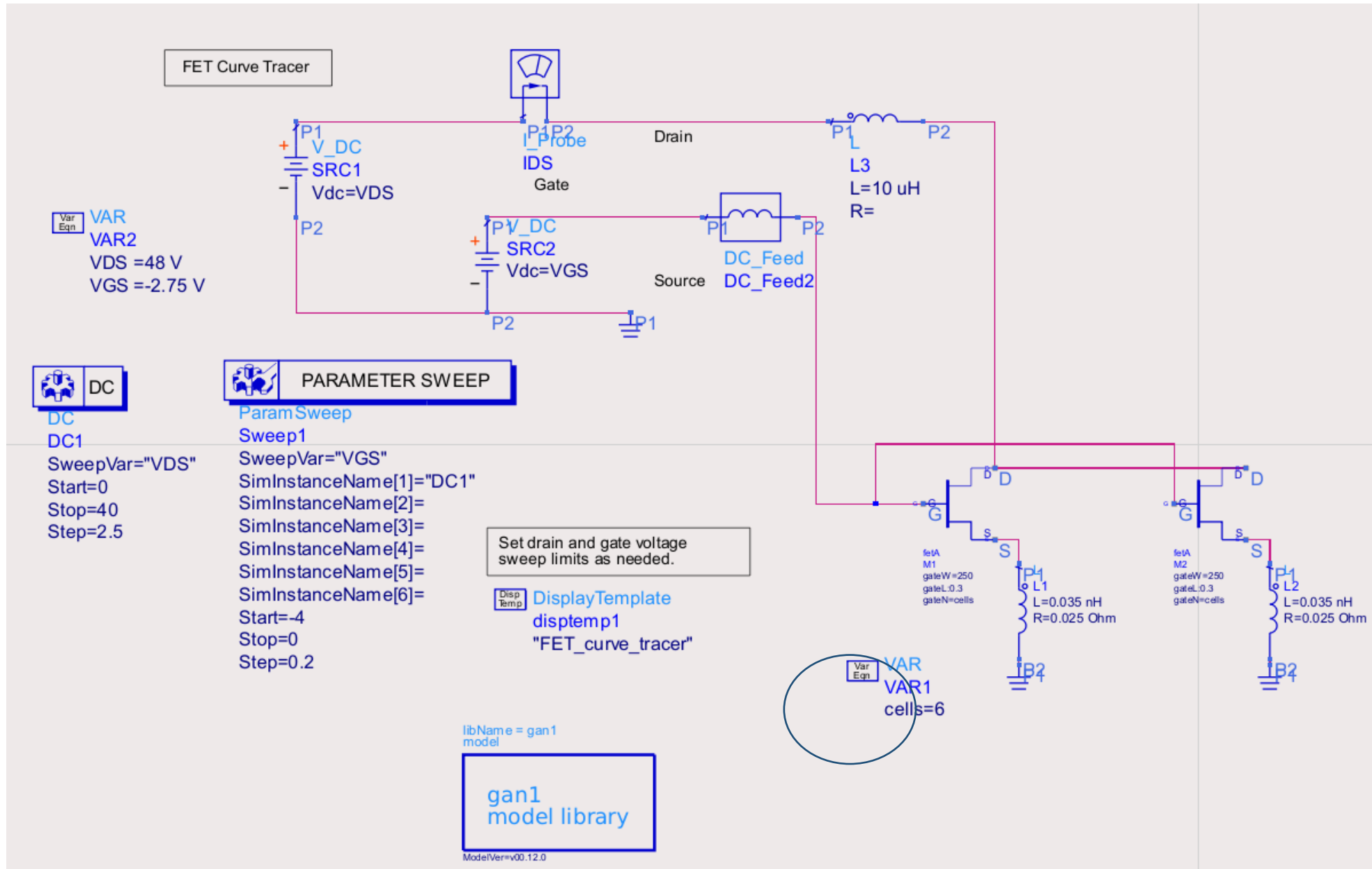
	Device Size		Load-line	
	Main Amp	Peaking Amp	Main Amp	Peaking Amp
Average Power	3.0 mm	4.8 mm	90 Ω @ 6.3 W	High Impedance
Peak Power	3.0 mm	4.8 mm	30 Ω @ 20 W	18 Ω @ 35.4 W

r^*	PAPR (dB)
1	-6
1.5	-8
2	-9.5
2.5	-10.9
3	-12

$r^*=P_p/P_c$ (ratio of peaking to carrier power)



DC Load-line Simulation base on Class B



DC Analysis on GaN1 Model 3.0 mm (Main Amp)

Vdd=40V

Average Power

m1
indep(m1)=5.000
plot_vs(I, V)=0.803
VGS=-1.625

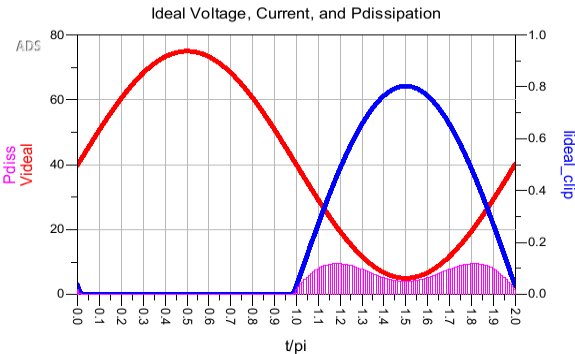
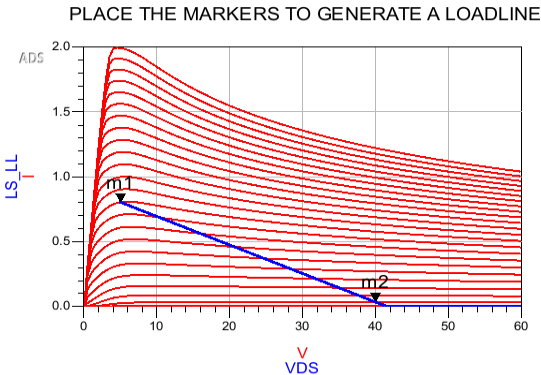
m2
indep(m2)=40.000
plot_vs(I, V)=0.033
VGS=-2.750

RL
87.156

Eqn Zin=50

Set these parameters to match the sweep variables in your design (I=output current, V=output voltage)

Eqn I=IDS.i
Eqn V=VDS
Eqn Vctrl=VGS



Pout_Max_W	n_max	Idc	Conduction_Angle	Duty_Cycle
7.03 =38.47 dBm	54.26	0.32	237.86	66.07

All results generated through calculation from first principles

Main Amp:

- Main alone operates on **2xRL** load line
- ½ MaxPout (Single Device)

Peak Power

m1
indep(m1)=5.000
plot_vs(I, V)=1.992
VGS=0.000

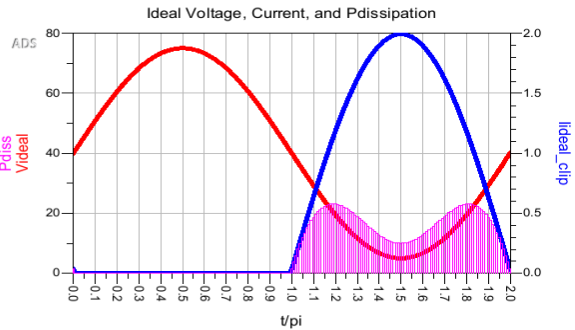
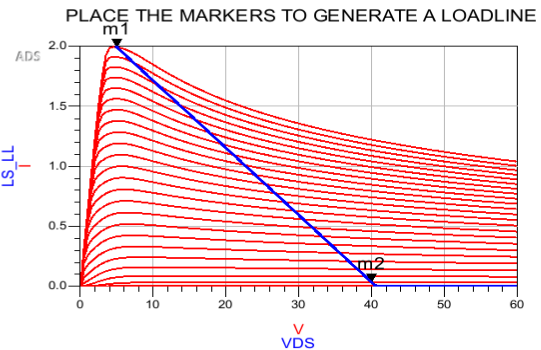
m2
indep(m2)=40.000
plot_vs(I, V)=0.033
VGS=-2.750

RL
35.138

Eqn Zin=50

Set these parameters to match the sweep variables in your design (I=output current, V=output voltage)

Eqn I=IDS.i
Eqn V=VDS
Eqn Vctrl=VGS



Pout_Max_W	n_max	Idc	Conduction_Angle	Duty_Cycle
17.43 =41.6 dBm	55.45	0.79	231.43	64.29

All results generated through calculation from first principles

DC Analysis on GaN1 Model 4.8 mm (Peaking Amp)

Vdd=40V

INTERACTIVE LOADLINE UTILITY FOR STANDARD CLASSES OF PA OPERATION

Average Power

m1
indep(m1)=5.000
plot_vs(I, V)=3.433
VGS=0.000

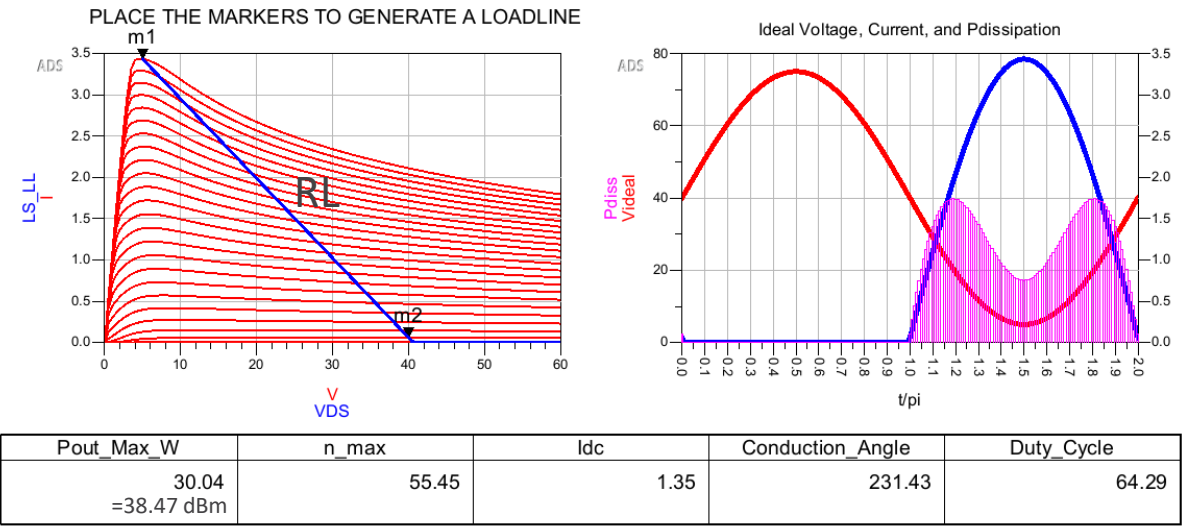
m2
indep(m2)=40.000
plot_vs(I, V)=0.057
VGS=-2.750

RL
20.391

Eqn Zin=50

Set these parameters to match
the sweep variables in your design
(I=output current, V=output voltage)

Eqn I=IDS.i
Eqn V=VDS
Eqn Vctrl=VGS

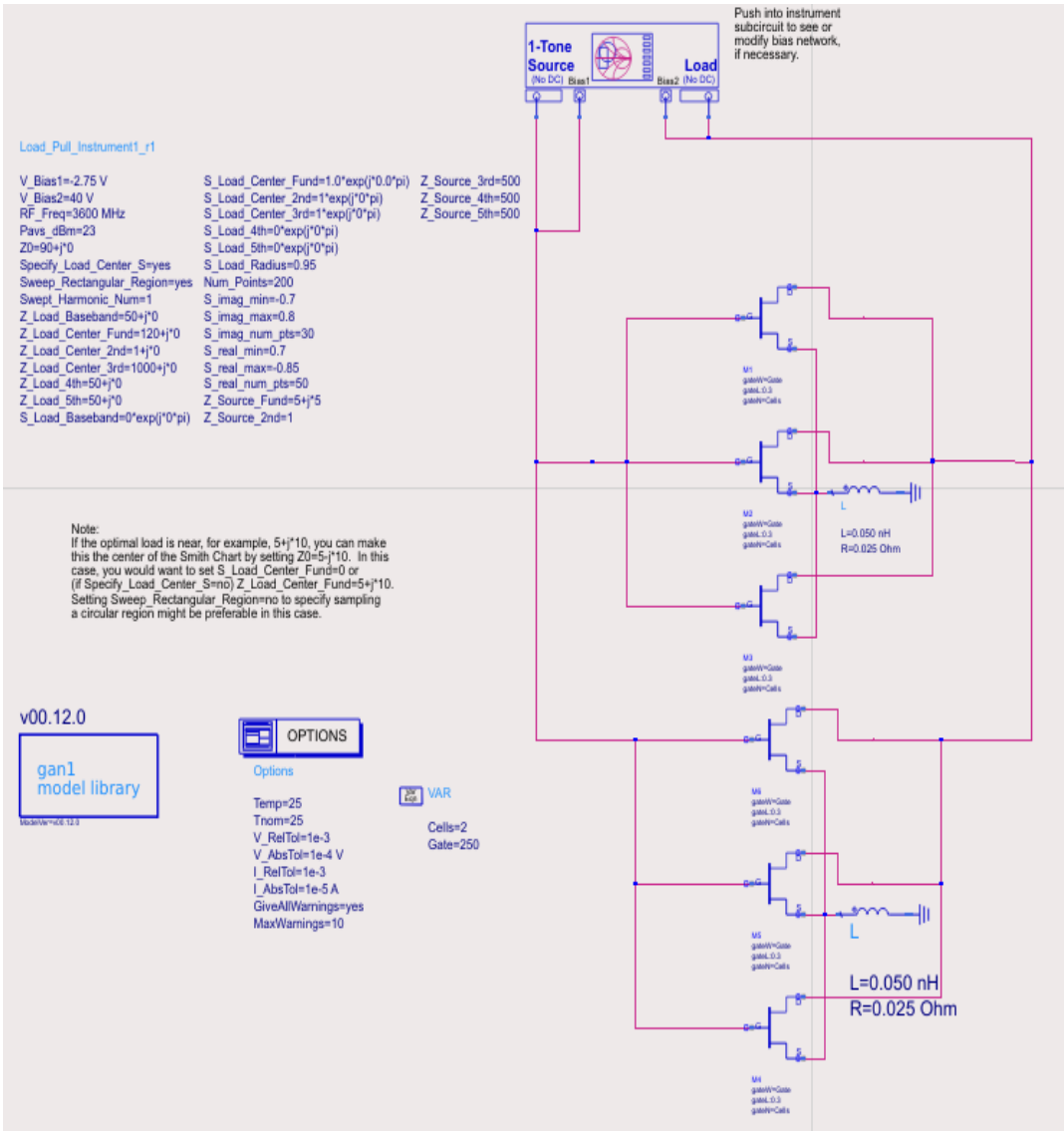


All results generated through calculation from first principles

Peaking Amp mode:

- Carrier and peaking each operate on **RL** load line
- 2 x MaxPout (Single Device)

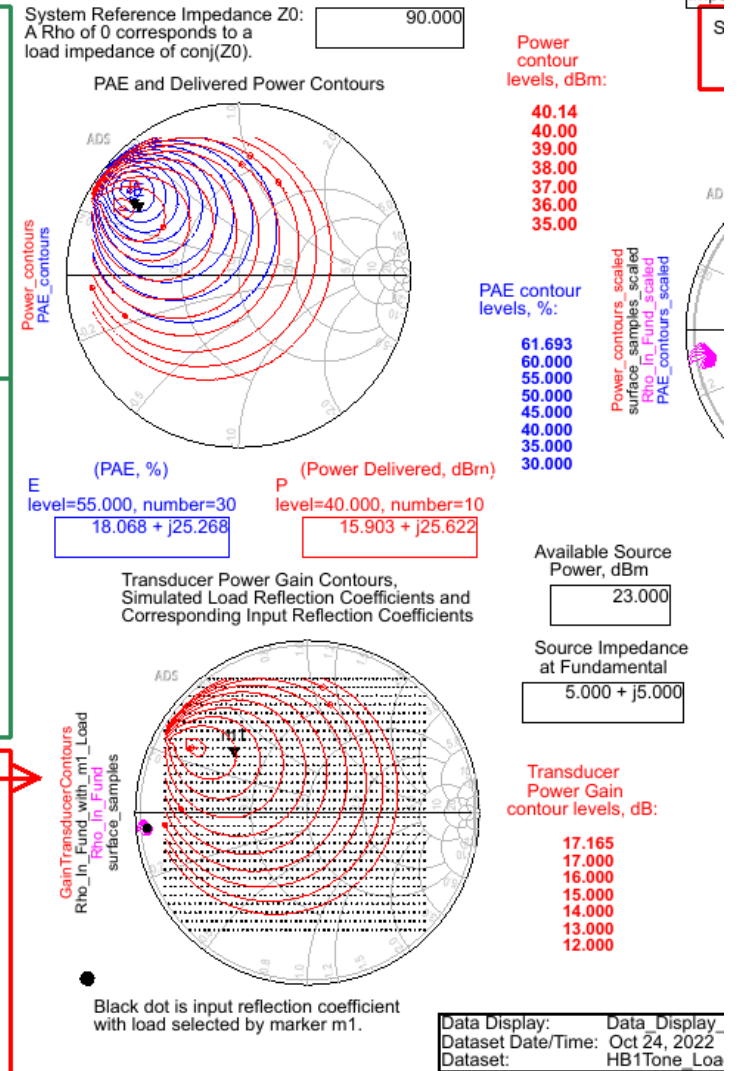
Load-Pull Simulation 3.0 mm (Main Amp)



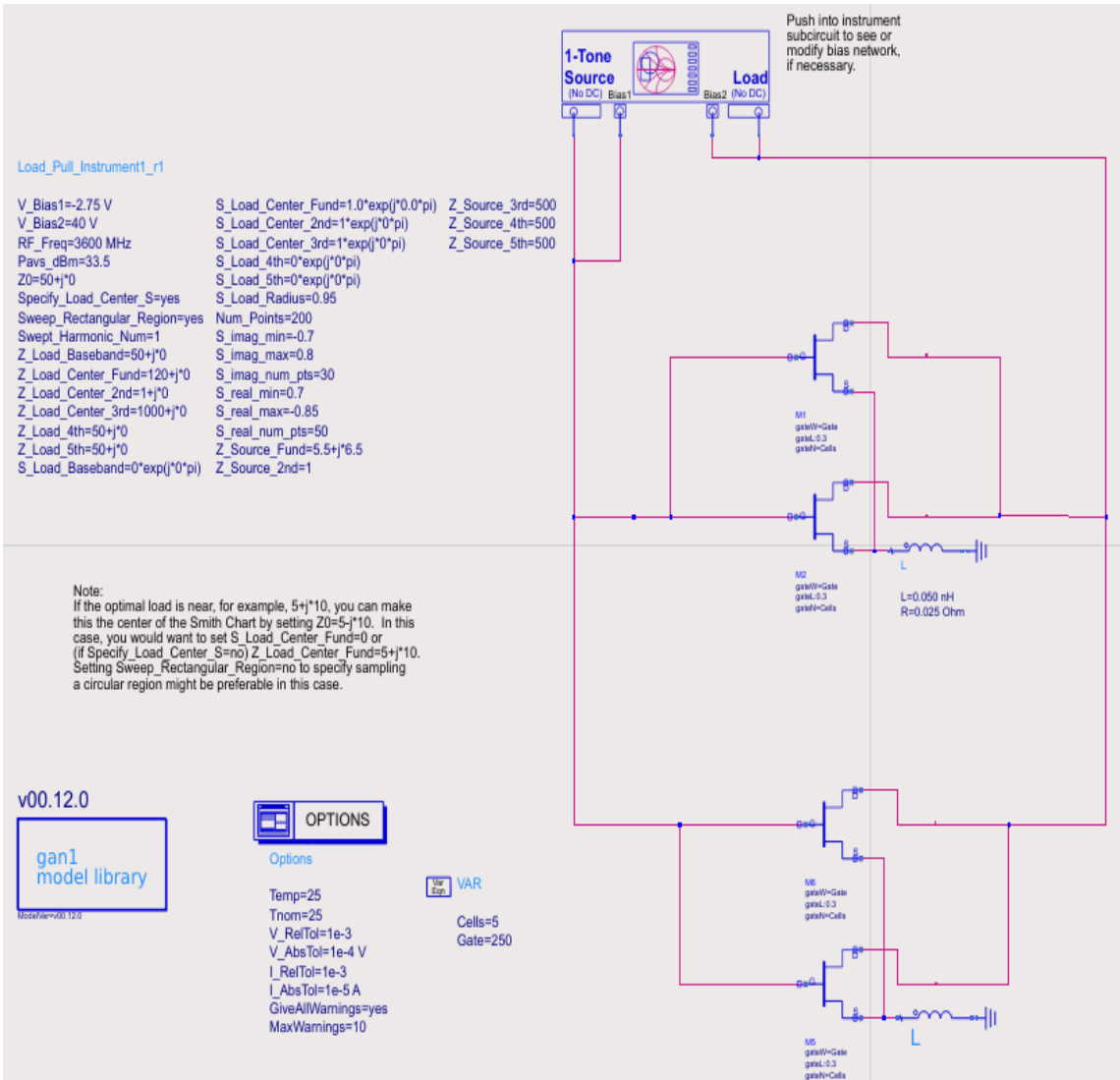
At load that gives maximum power (and gain):		
BiasCurrent_at_MaxPower	Zload_at_MaxPower	MaxPowerRho
0.468	12.854 + j23.927	0.765 / 149.673
PAE_at_MaxPower		
55.100		
Z_In_at_MaxPower	Gain_at_MaxPower	
0.988 - j4.679	17.182	
Pdel_dBm_Max		
40.182		

At load that gives maximum PAE:		
BiasCurrent_at_MaxPAE	Zload_at_MaxPAE	MaxPAE_Rho
0.351	11.363 + j30.476	0.797 / 142.082
PAEmax		
61.755		
Z_In_at_MaxPAE	Gain_at_MaxPAE	
0.948 - j3.770	16.431	
Pdel_dBm_at_MaxPAE		
39.431		

At load selected by marker m1:		
BiasCurrent_at_m1	Zload_at_m1	Rho_at_m1
0.413	28.692 + j27.593	0.552 / 142.681
PAE_at_m1		
47.984		
Z_In_at_m1	Gain_at_m1	
1.729 - j4.417	16.090	
Pdel_dBm_at_m1		
39.090		



Load-Pull Simulation 5.0 mm (Peaking Amp)



At load that gives maximum power (and gain):

BiasCurrent_at_MaxPower	Zload_at_MaxPower	MaxPowerRho
1.438	8.536 + j9.970	0.718 / 156.815

PAE_at_MaxPower	Gain_at_MaxPower
64.839	12.329

Z_In_at_MaxPower	Pdel_dBm_Max
1.133 - j3.032	45.829

At load that gives maximum PAE:

BiasCurrent_at_MaxPAE	Zload_at_MaxPAE	MaxPAE_Rho
1.151	8.134 + j13.786	0.738 / 148.433

PAEmax	Gain_at_MaxPAE
72.570	11.832

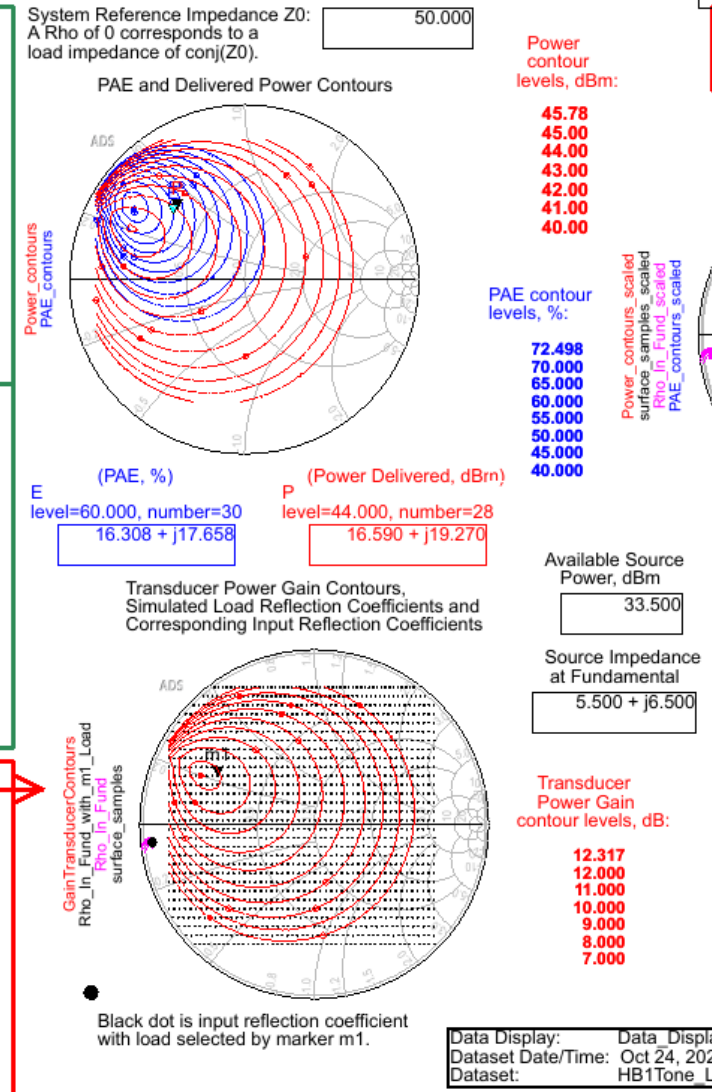
Z_In_at_MaxPAE	Pdel_dBm_at_MaxPAE
0.945 - j2.626	45.332

At load selected by marker m1:

BiasCurrent_at_m1	Zload_at_m1	Rho_at_m1
1.324	11.866 + j11.176	0.632 / 153.426

PAE at m1	Gain at m1
66.685	12.108

Z_In_at_m1	Pdel_dBm_at_m1
1.185 - j2.768	45.608

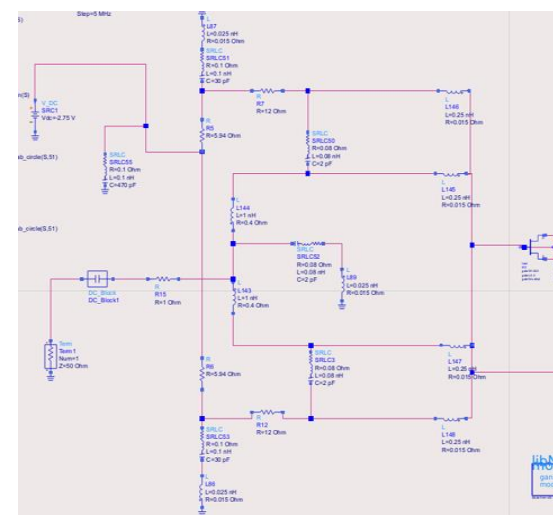
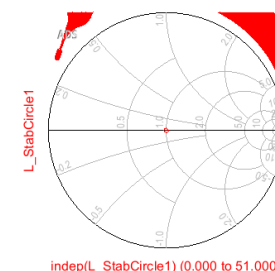
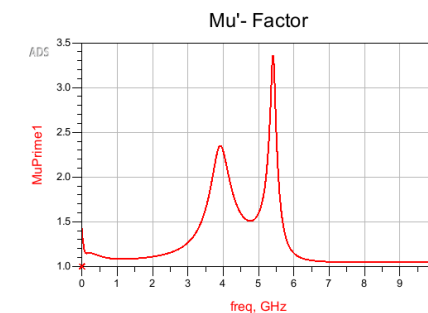
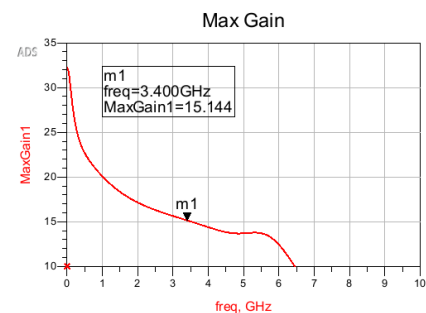
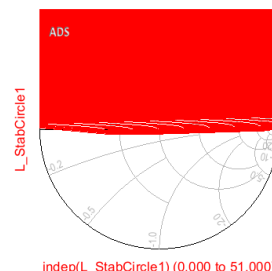
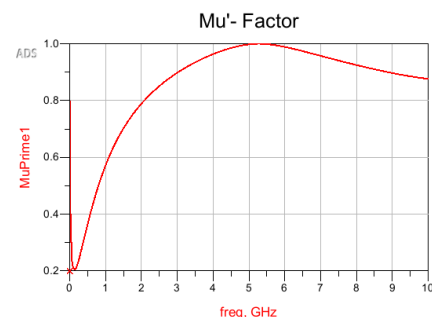
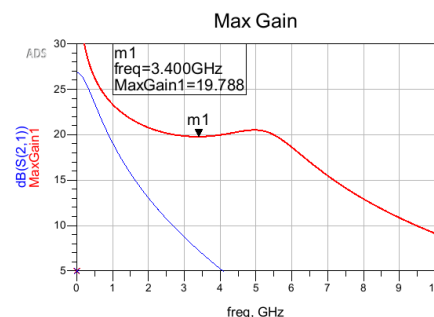
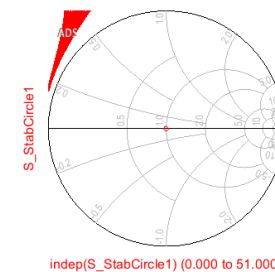
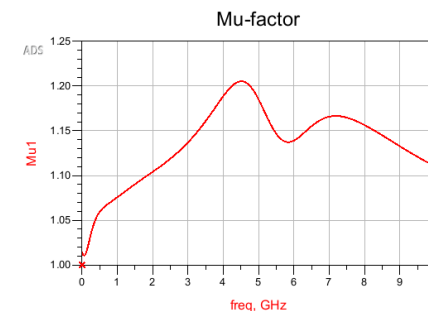
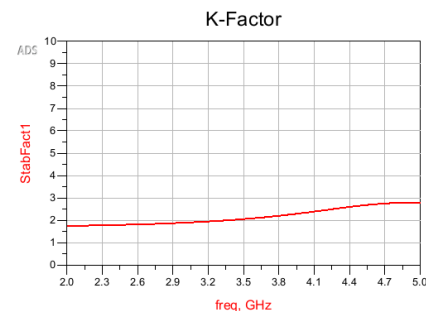
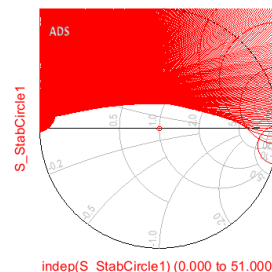
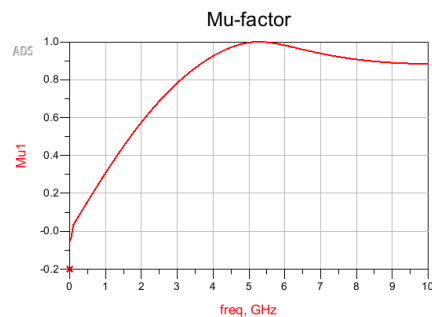
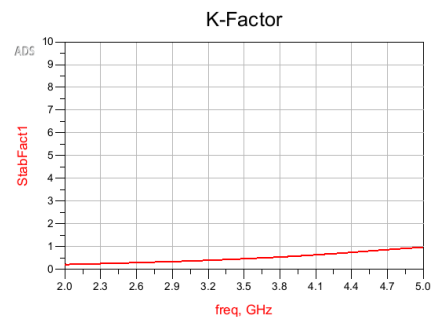


Stability Simulation GaN1 3.0 mm Device

VDD=48V; Vgate=-2.75V (With ad Without Stability network)

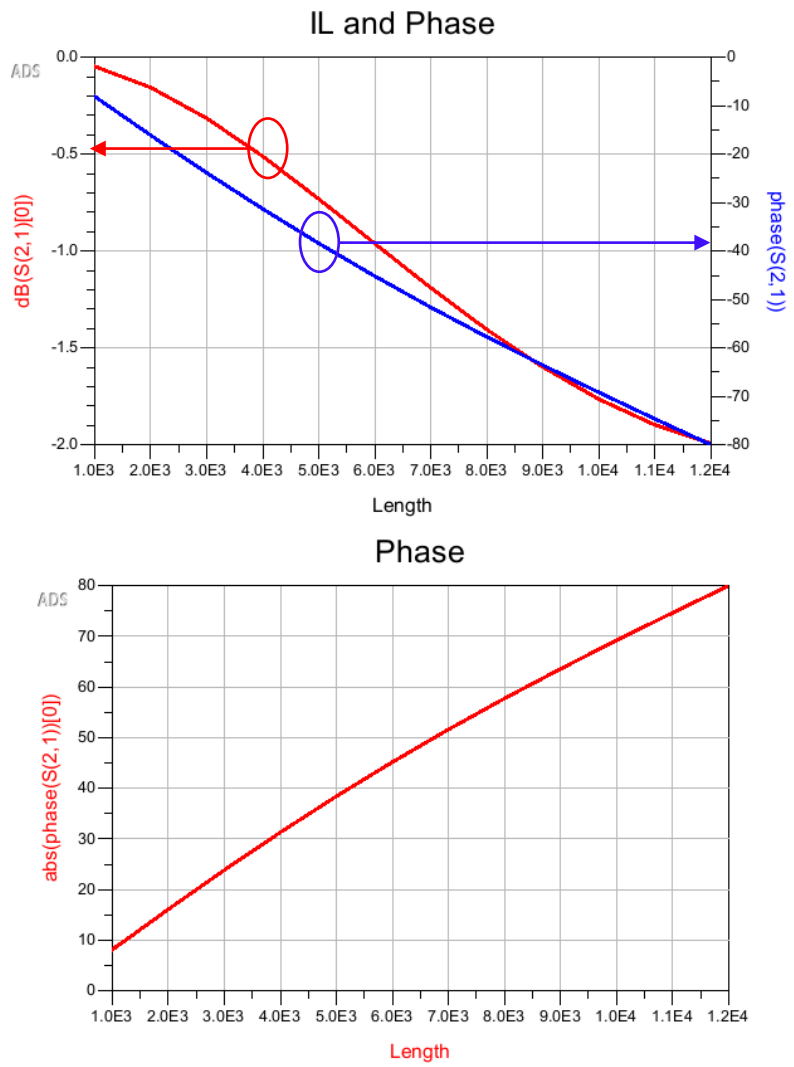
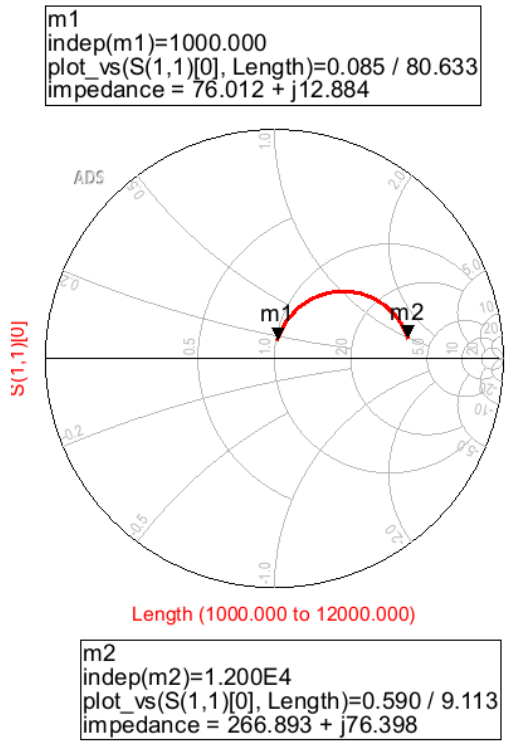
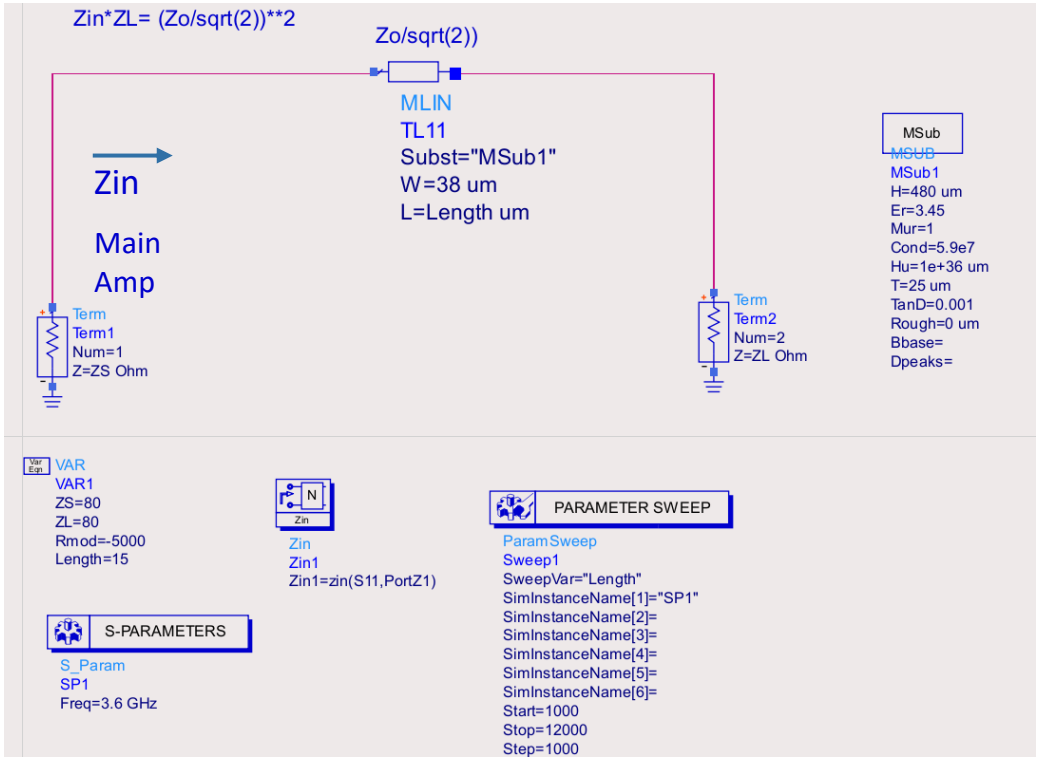
With Stability network

Without Stability network



Output Combining Network Design

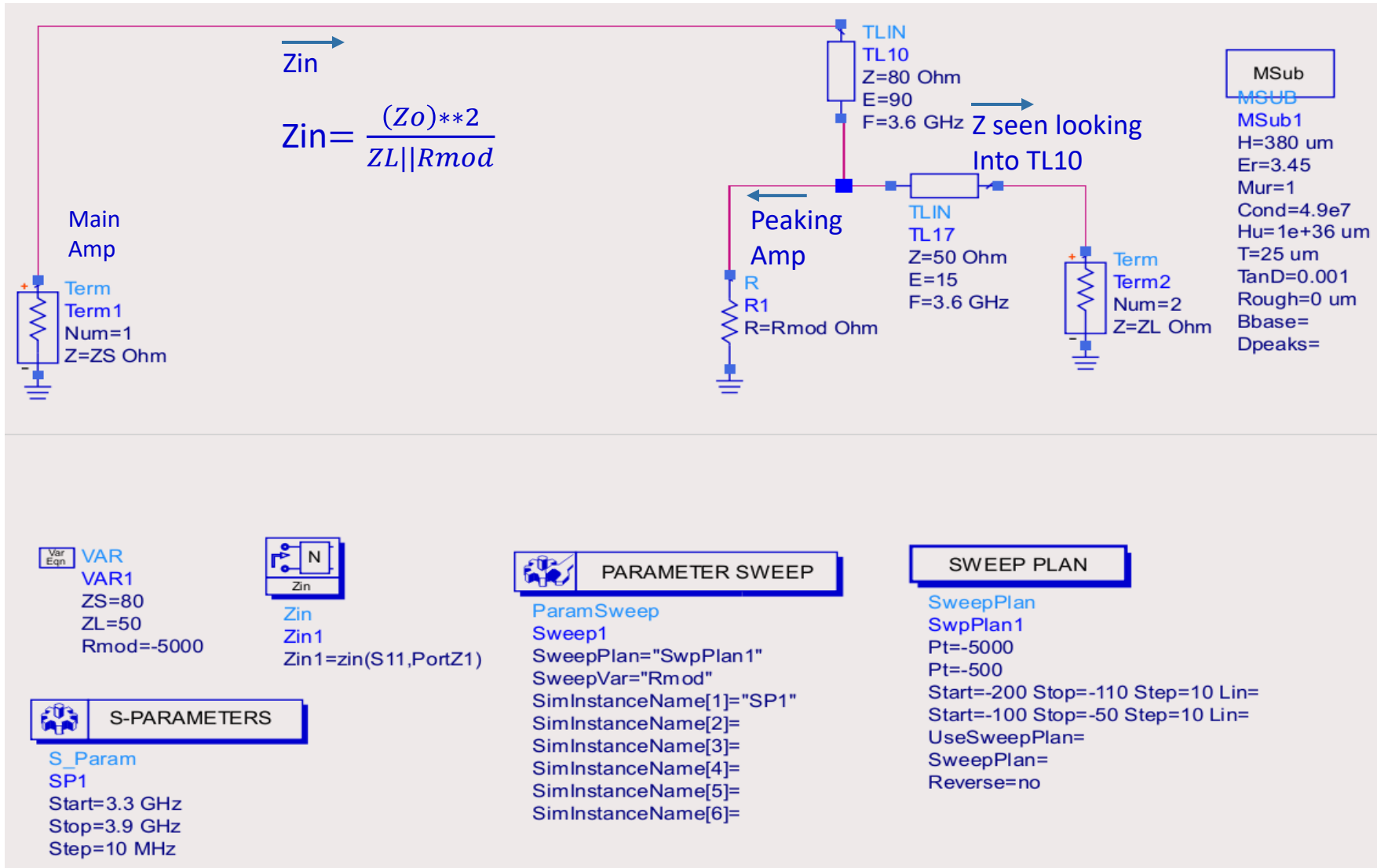
Final Stage Doherty Power Combiner Design



The Transmission line transform 76 ohms to 266 ohms impedance.

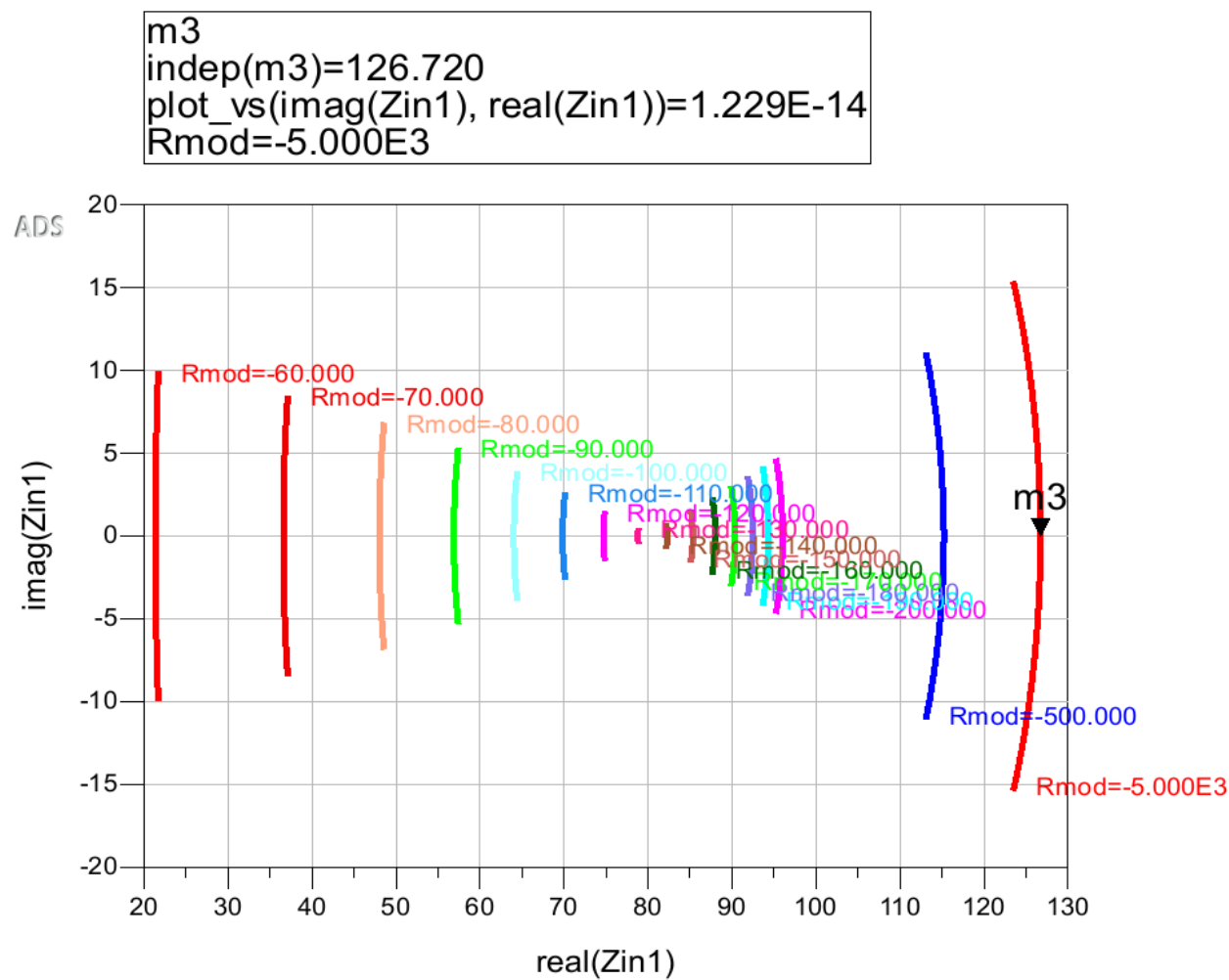
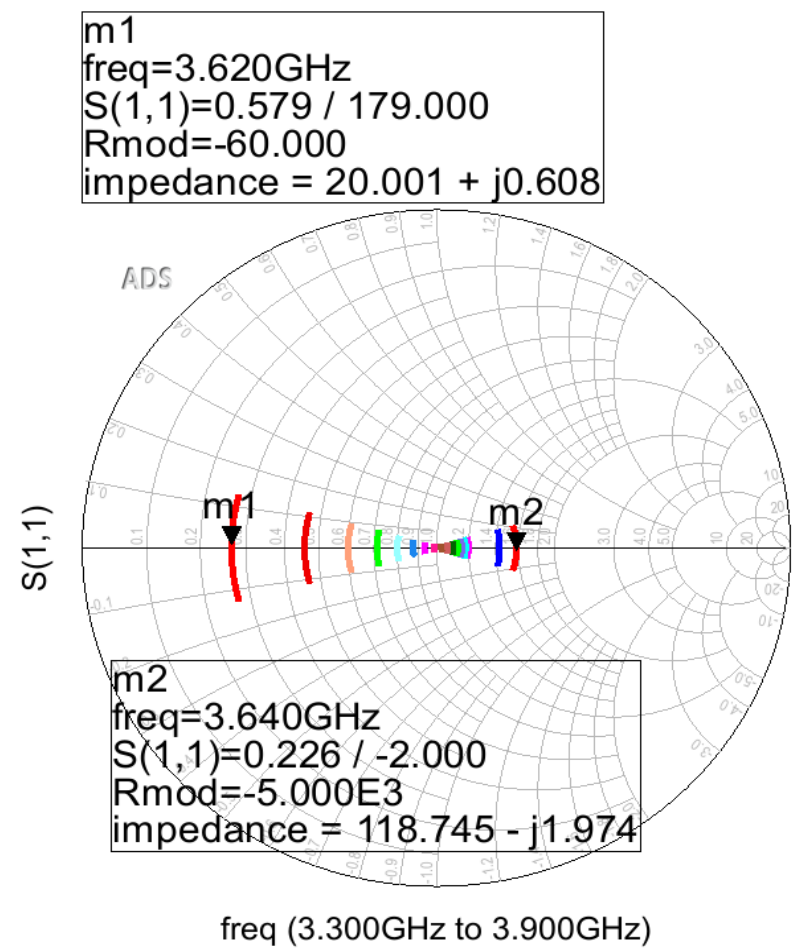
Final Stage Doherty Power Combiner Analysis

Sweeping the peaking Amp impedance



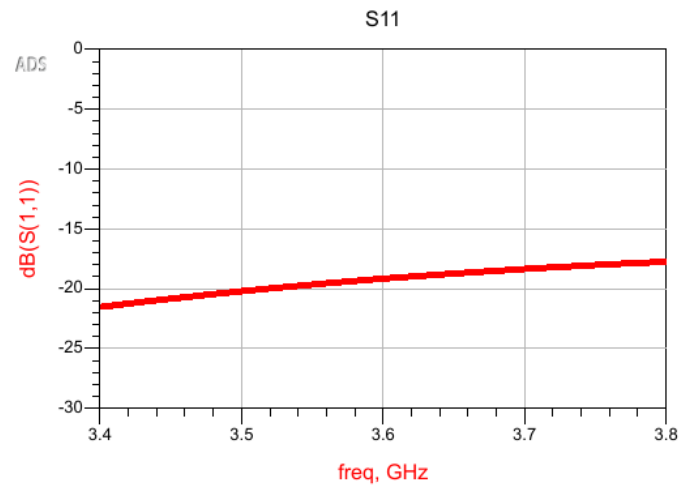
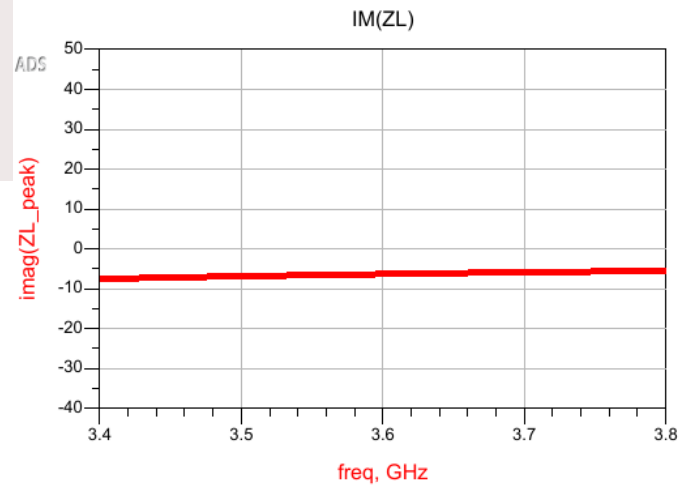
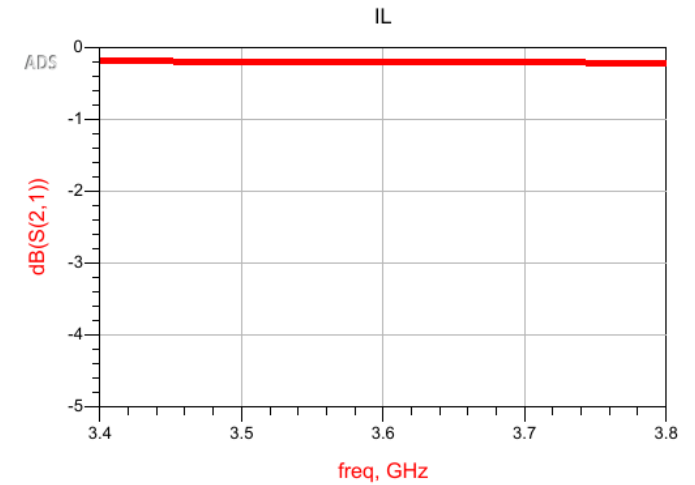
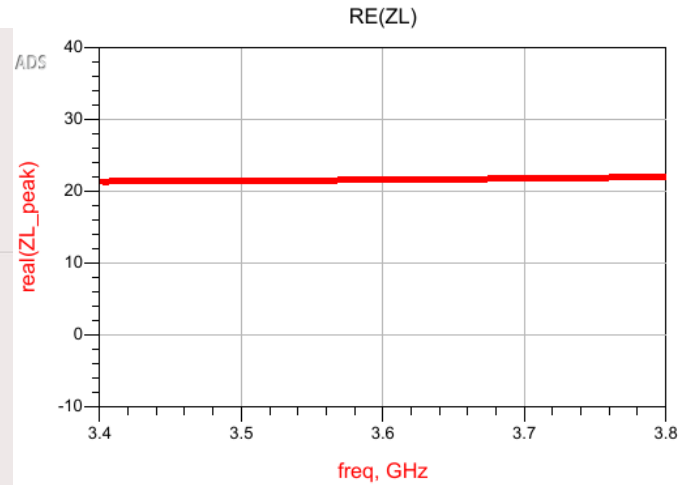
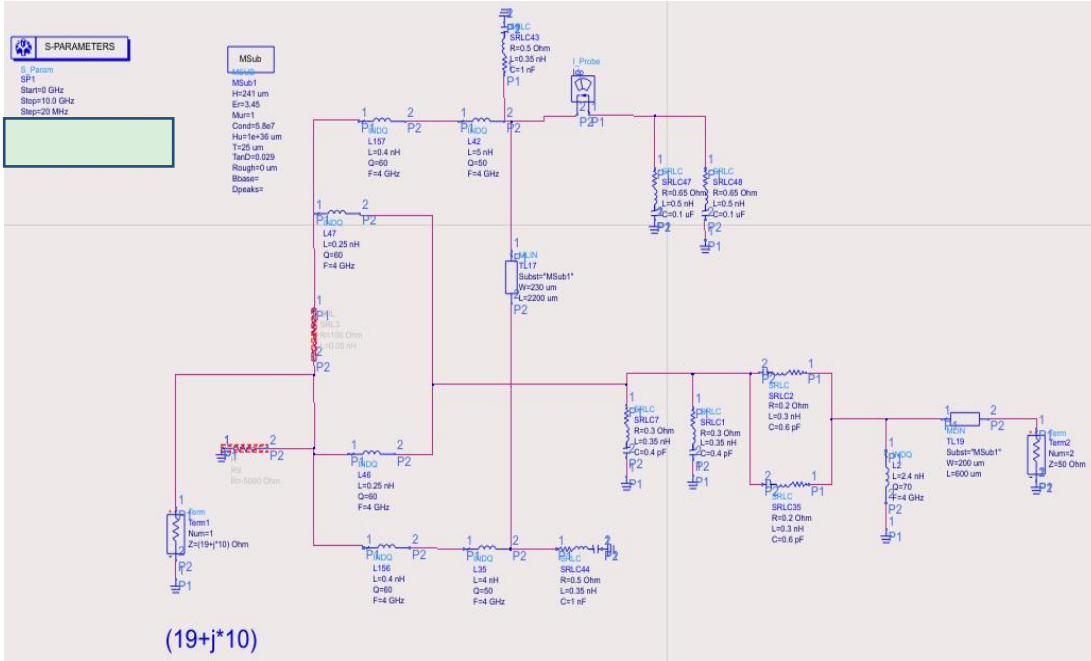
Rmod=-5000 Ohms model
Peaking amp is OFF
Rmod=-50 Ohms model
Peaking amp is ON

Final Stage Doherty Power Combiner Analysis



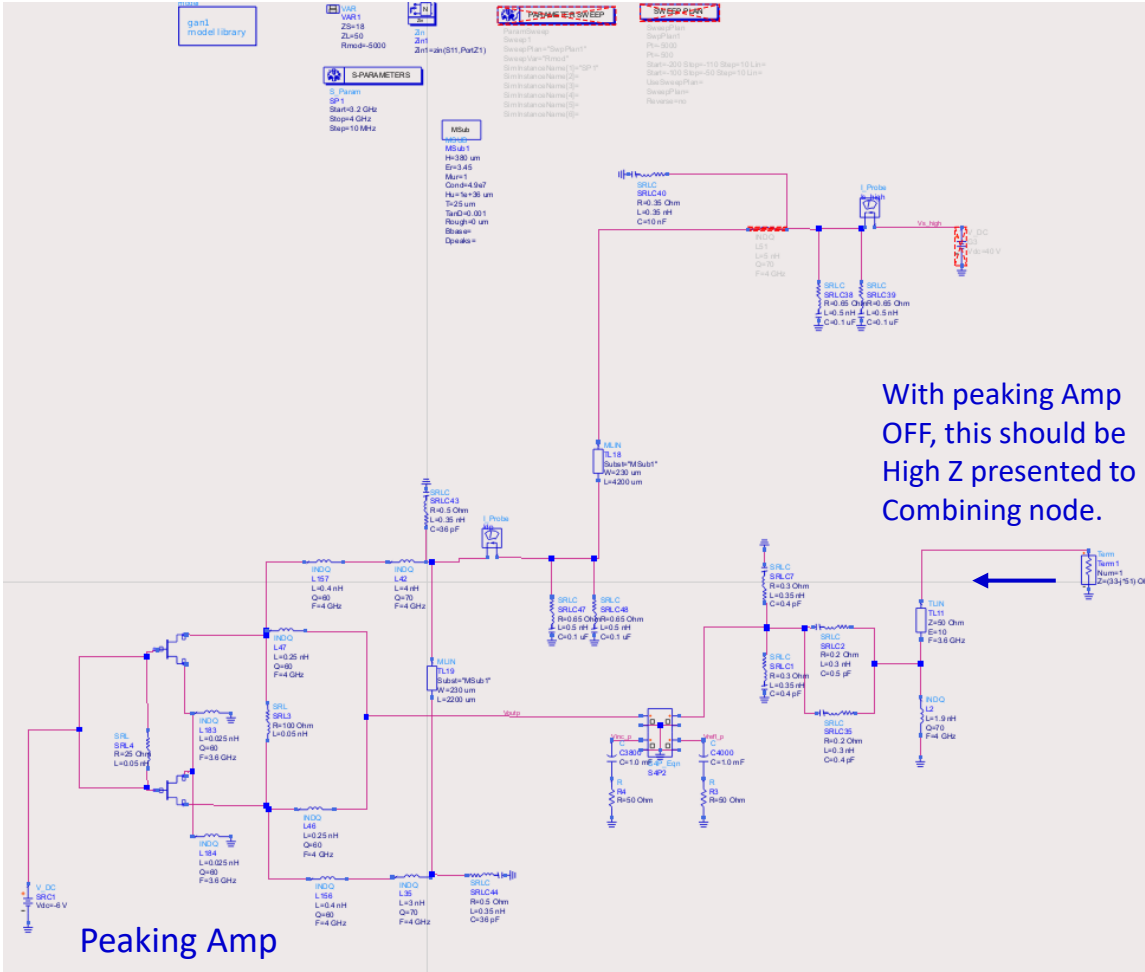
Zo = 50 Ohms, when Rmod = -5000 (peaking amplifier off), Zin = 118 Ohms. When Rmod = -60 (peaking amplifier on), Zin = 20 Ohms.

Peaking Amp Matching Network and Load-line

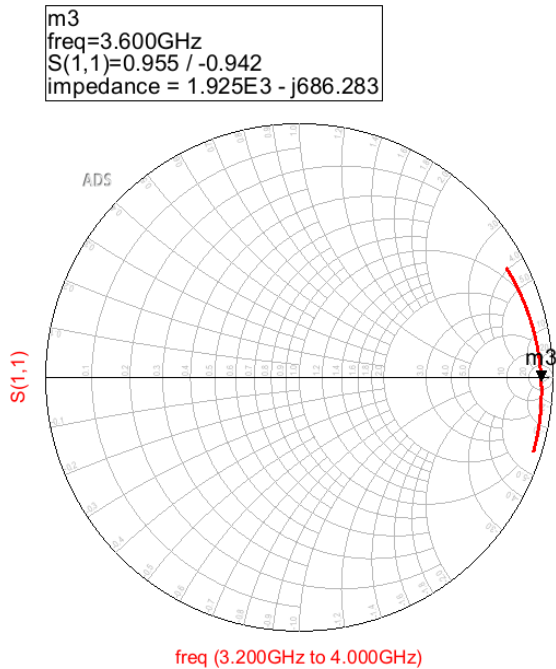
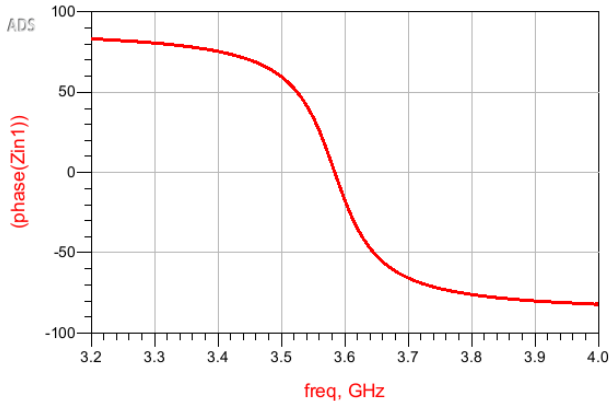
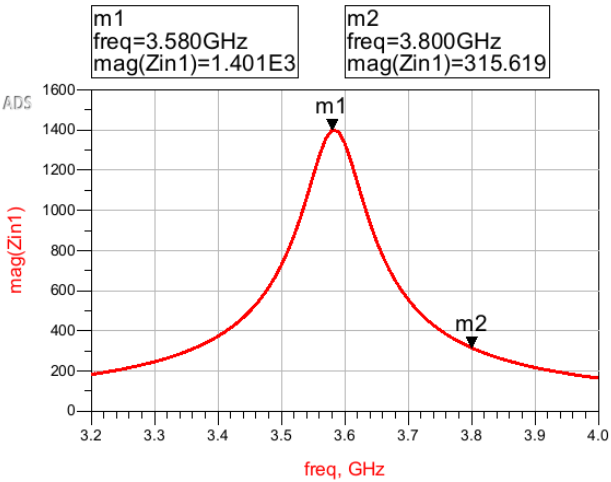


Final Stage Doherty Power Combiner Analysis

Impedance presented to the combining node by the peaking amplifier when it is biased off



This should be a relatively high impedance



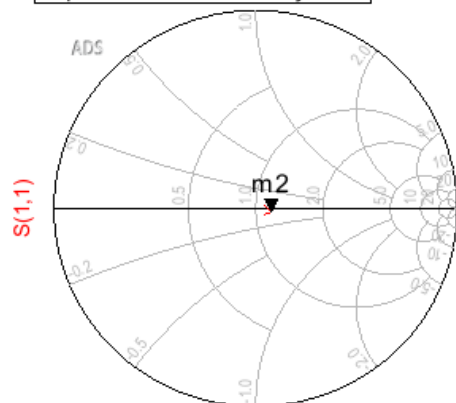
Main Amp and Peaking Amp (OFF) with power combining network



Final Stage DPA Output Matching Network

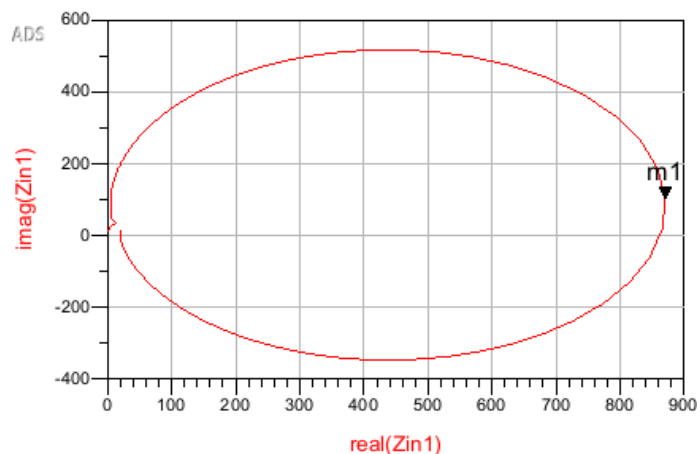
Main Amp and Peaking Amp (OFF) with power combining network

m2
freq=3.600GHz
S(1,1)=0.079 / -11.213
impedance = 87.559 - j2.708

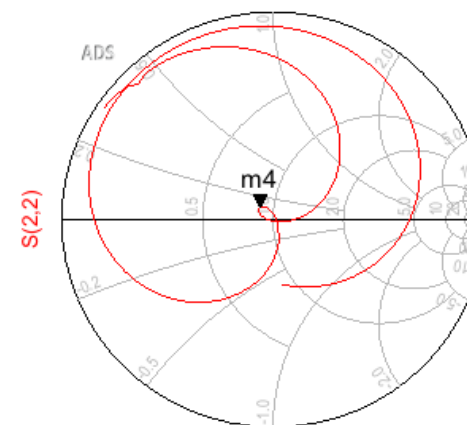


freq (3.300GHz to 3.900GHz)

m1
indep(m1)=870.390
plot_vs(imag(Zin1), real(Zin1))=101.572

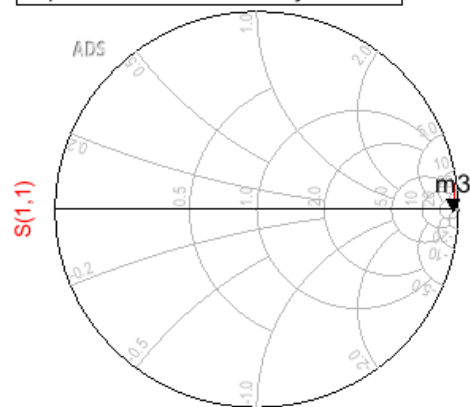


m4
freq=3.600GHz
S(2,2)=0.086 / 136.124
impedance = 43.871 + j5.265

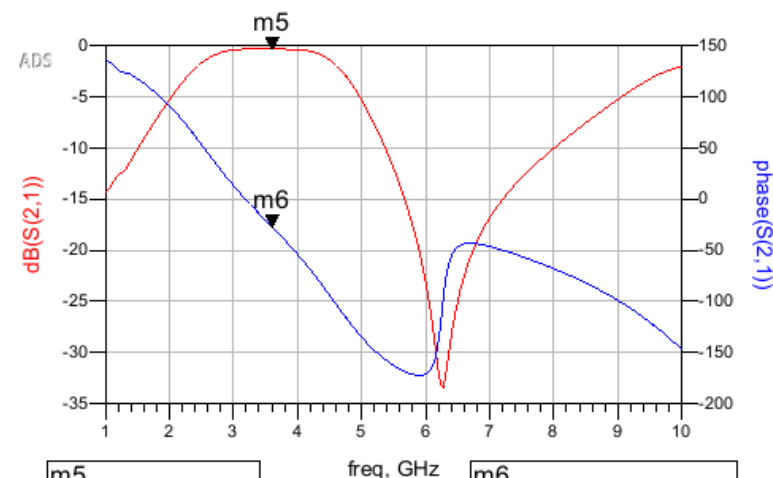
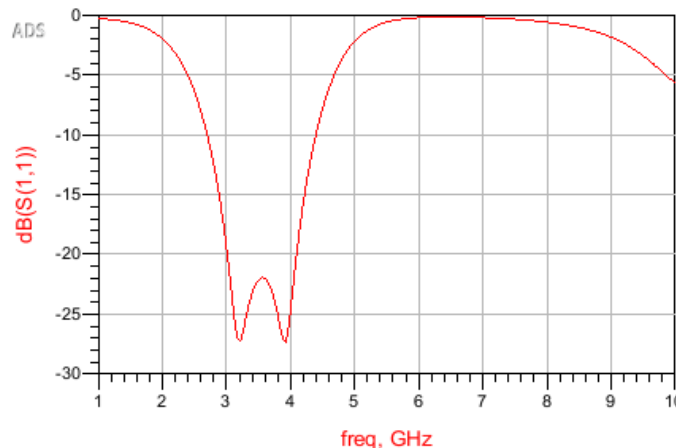


freq (1.000GHz to 10.00GHz)

m3
freq=7.200GHz
S(1,1)=0.973 / -0.711
impedance = 4.602E3 - j2.127E3



freq (6.600GHz to 7.200GHz)



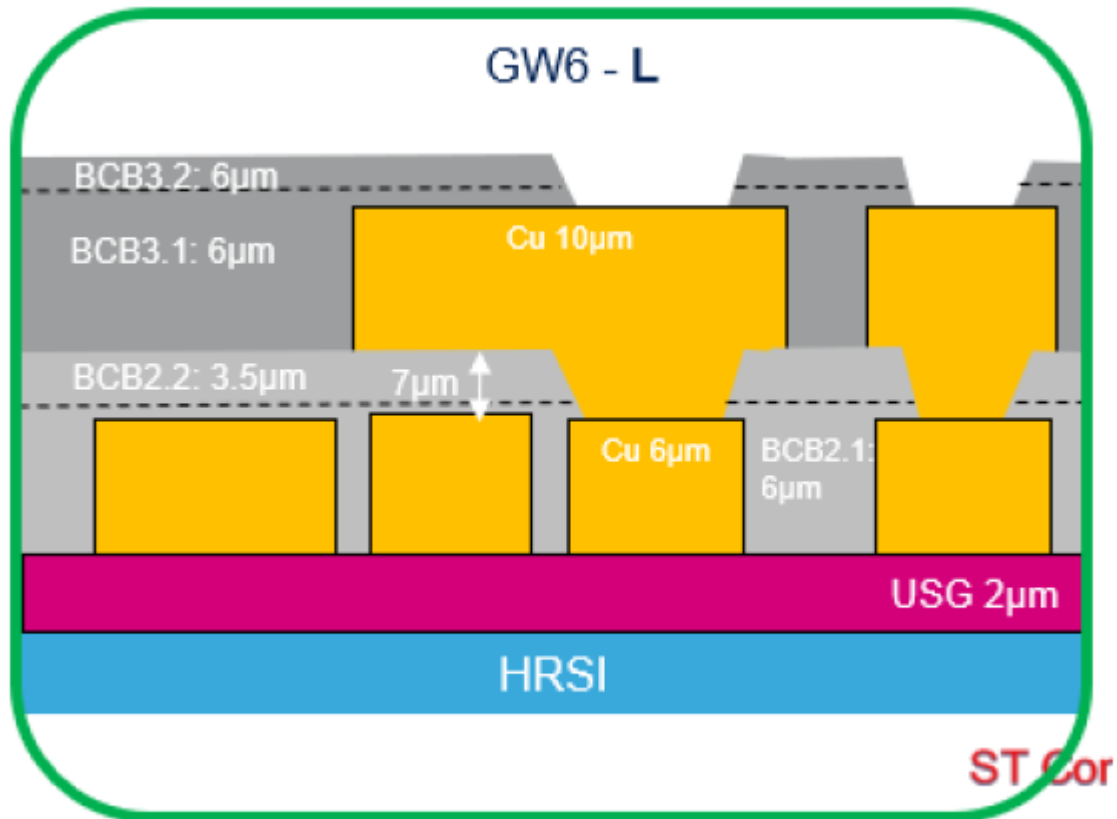
m5
freq=3.600GHz
dB(S(2,1))=-0.370

m6
freq=3.600GHz
phase(S(2,1))=-27.242

Impedance Inverter Design

RLC06E-ST IPD

RLC06E Process Technology by ST



IPD Technology

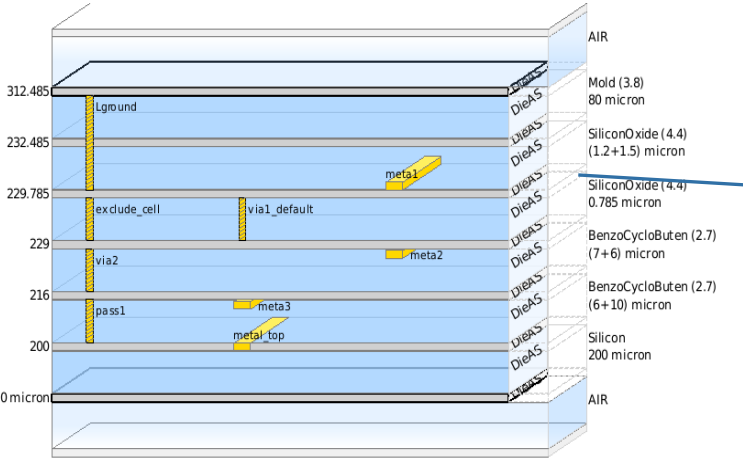
- RLC06E

Options

- L
- CSP u-bumping

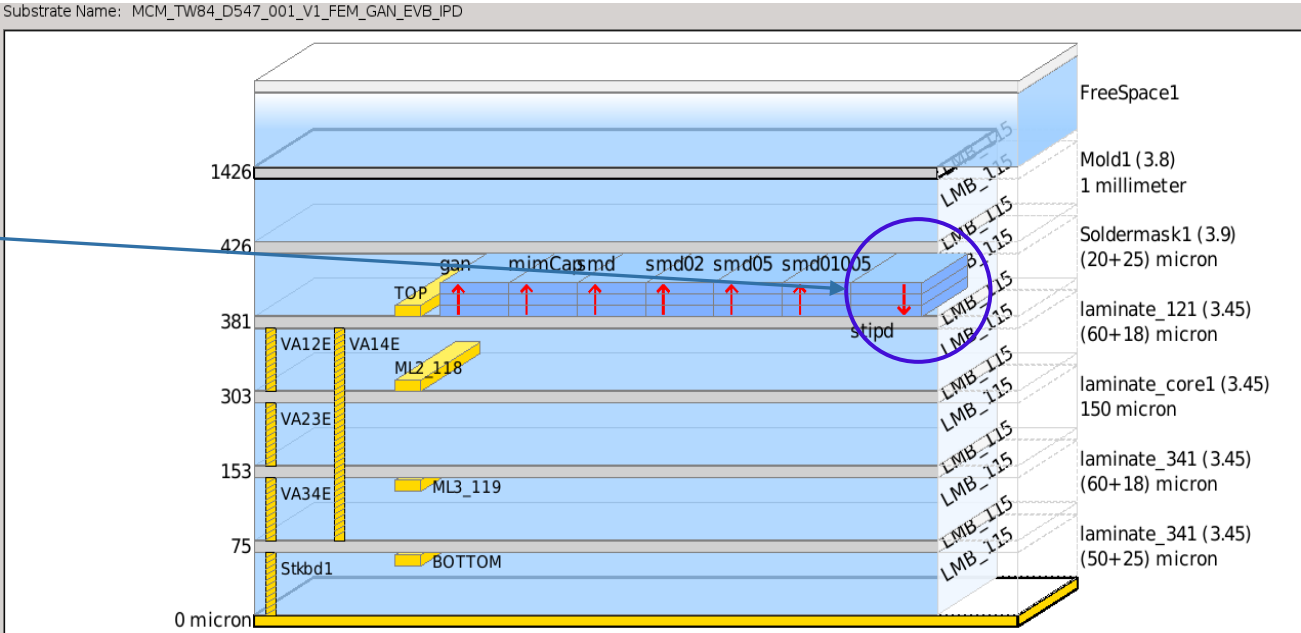
IPD Nested Substrate Stack-up (RLC06E Process)

IPD substrate



Substrate Layer Stackup				Substrate Vias				
Type	Name	Material	Thickness	Type	Name	Top	Bottom	Material
Dielectric		AIR		Conductor Via	pass1 (12)	meta3 (15)	metal_top (...)	TiNiAu
Dielectric		Mold	80 um	Conductor Via	via2 (14)	meta2 (9)	meta3 (15)	Electrolytic...
Dielectric		SiliconOxide	1.2 um	Conductor Via	exclude_cell (...)	meta1 (11)	meta2 (9)	Electrolytic...
3 Conductor Layer	meta1 (11)	Aluminium	1.5 um	Conductor Via	via1_default (...)	meta1 (11)	meta2 (9)	Electrolytic...
Dielectric		SiliconOxide	0.785 um	Conductor Via	Lground (30)	Interface 1	meta1 (11)	Electrolytic...
4 Conductor Layer	meta2 (9)	Electrolytic_Copper	6 um					
Dielectric		BenzoCycloButen	7 um					
5 Conductor Layer	meta3 (15)	Electrolytic_Copper	10 um					
Dielectric		BenzoCycloButen	6 um					
6 Conductor Layer	metal_top (13)	TiNiAu	1 um					
Dielectric		Silicon	200 um					
Dielectric		AIR						

Nested IPD to Laminate substrate



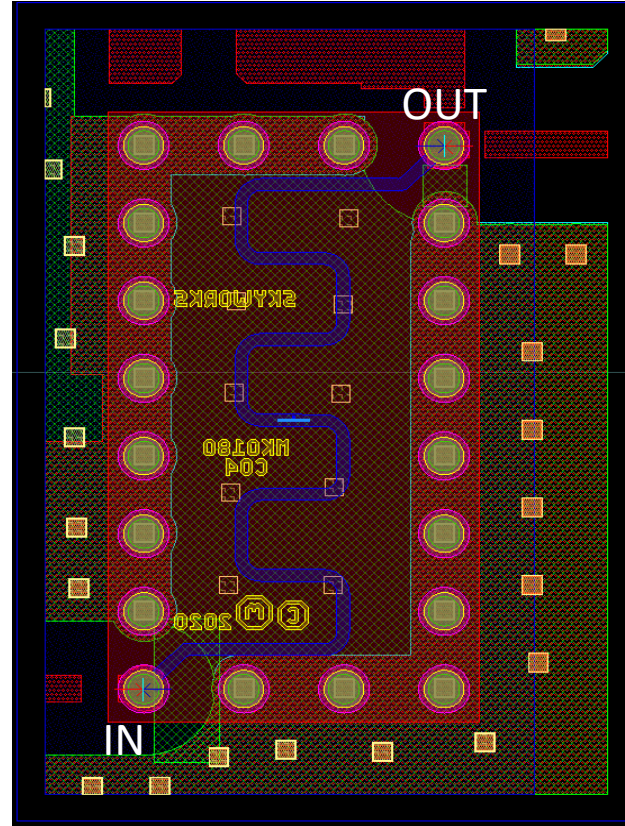
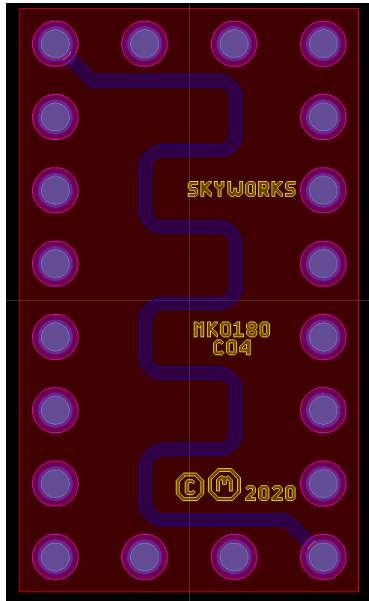
Substrate Layer Stackup				Substrate Vias				
Type	Name	Material	Thickness	Type	Name	Top	Bottom	Material
Dielectric		FreeSpace1		Conductor Via	VA23E (126)	ML2_118 (...)	ML3_119 (...)	VA23E1
Dielectric		Mold1	1 mm	Conductor Via	VA34E (127)	ML3_119 (...)	BOTTOM (...)	VA34E1
Dielectric		Soldermask1	20 um	Conductor Via	VA12E (125)	TOP (117)	ML2_118 (...)	VA12E1
3 Conductor Layer	TOP (117)	TOP1	25 um	Conductor Via	VA14E (143)	TOP (117)	BOTTOM (...)	VA14E1
Dielectric		laminate_121	60 um	Conductor Via	Stkbd1 (174)	BOTTOM (...)	Bottom Co...	VA14E1
4 Conductor Layer	ML2_118 (118)	ML2_1181	18 um					
Dielectric		laminate_core1	150 um					
5 Conductor Layer	ML3_119 (119)	ML3_1191	18 um					
Dielectric		laminate_341	60 um					
6 Conductor Layer	BOTTOM (122)	BOTTOM1	25 um					

MK0180 Version C04 on TW84_D547_001_V1

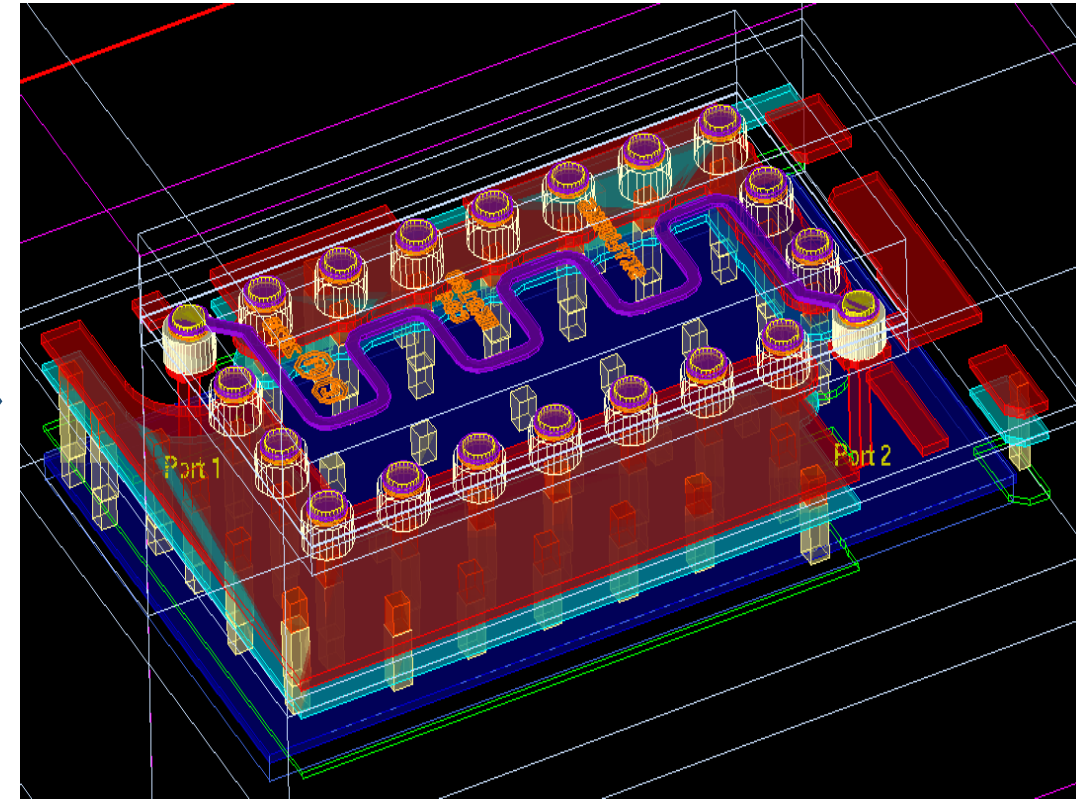
Nested IPD to Laminate

Ground reference: L3

ST IPD



3D View of Nested IPD to Laminate



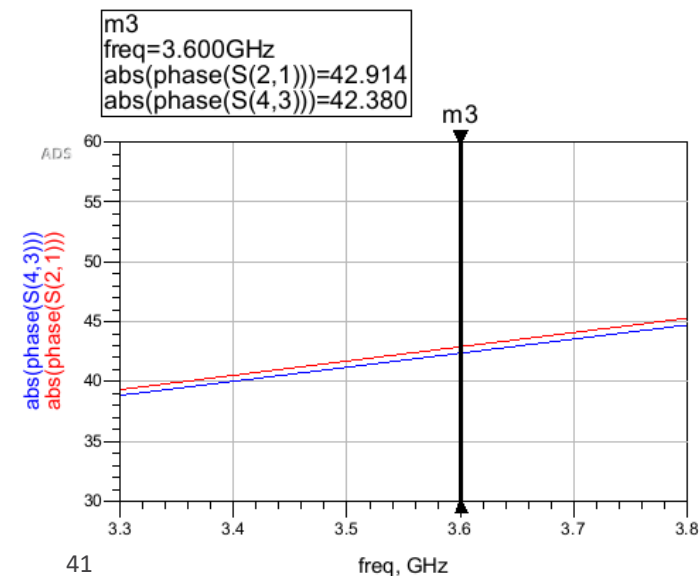
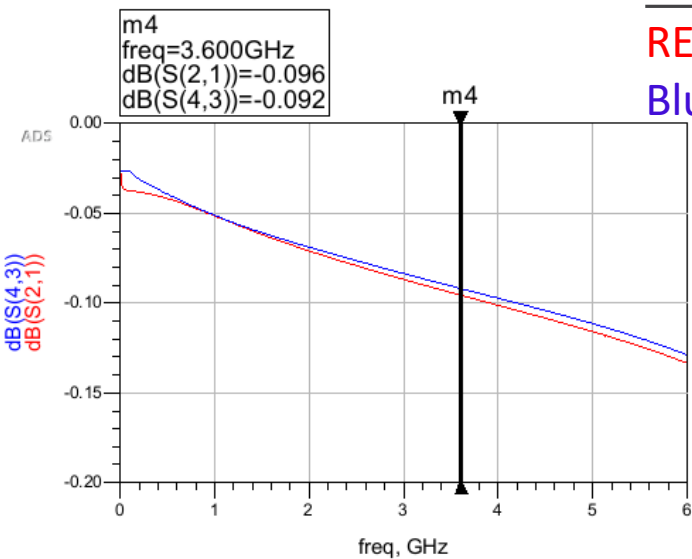
- High impedance $> 75 \Omega$
- Low insertion loss
- Less tolerance variation
- Better thermal management

MK0180 Version C04 on TW84_D547_001_V1

Comparison of SKY vs ST EM simulations

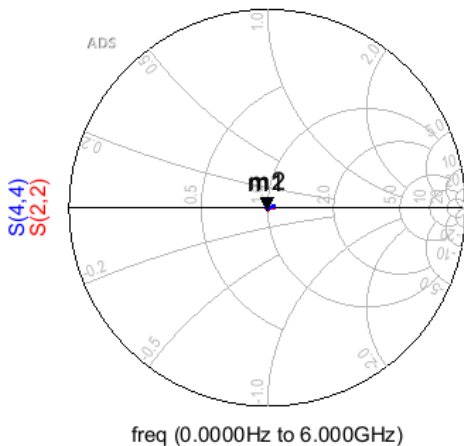
RED: SKY Sim reference GND is L3

Blue: ST Sim reference GND is L3



m2
freq=3.600GHz
S(4,4)=0.006 / -77.230
impedance = 75.183 - j0.827

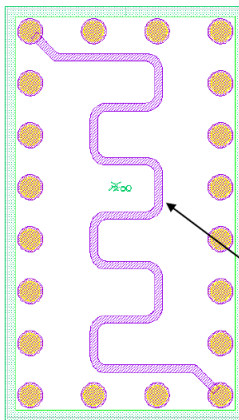
m1
freq=3.600GHz
S(2,2)=0.013 / -108.181
impedance = 74.383 - j1.805



OPTION 2

GND on L2

BUMP SIDE VIEW



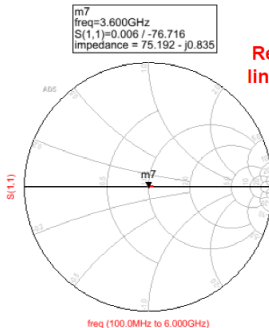
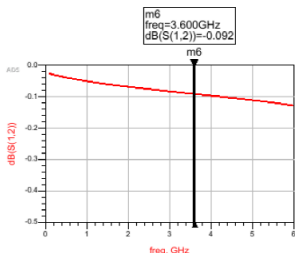
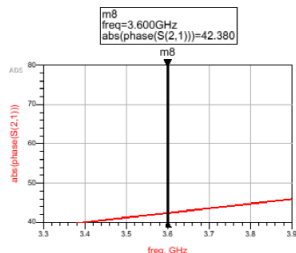
Die size after sawing
1310 x 2280 µm²

	42 Deg	46 Deg	50 Deg	55 Deg	60 Deg
65 Ohms			A04	A05	
70 Ohms		B04	B05		
75 Ohms	C04	C05	C01	C02	C03
80 Ohms		D03	D04		

ST Simulation

EM simulation results

Line in M3 only



Results obtained when
line load by 75Ω IN/OUT

ST Confidential

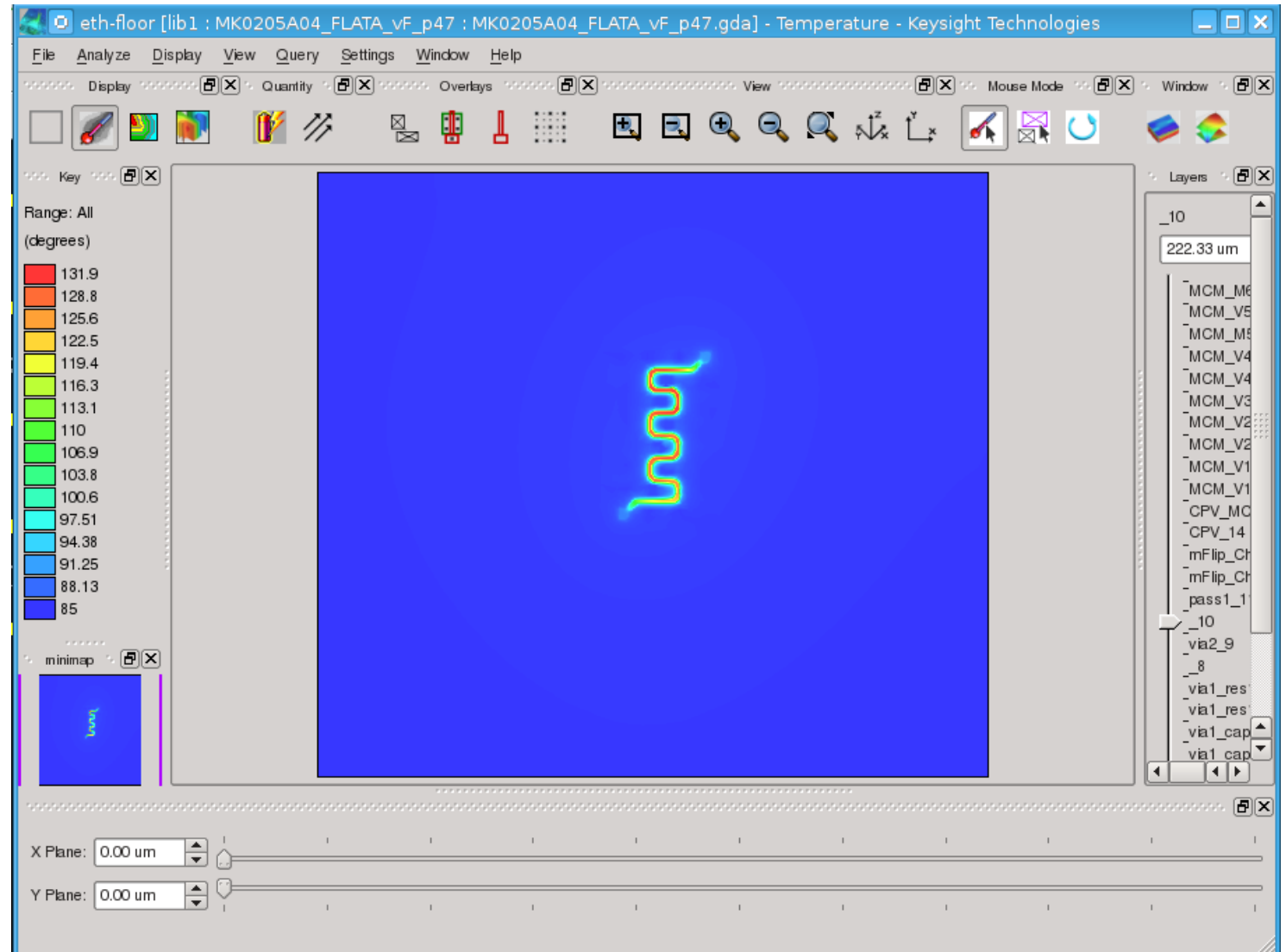
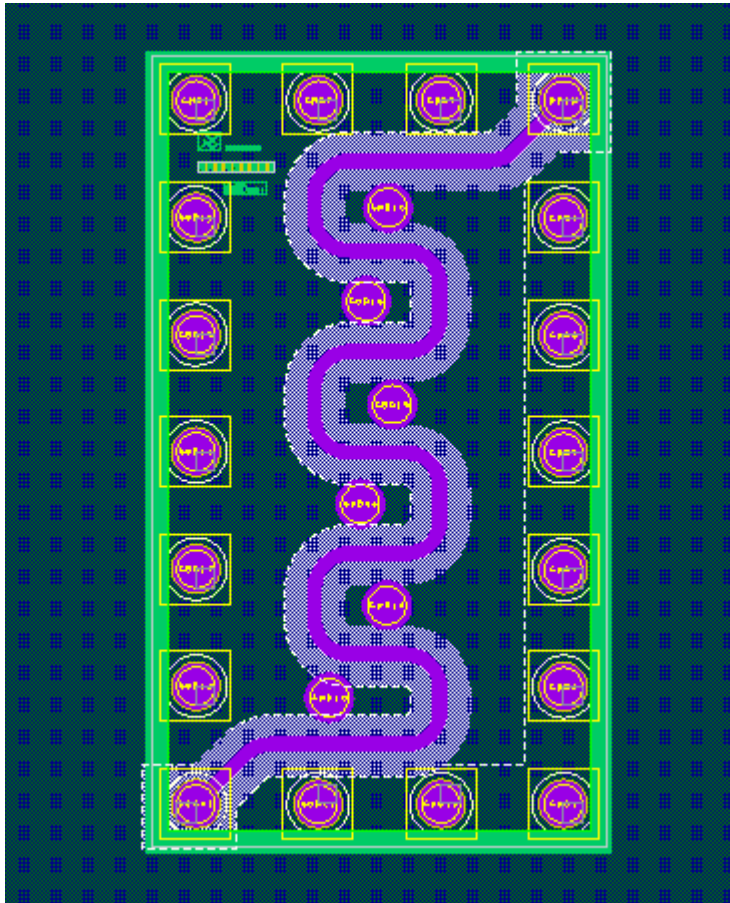
SKYWORKS®

ST IPD

ETH Thermal Simulation

38 um; GND REF=Met4

Pdiss 0.47 watt



Input Matching Network IPD Design

HIPD1-Process

IPD Input Matching Design

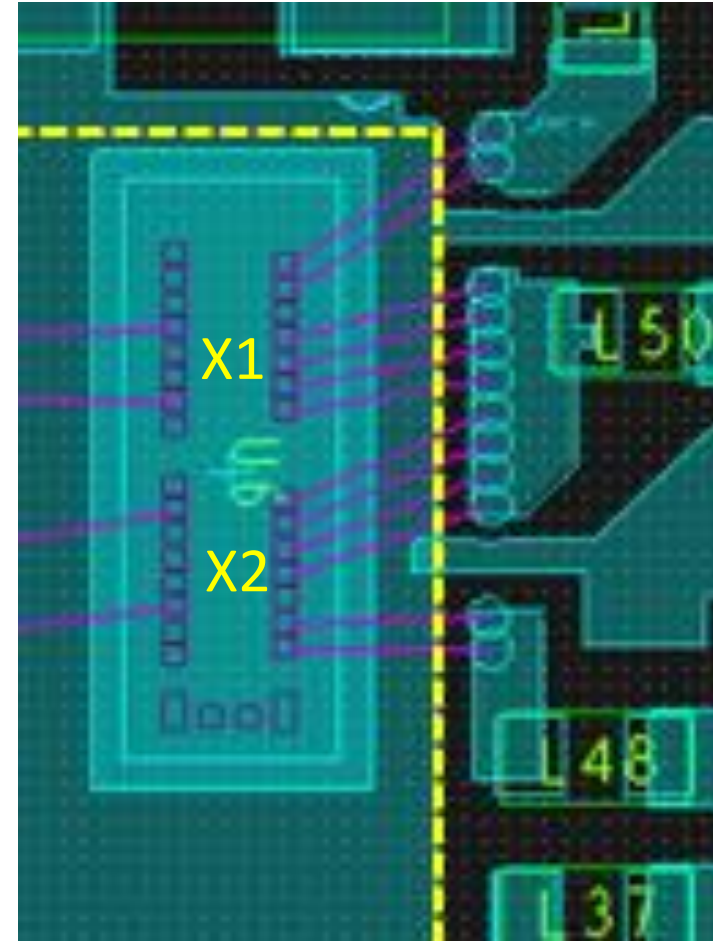
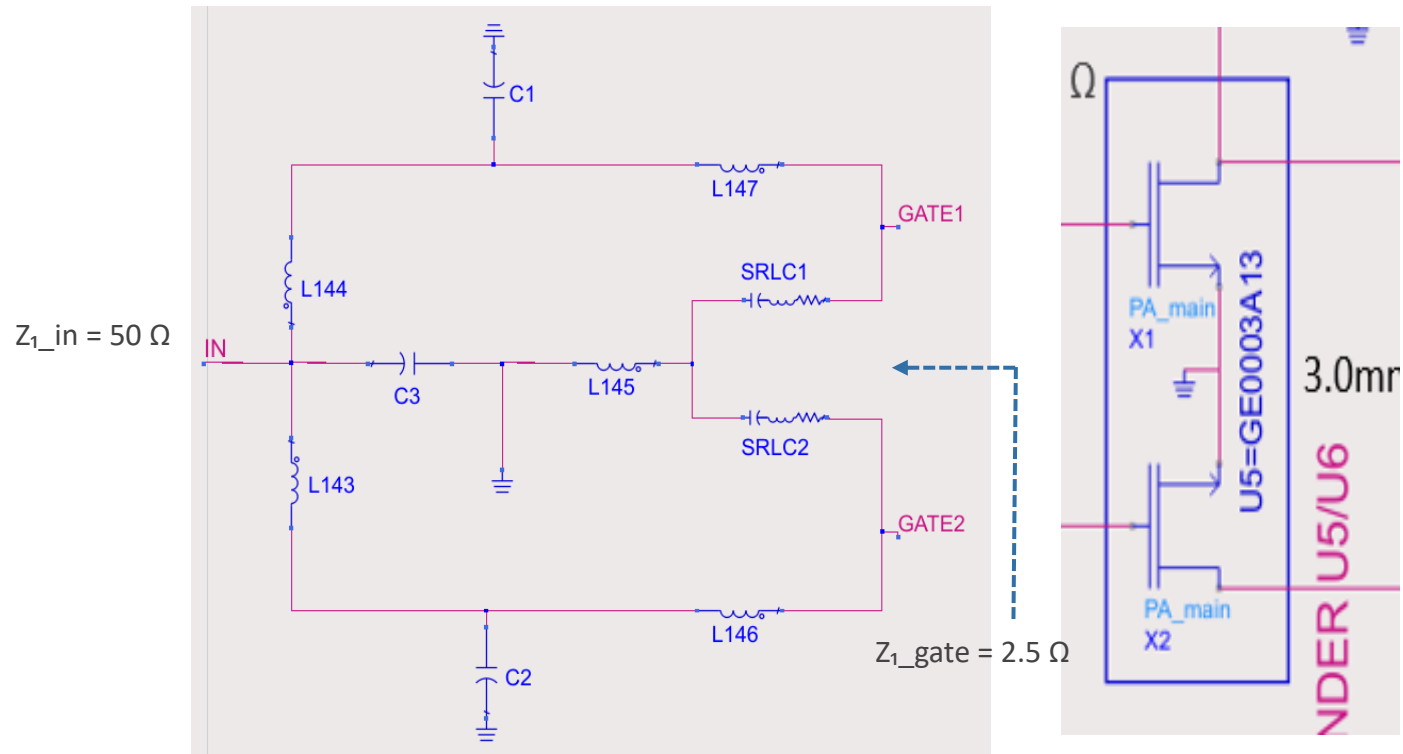
❑ HIPD1 Overview

- Utilize Hipd1 process, Skyworks internal GaAs IPD solution.
- HIPD1 die cost is cheaper than of WIN Semi.
- Provide gate matching impedance to the input coupler.
- Minimum insertion loss

❑ Frequency Band

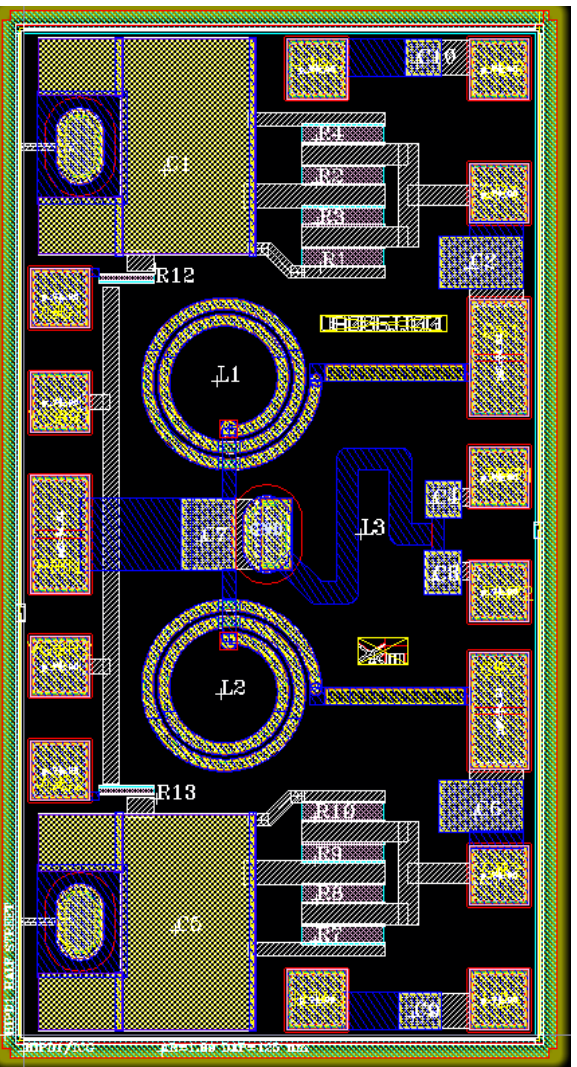
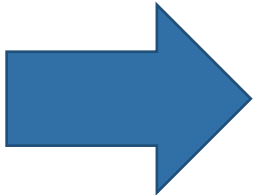
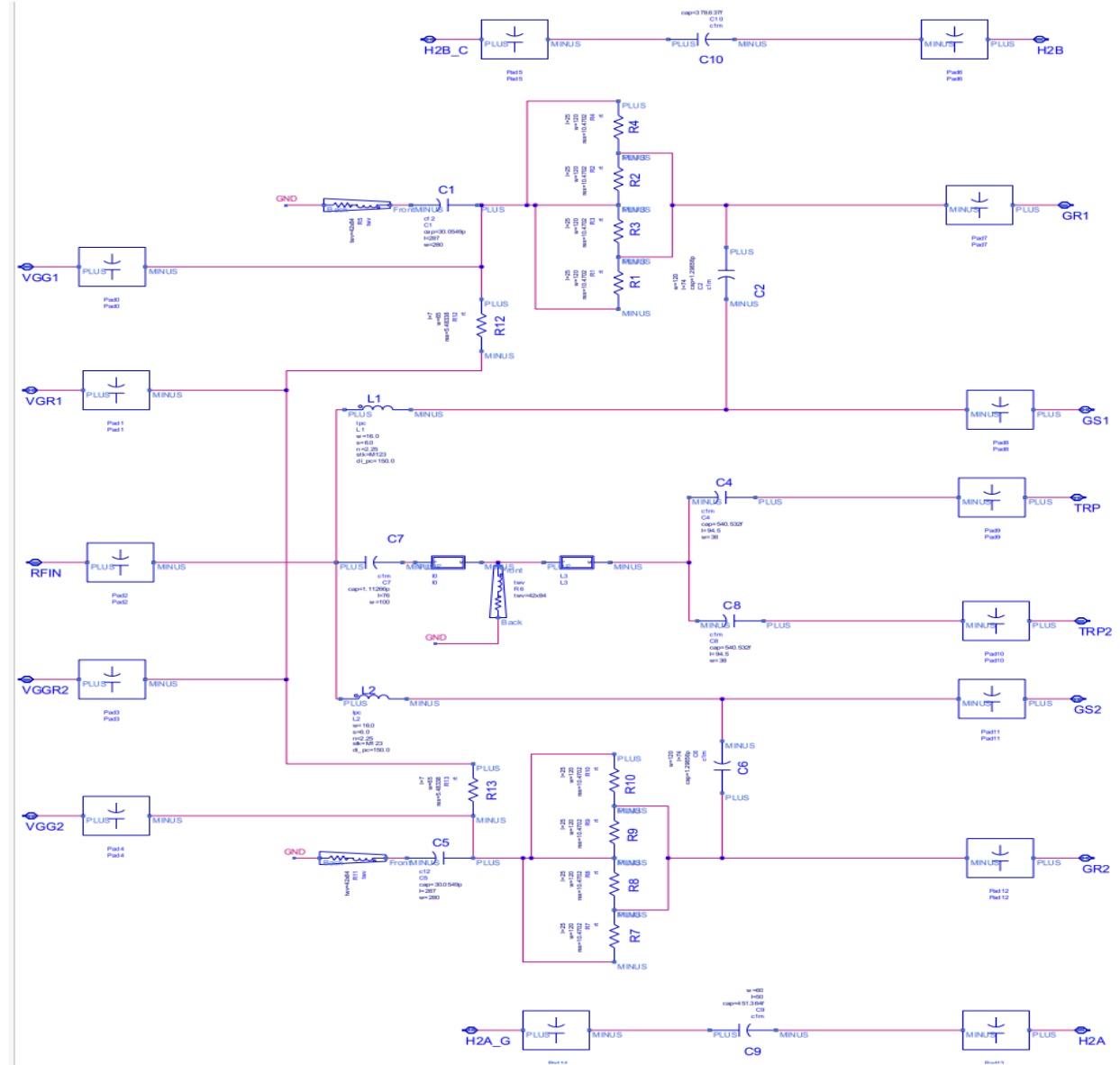
- 3.4 to 3.8 GHz

Input Matching Network Architecture



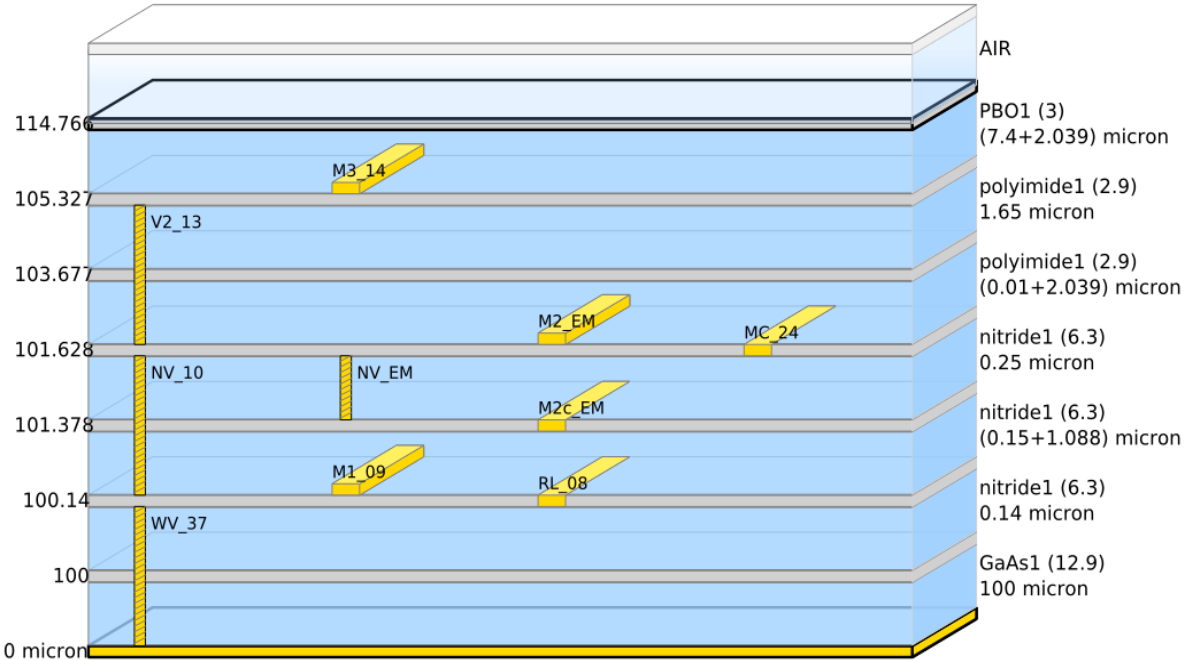
GaN Device Layout

IPD Schematic and Layout



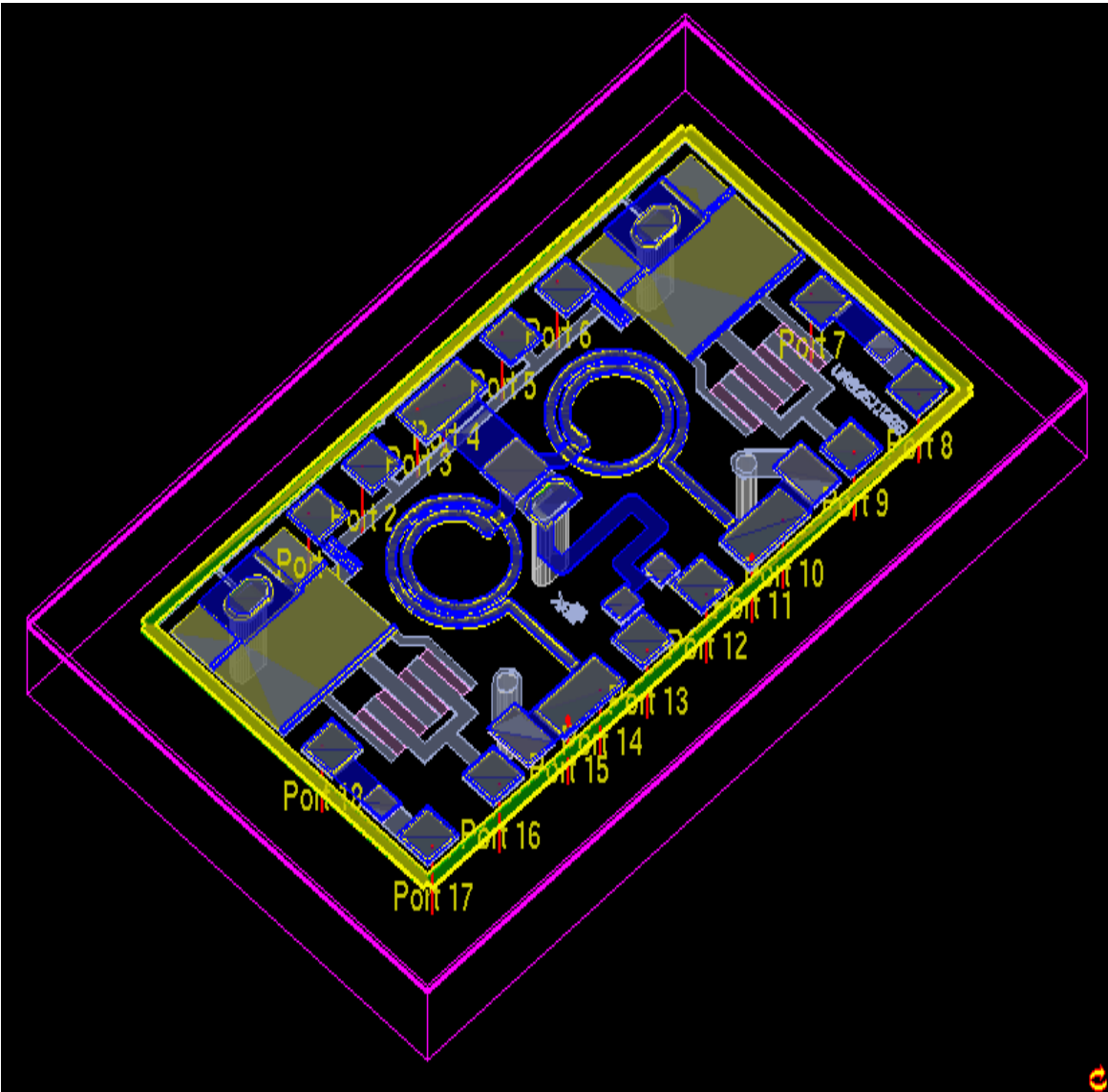
Die size: 0.81x1.528mm

HIPD1 Momentum EM Simulation Stack-Up



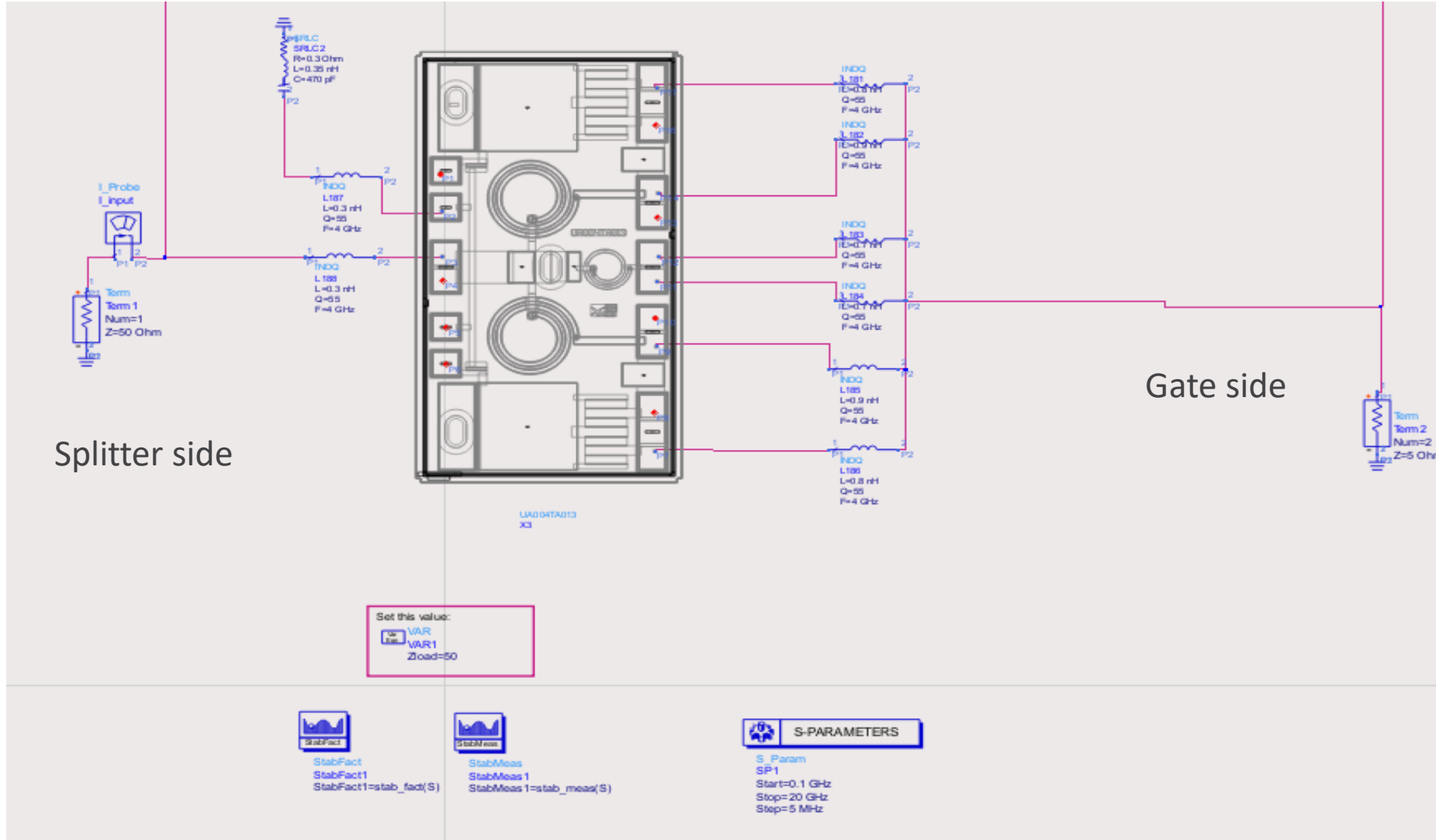
Derived Layers

Layer Display Properties						
Layers		Purposes		Display Order		
Layer Name	Num	Library	Process Role	Bindi	Derived Layer Operation	
M12_EM	42	hipd1	Conductor			
NV_EM	43	hipd1	Conductor		DIFF("PV_EM", "MC_24")	
PV_EM	44	hipd1	Conductor		AND("CAP", "V2_13")	
TXTWV	46	hipd1	Conductor			
VP_15	47	hipd1	Conductor			
MP_19	48	hipd1	Conductor			
MT_70	50	hipd1	Conductor			
M2c_EM	70	hipd1	Conductor		AND("M2_11", "CAP")	
M2_EM	95	hipd1	Conductor		DIFF("M2_11", "CAP")	



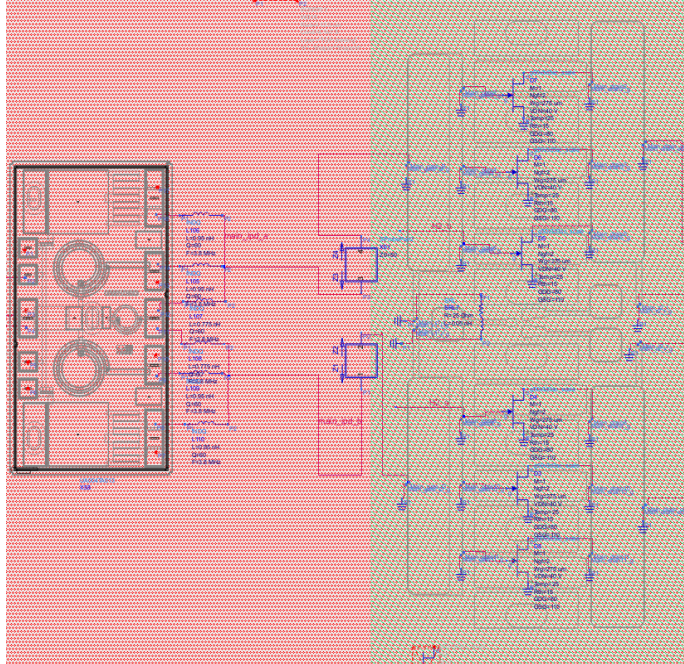
UA004TA013 IPD EM verification

Pre-offset match are not included

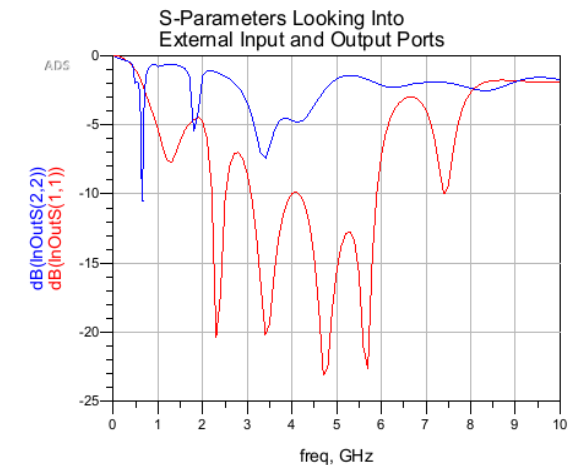
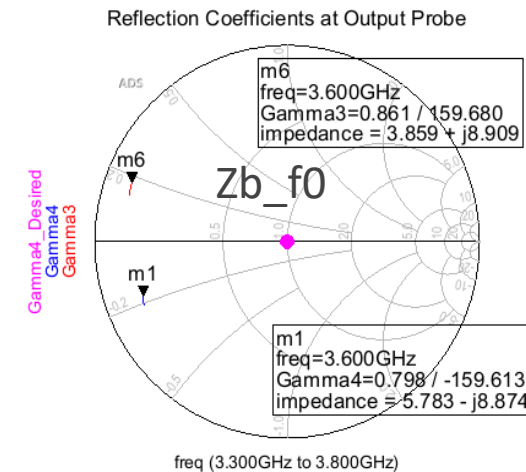
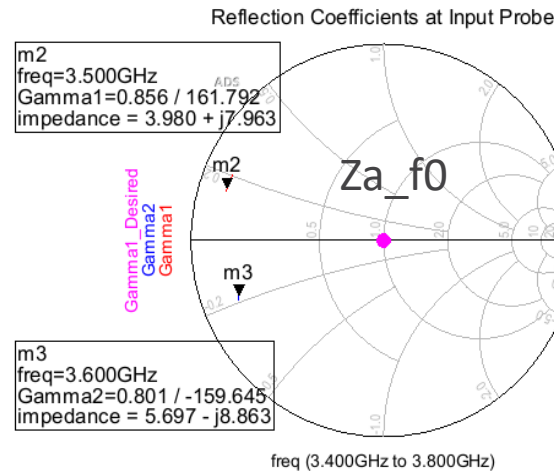
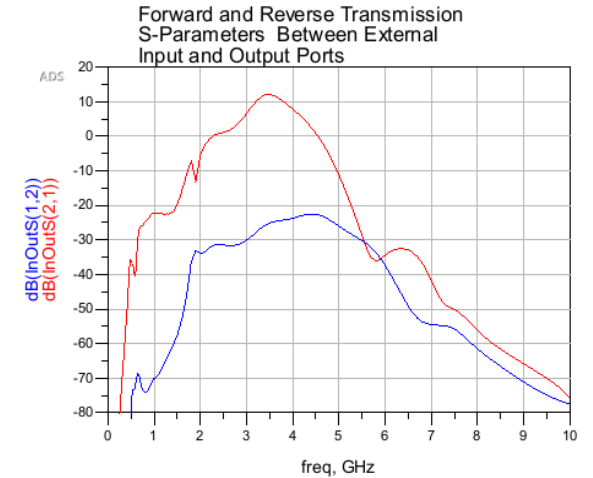
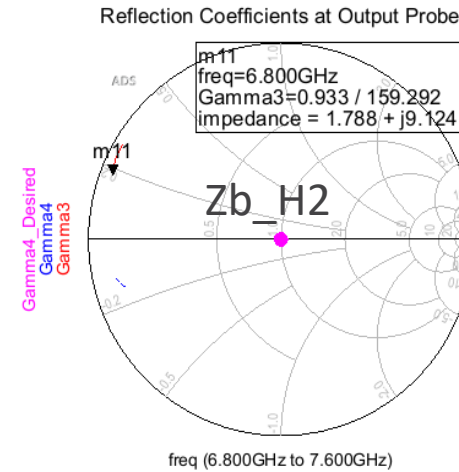
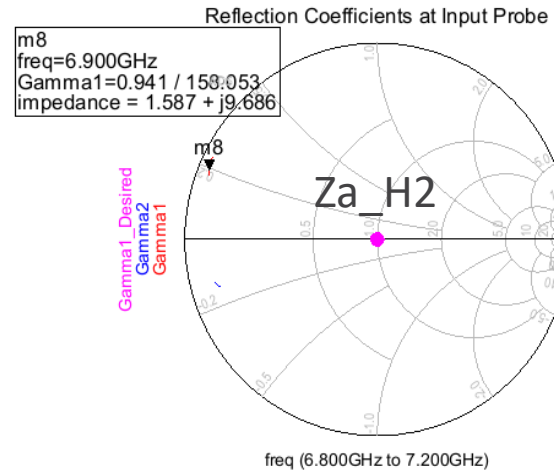


Sprobe at the Gate of the Main Amp

Gate Impedance with single H2-traps

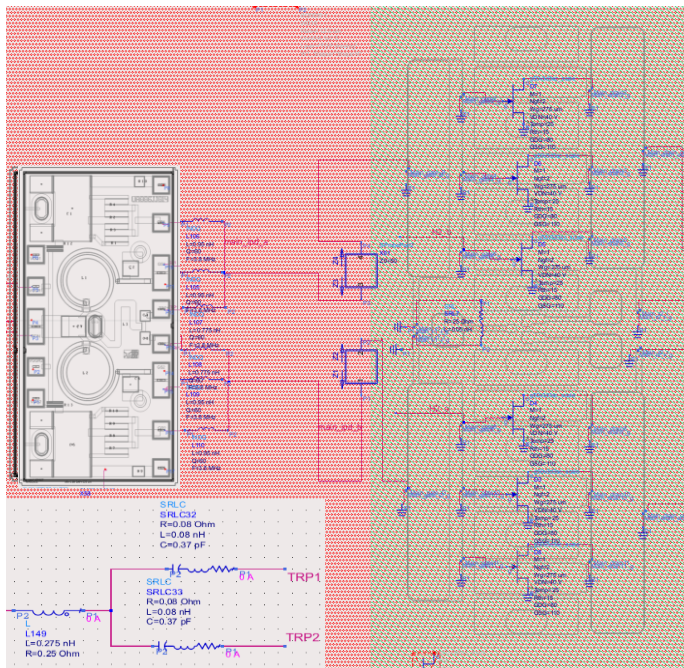


Gamma1=looking at the IPD
 Gamma2=looking at the Gate
 Gamma3=looking at the IPD
 Gamma4=looking at the Gate



Sprobe at the Gate of the Main Amp

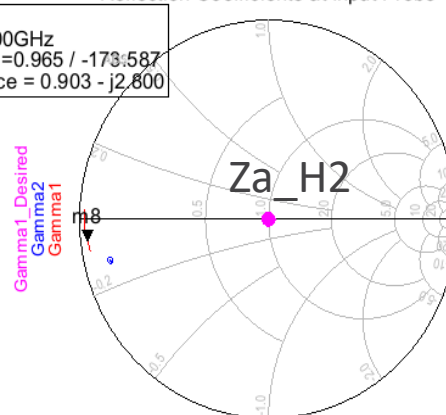
Gate impedance with two-H2 traps



Gamma1=looking at the IPD
 Gamma2=looking at the Gate
 Gamma3=looking at the IPD
 Gamma4=looking at the Gate

Reflection Coefficients at Input Probe

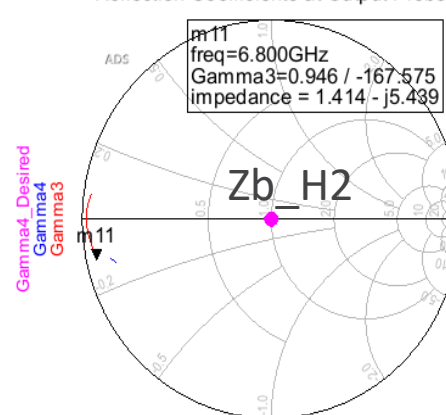
m8
 freq=6.900GHz
 Gamma1=0.965 / -173.587
 impedance = 0.903 - j2.600



freq (6.800GHz to 7.200GHz)

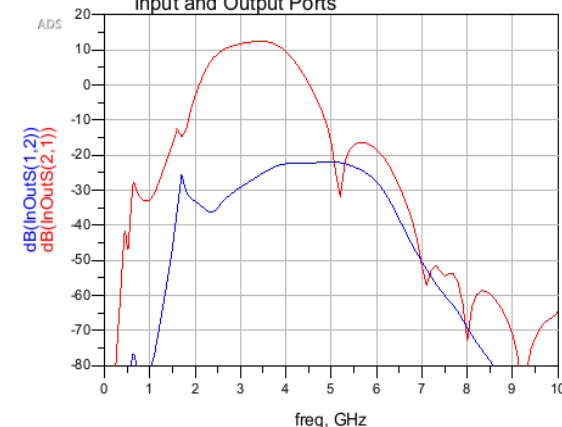
Reflection Coefficients at Output Probe

m11
 freq=6.800GHz
 Gamma3=0.946 / -167.575
 impedance = 1.414 - j5.439



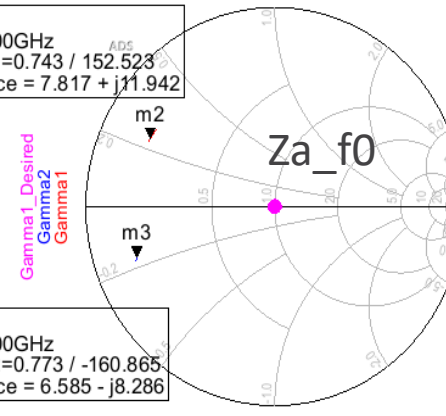
freq (6.800GHz to 7.600GHz)

Forward and Reverse Transmission S-Parameters Between External Input and Output Ports



Reflection Coefficients at Input Probe

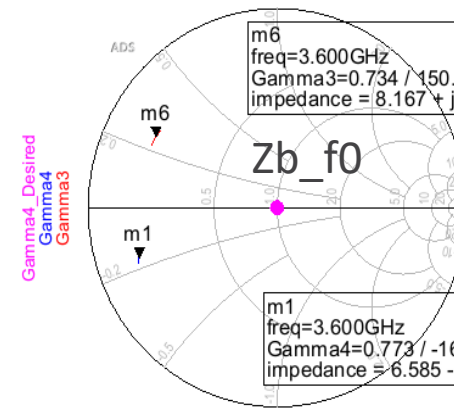
m2
 freq=3.500GHz
 Gamma1=0.743 / 152.523
 impedance = 7.817 + j11.942



freq (3.400GHz to 3.800GHz)

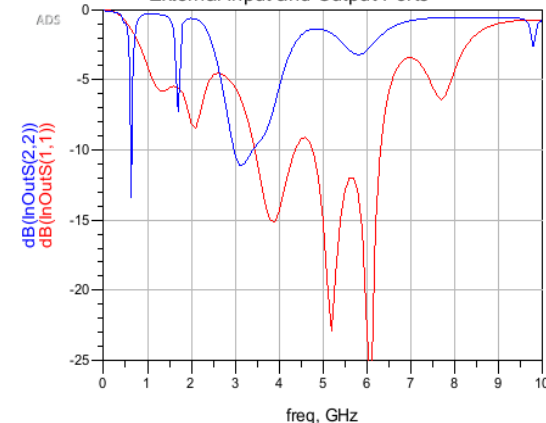
Reflection Coefficients at Output Probe

m6
 freq=3.600GHz
 Gamma3=0.734 / 150.942
 impedance = 8.167 + j12.634



freq (3.300GHz to 3.800GHz)

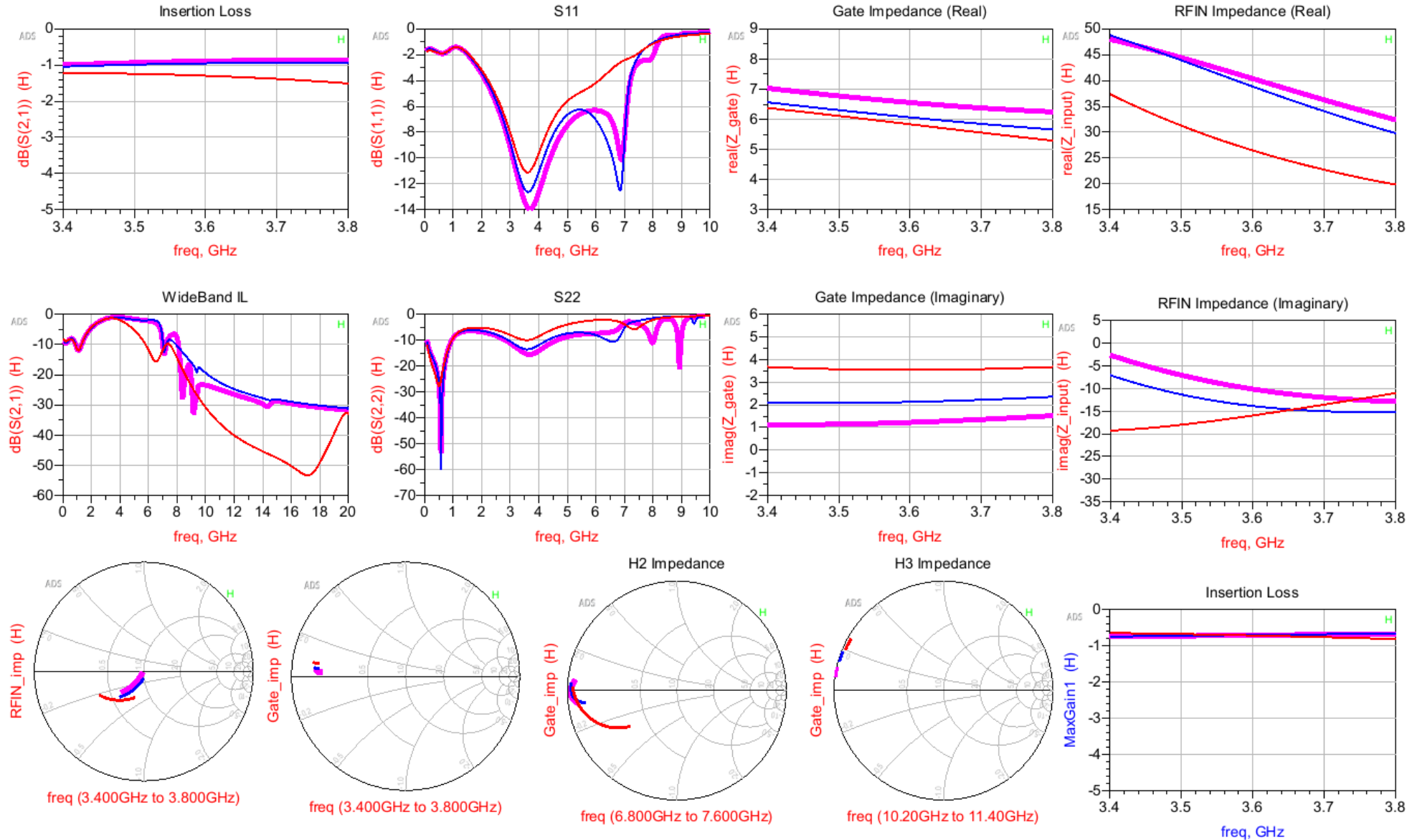
S-Parameters Looking Into External Input and Output Ports



UA006 IPD Simulation

Blue=UA006JJ008; Purple= UA006JJ008 (2Traps); Red=UA004TA013

(Zgate=7+j4Ω)



Power Splitter Design

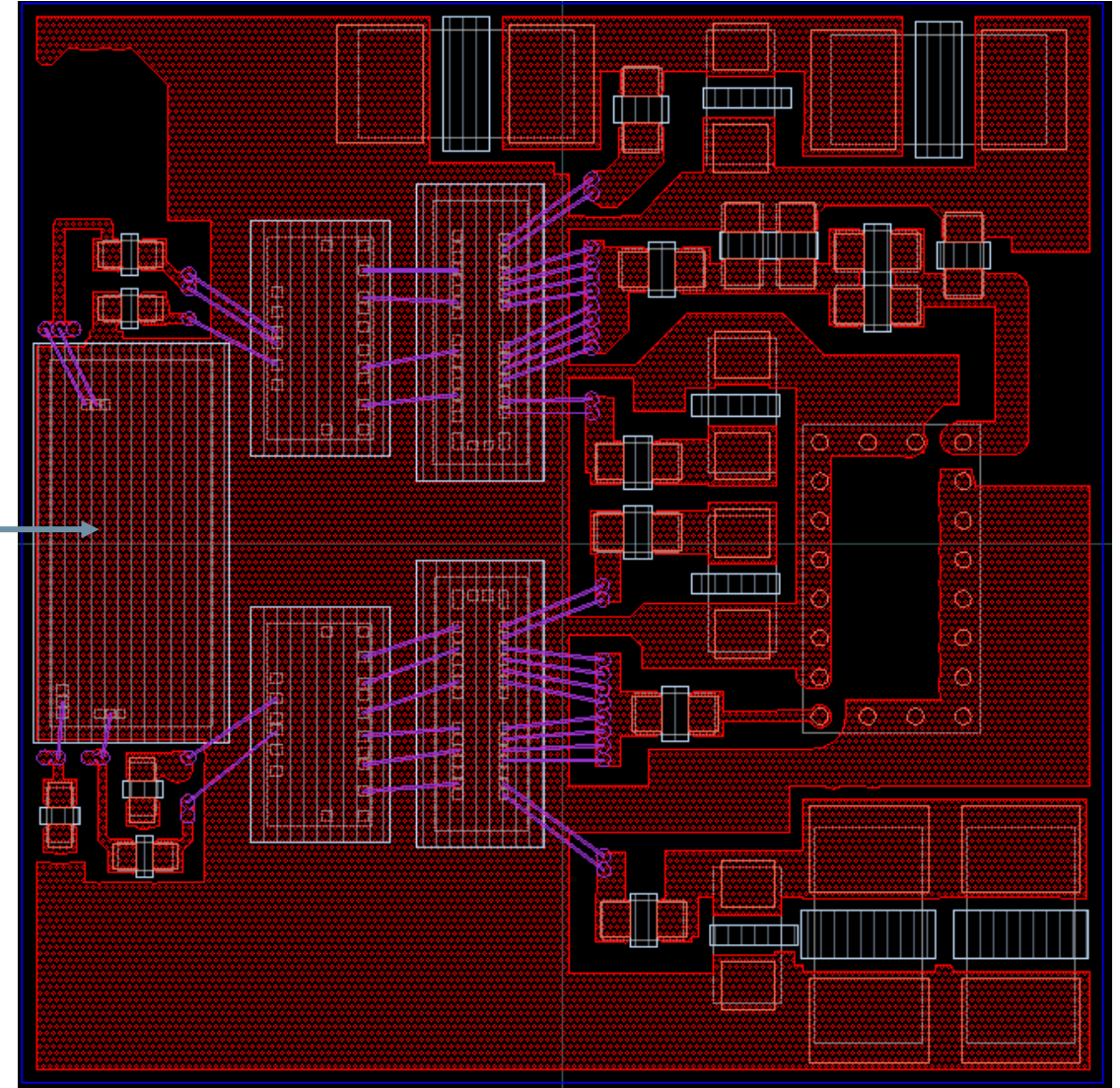
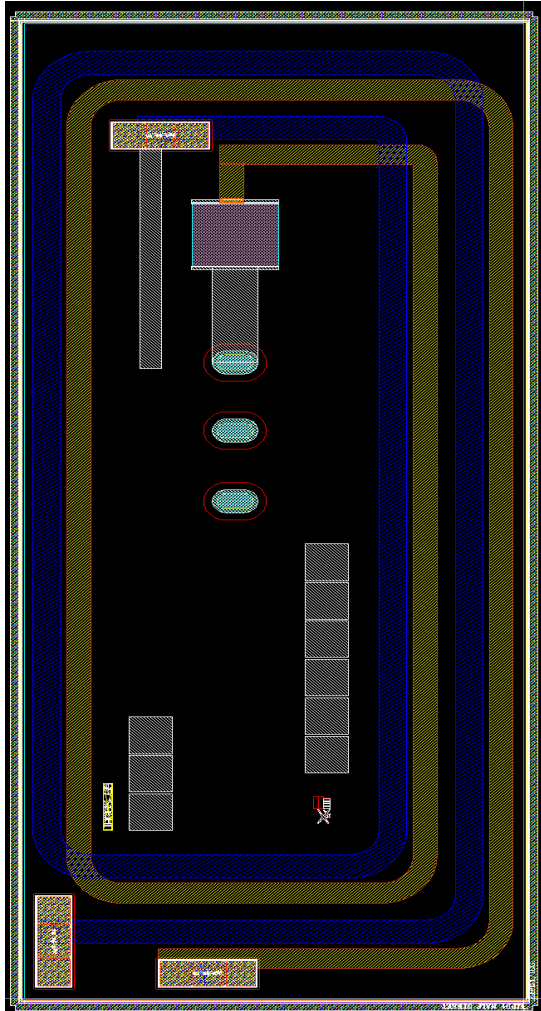
LIPD1

Power Splitter Coupler Design

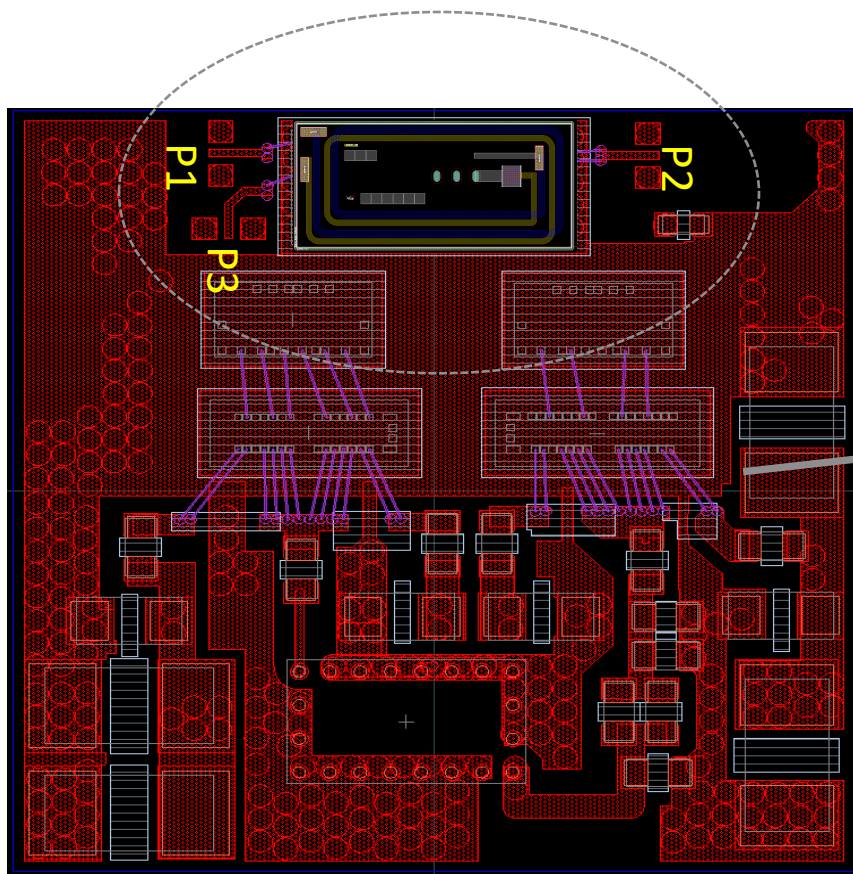
- UA005 – LIPD1 Process
 - delta dB between direct and coupled port
 - 90° phase delta between direct and coupled port
- Frequency Bands
 - 3.4 to 3.8 GHz

1st Generation Splitter Design: UA005_05 (1217x2727um)

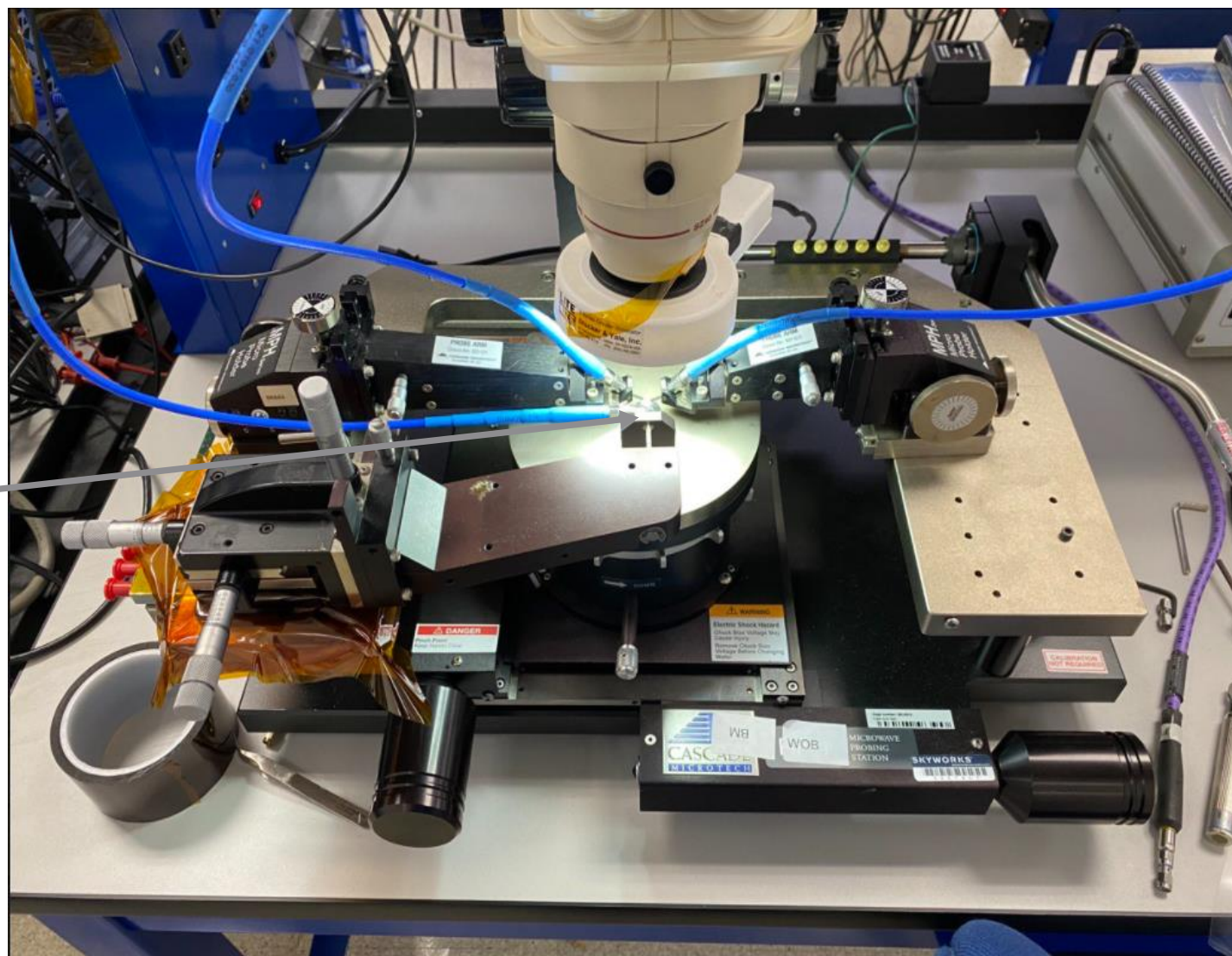
Splitter Design



UA005_05 Probe Measurements

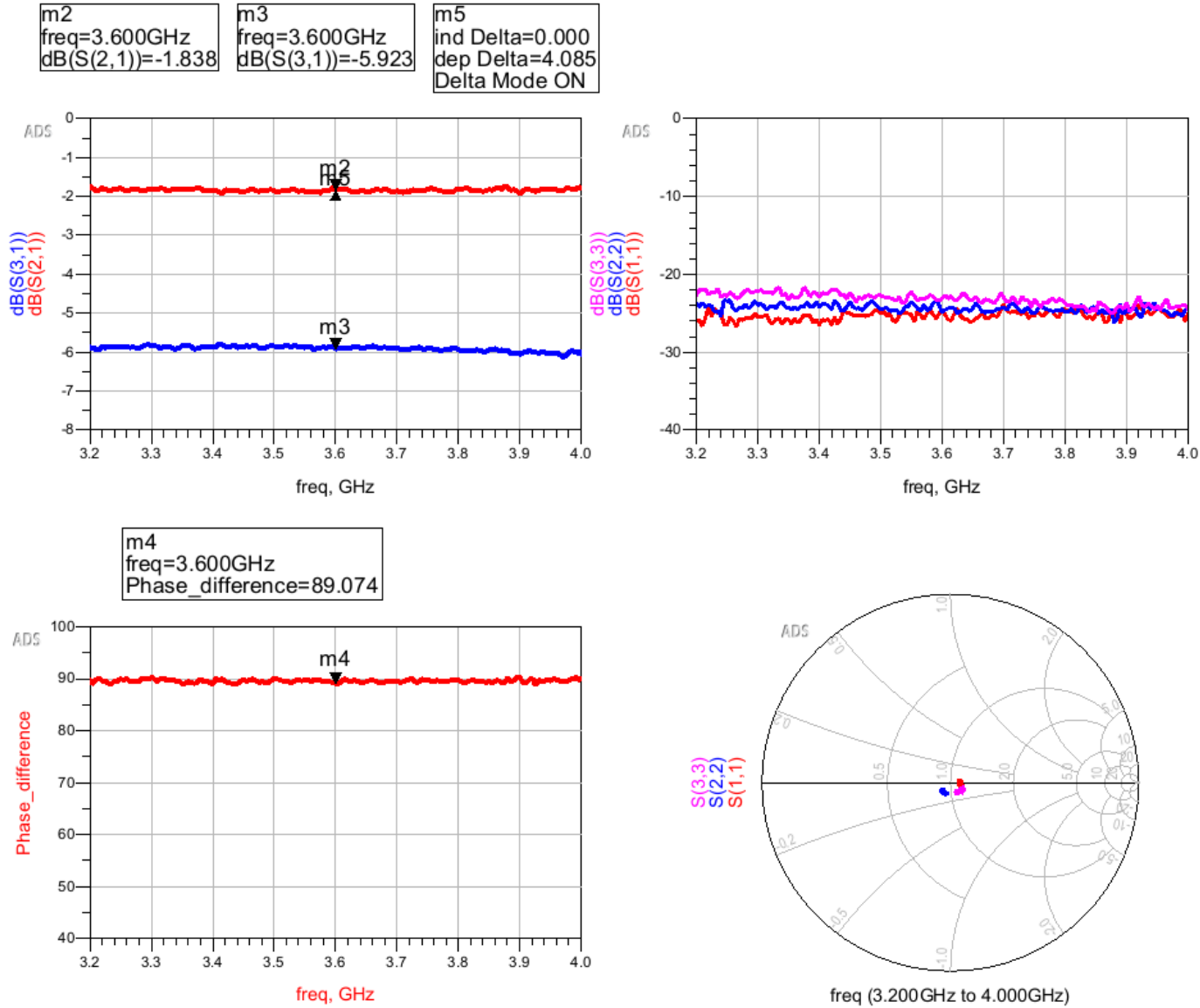


TW85_D996_001_V8



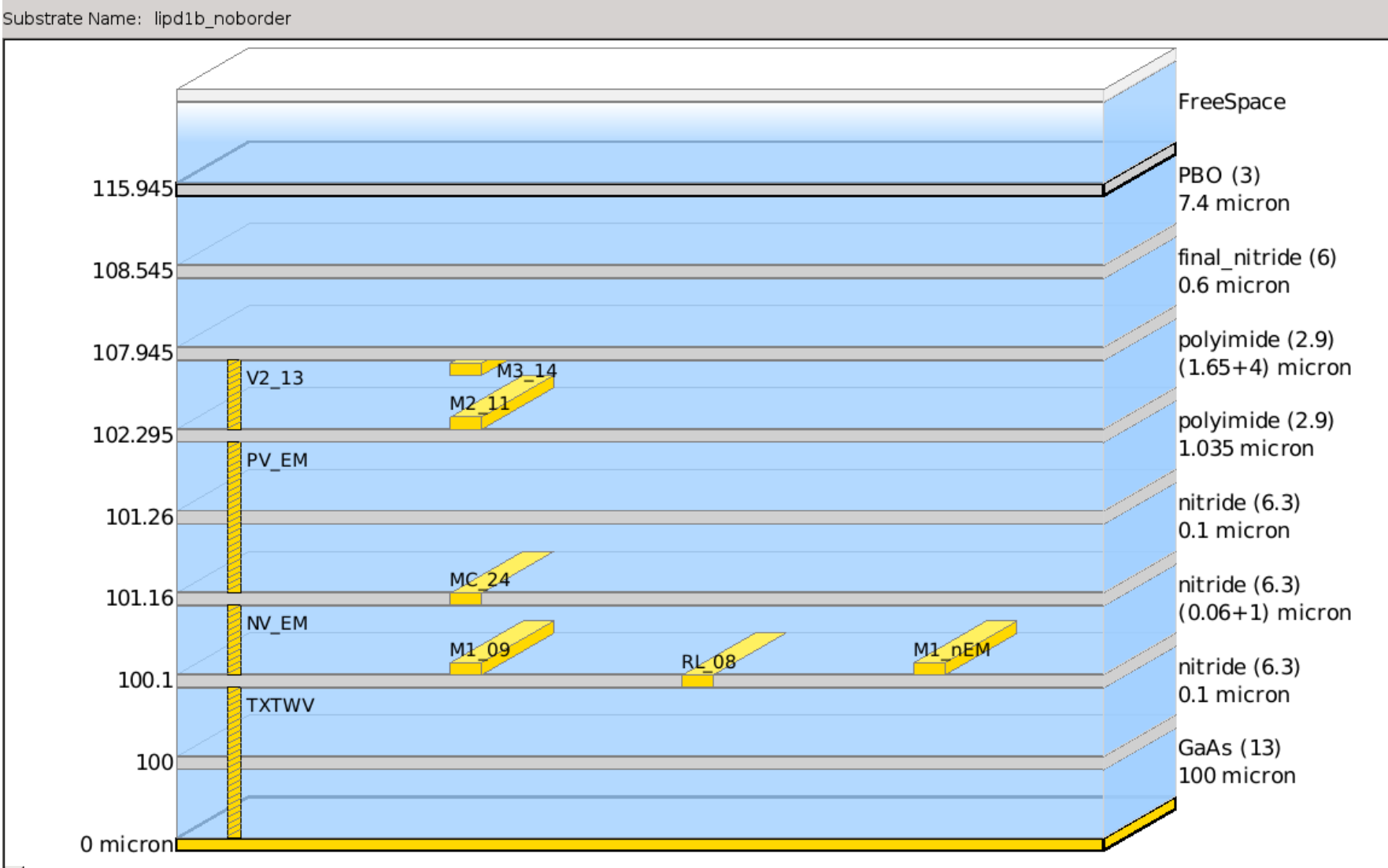
Probe of UA005_05

3.2 – 4.0 GHz



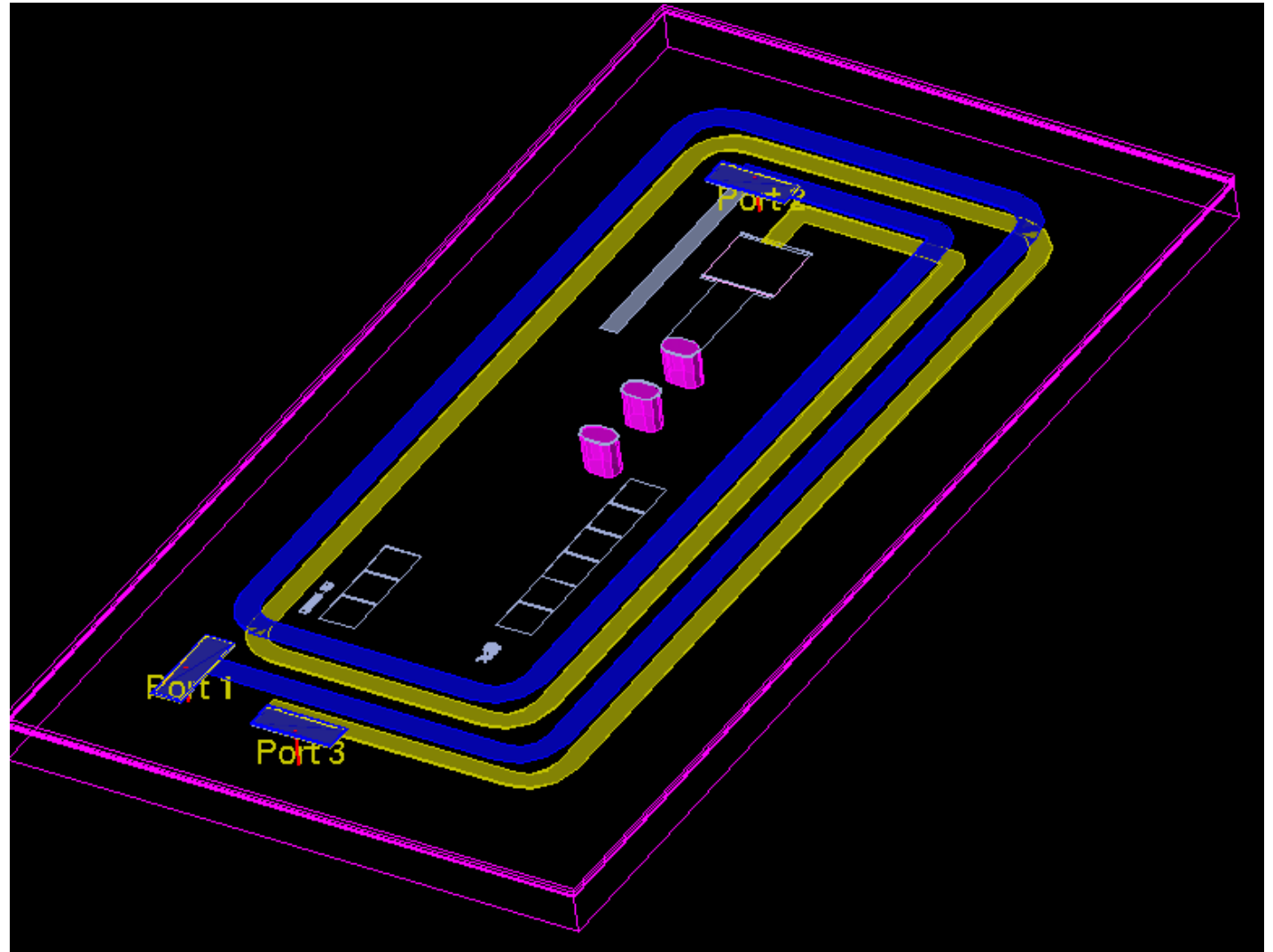
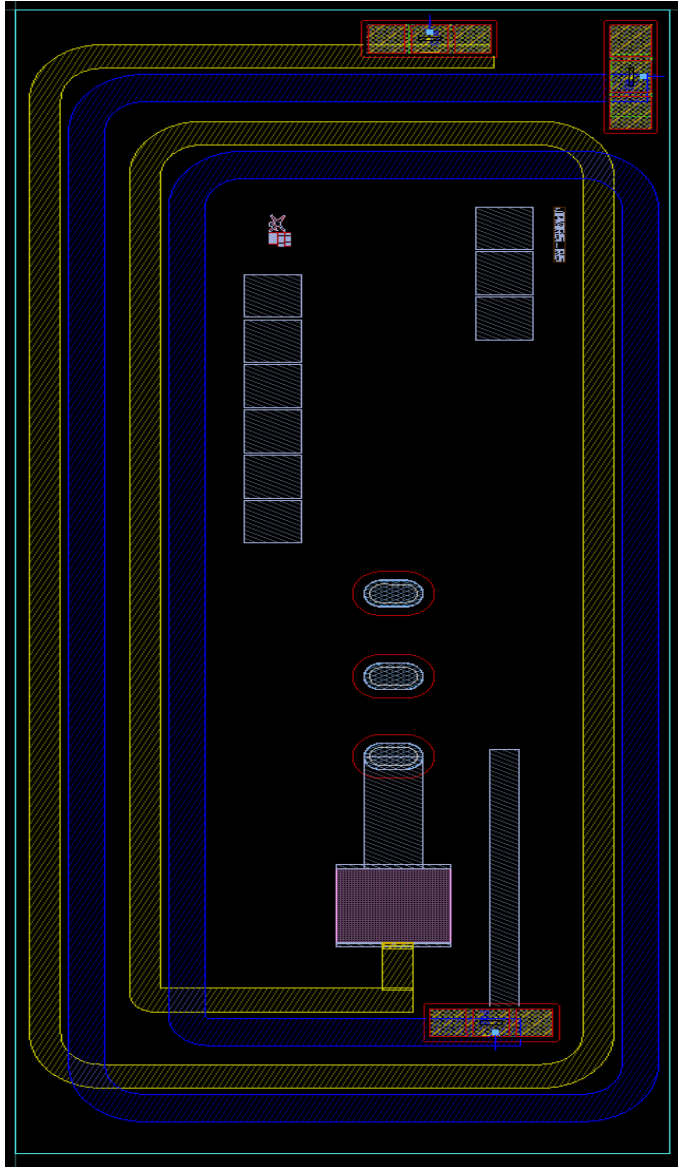
LIPD1 Substrate Momentum Stack-up

Modified by Kai Kwok



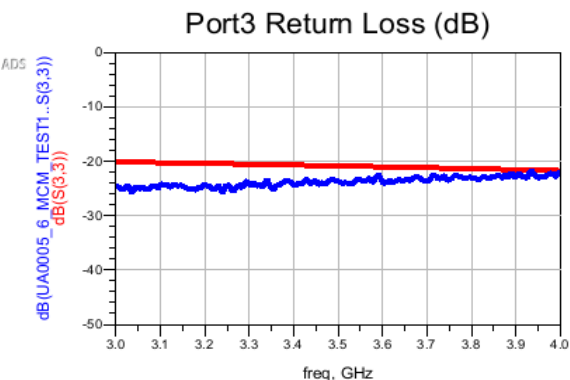
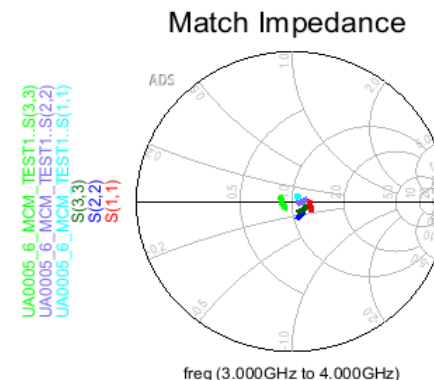
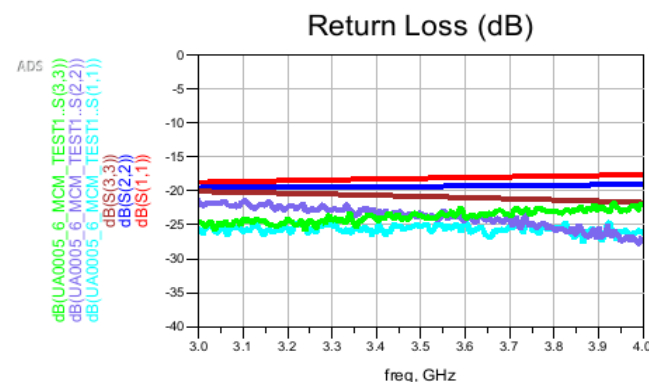
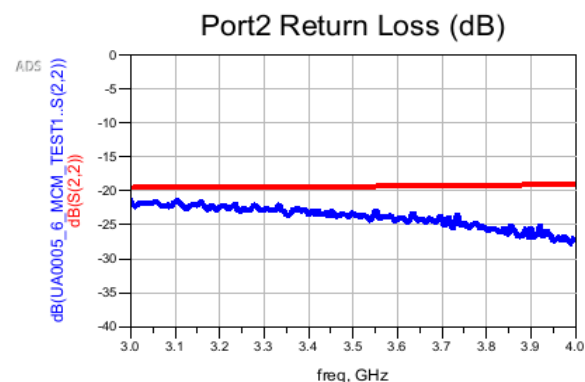
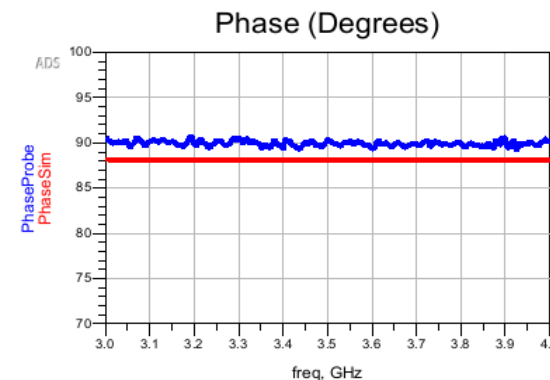
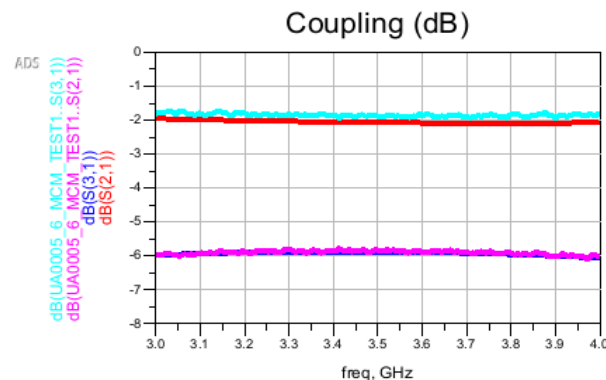
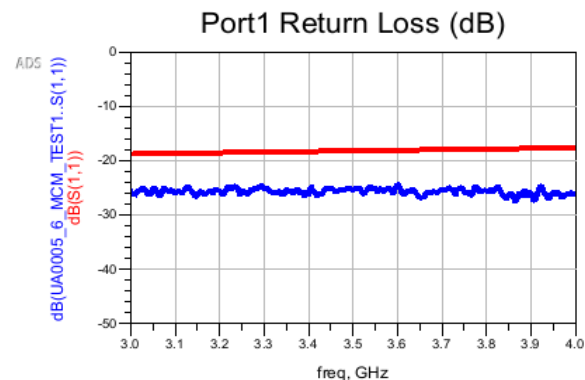
UA005_05

EM Validation Using Momentum



Probe of UA005_05 vs UA005_05 die MoM Sim

3.2 – 4.0 GHz (Red=MoM Sim; Blue=Measurement)



Datasets

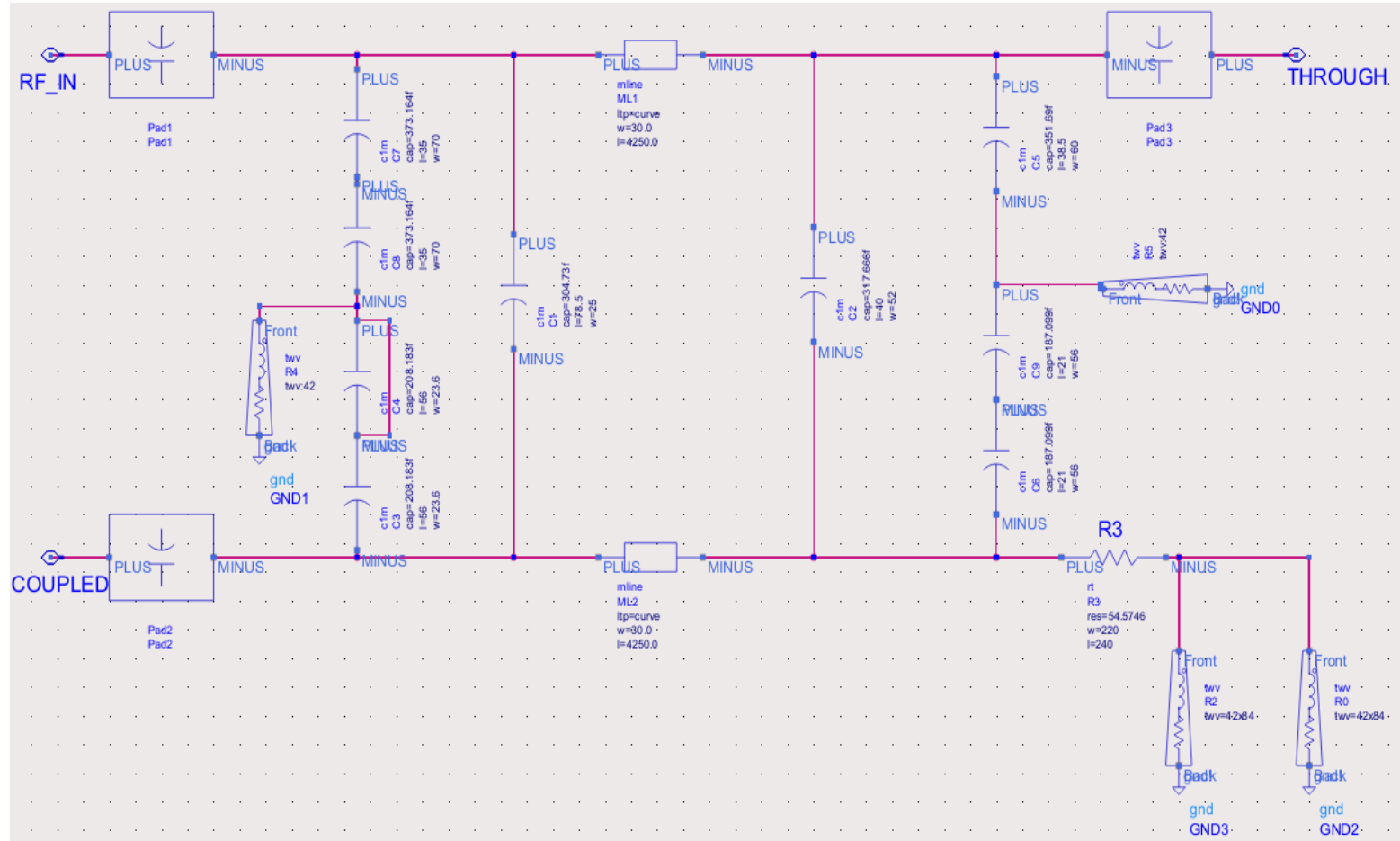
Probe=UA0005_6_MCM_TEST1

MoM=sim_UA005_05_die_only_MoM_020722

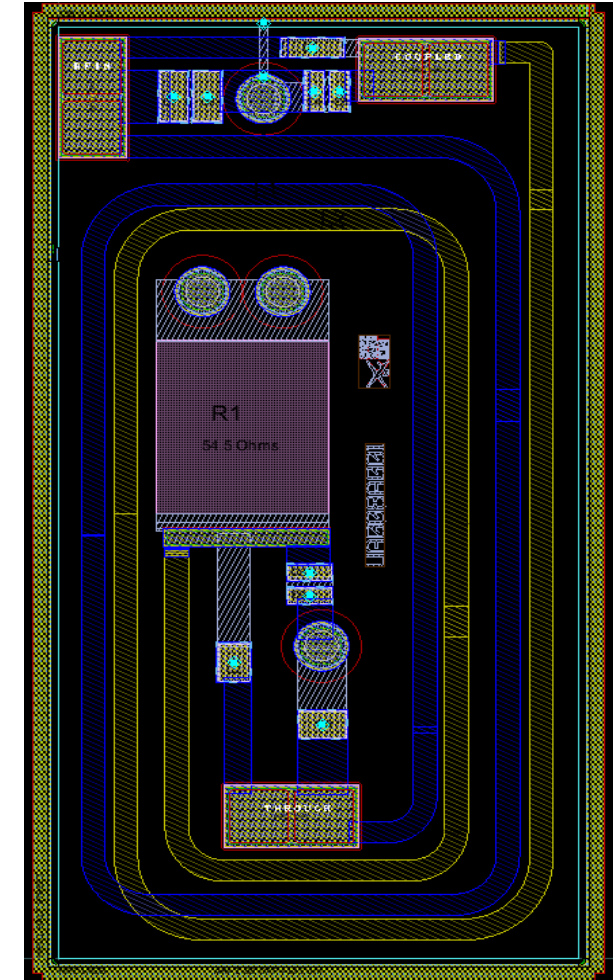
2nd Generation Design Power Splitter

UA008A001

Power Splitter Schematic

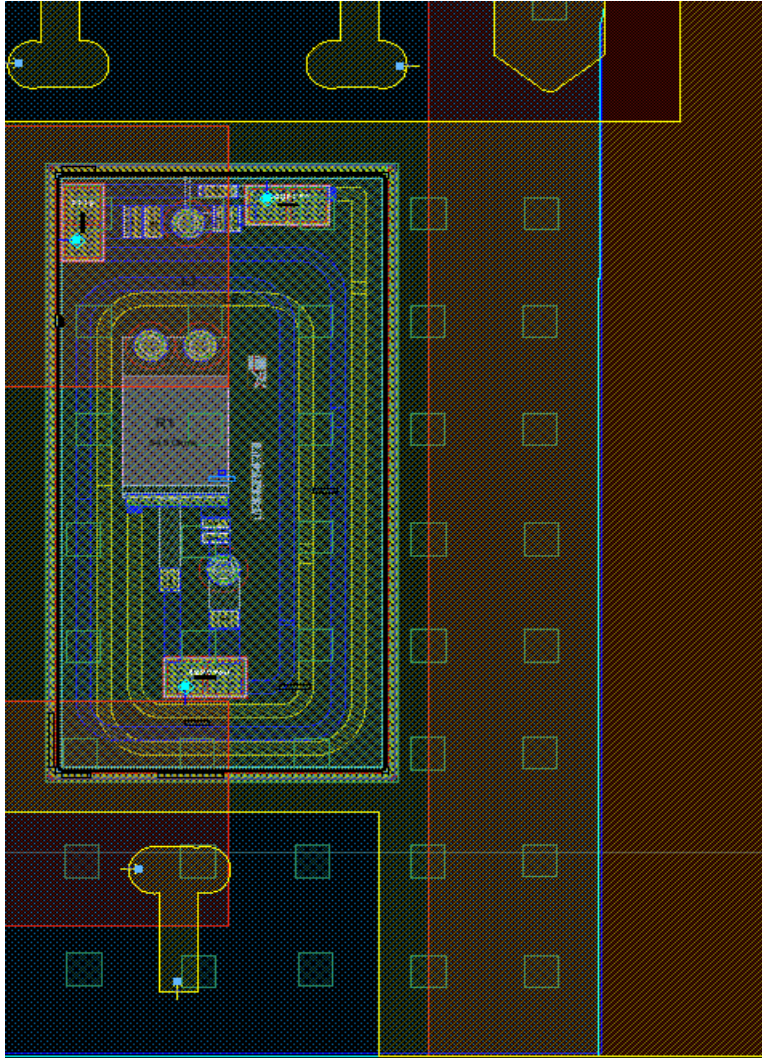


0.730mmx1.375mm

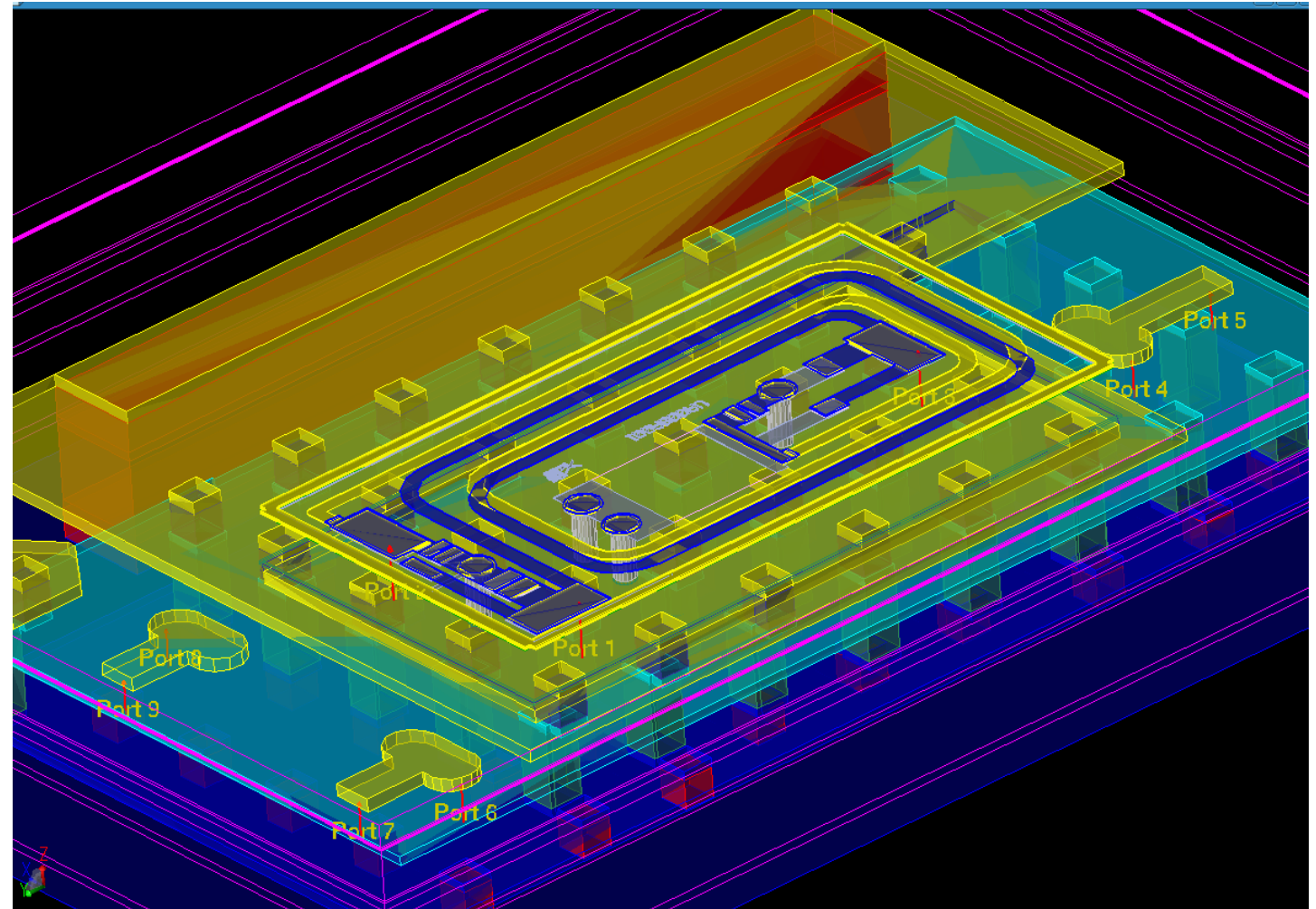


sim_MoM_TW85_D99x_001_UA008A001

UA008A001 die + 4L MCM simulation

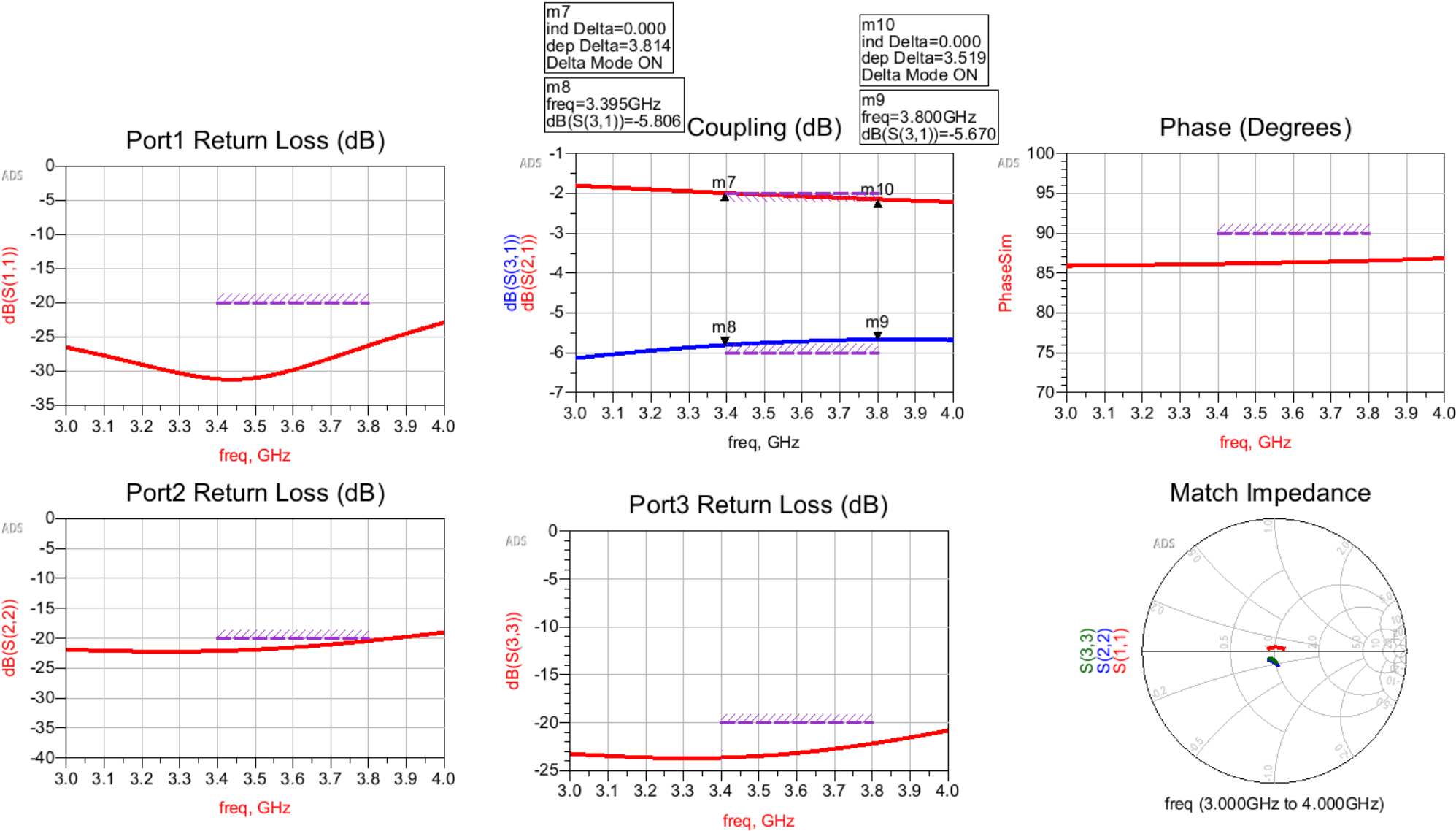


sim_MoM_TW85_D99x_001_UA008A001



UA008A003

3-Ports MoM Simulation



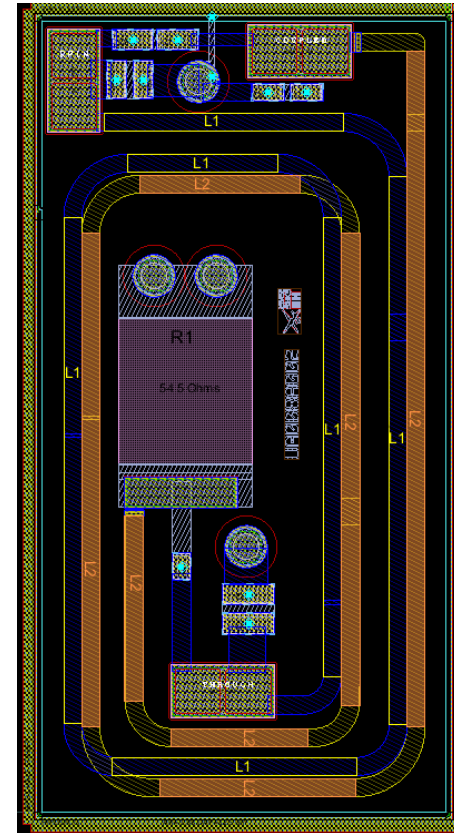
Thermal Analysis Using ETH in Splitter Passive Device

```
#-----
# TECHNOLOGY: THERMAL LAYER DEFINITIONS
#-----
# {name thickness(m) bkgnd-material {layer1 material1}...{lyrN matN}}

set unusedList {
  { backcopper 4.000e-6 Air {BORDR Cu} }
  { backgold 1.000e-6 Air {BORDR Au} }
}

set layerList {
  { backside 4.000e-6 MOLD {BORDR Cu} }
  { substrate 100.000e-6 MOLD {WV_37 Cu} {BORDR GaAs} }
  { res 0.055e-6 MOLD {CV_42 Au} {RL_08 TaN} {BORDR Si3N4} }
  { metal1 1.000e-6 MOLD {M1_09 Au} {BORDR Polyi} }
  { vial2a 0.060e-6 MOLD {NV_10 Au} {BORDR Si3N4} }
  { vial2b 1.360e-6 MOLD {V1_16 AuV} {BORDR Polyi} }
  { metal2 2.000e-6 MOLD {M2_11 Au} {BORDR Polyi} }
  { via23 2.200e-6 MOLD {V2_13 AuV} {BORDR Polyi} }
  { metal3 2.000e-6 MOLD {M3_14 Au} {BORDR Polyi} }
  { topcoat 7.400e-6 MOLD {SP_38 Au} }
}
```

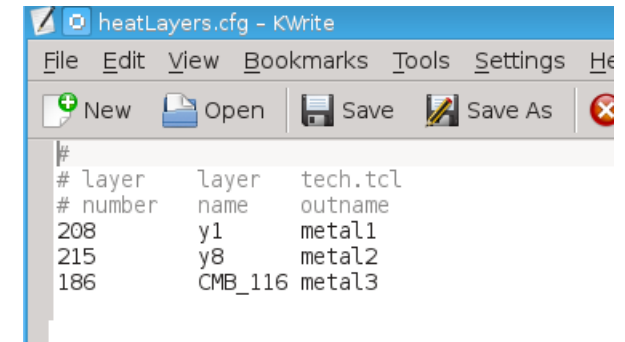
HIPD1 Process



Line Width:

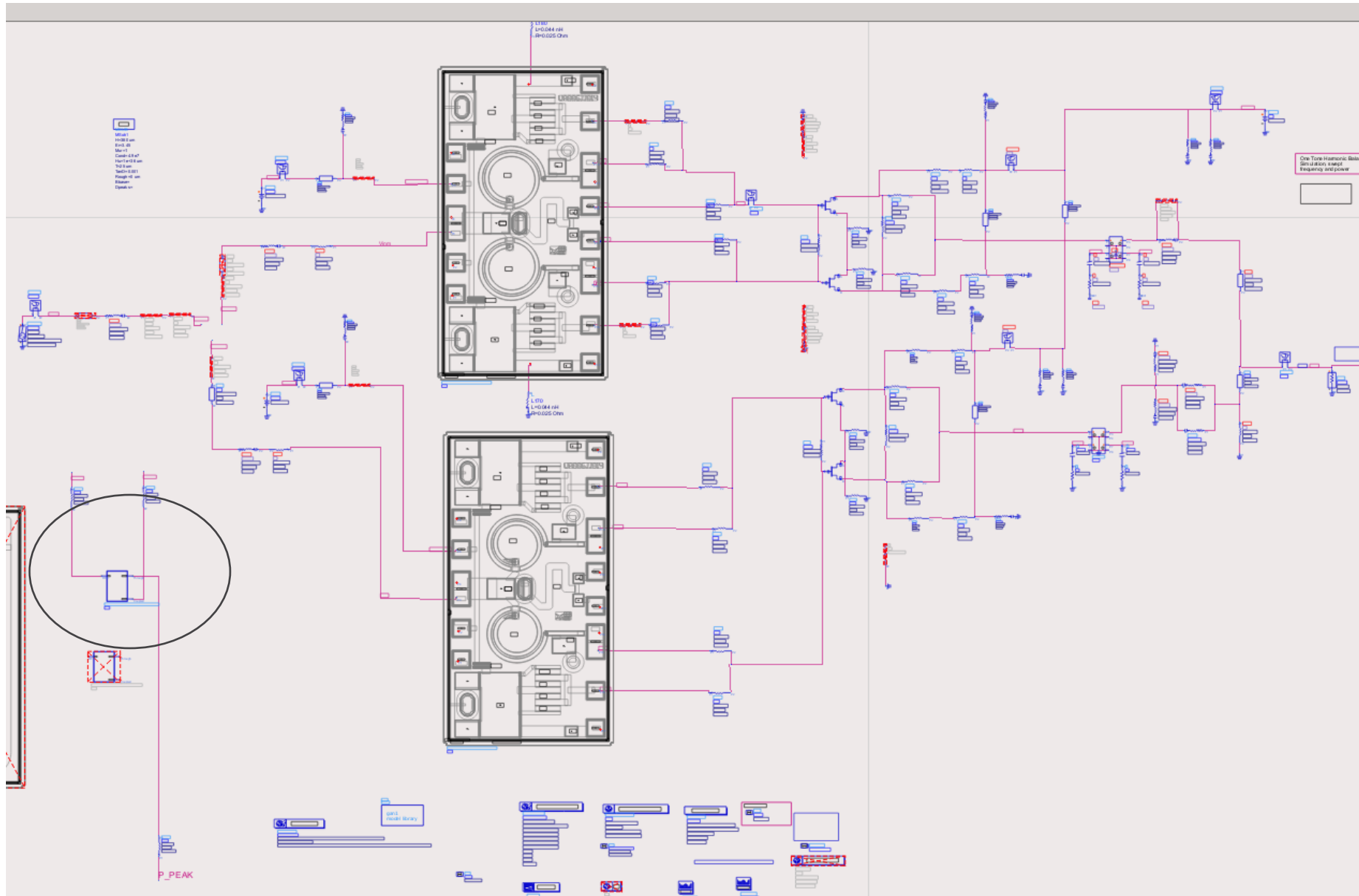
M3 = 30 um

M2 = 30 um



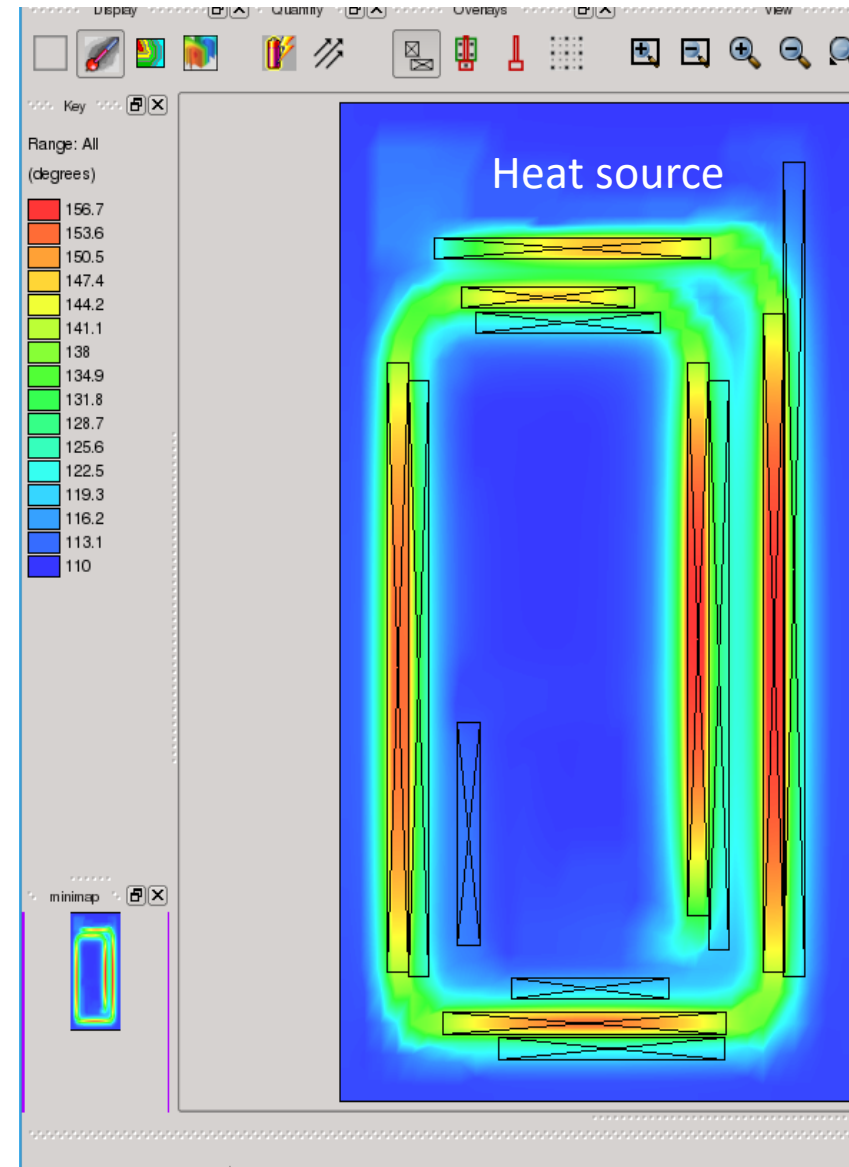
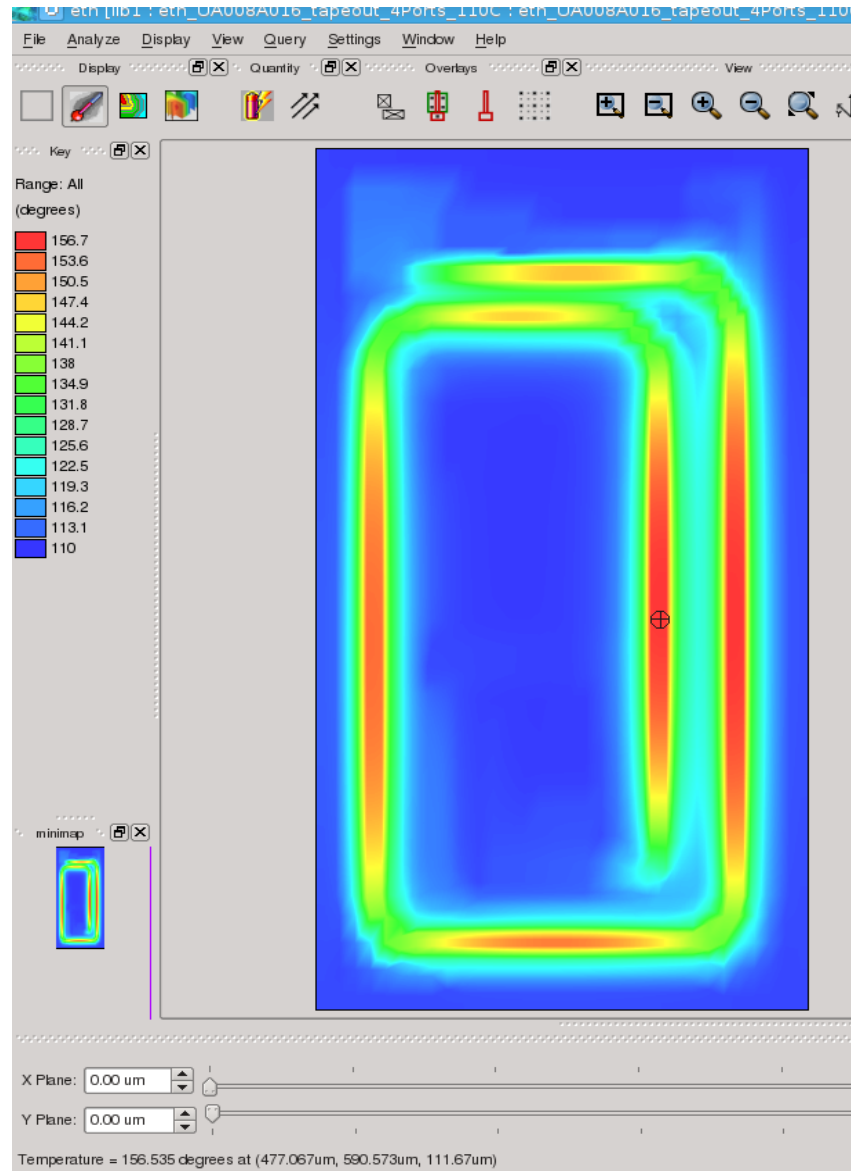
Thermal Analysis Using ETH in Splitter Passive Device

Ambient Temp: 110 C; Pin=Sweep



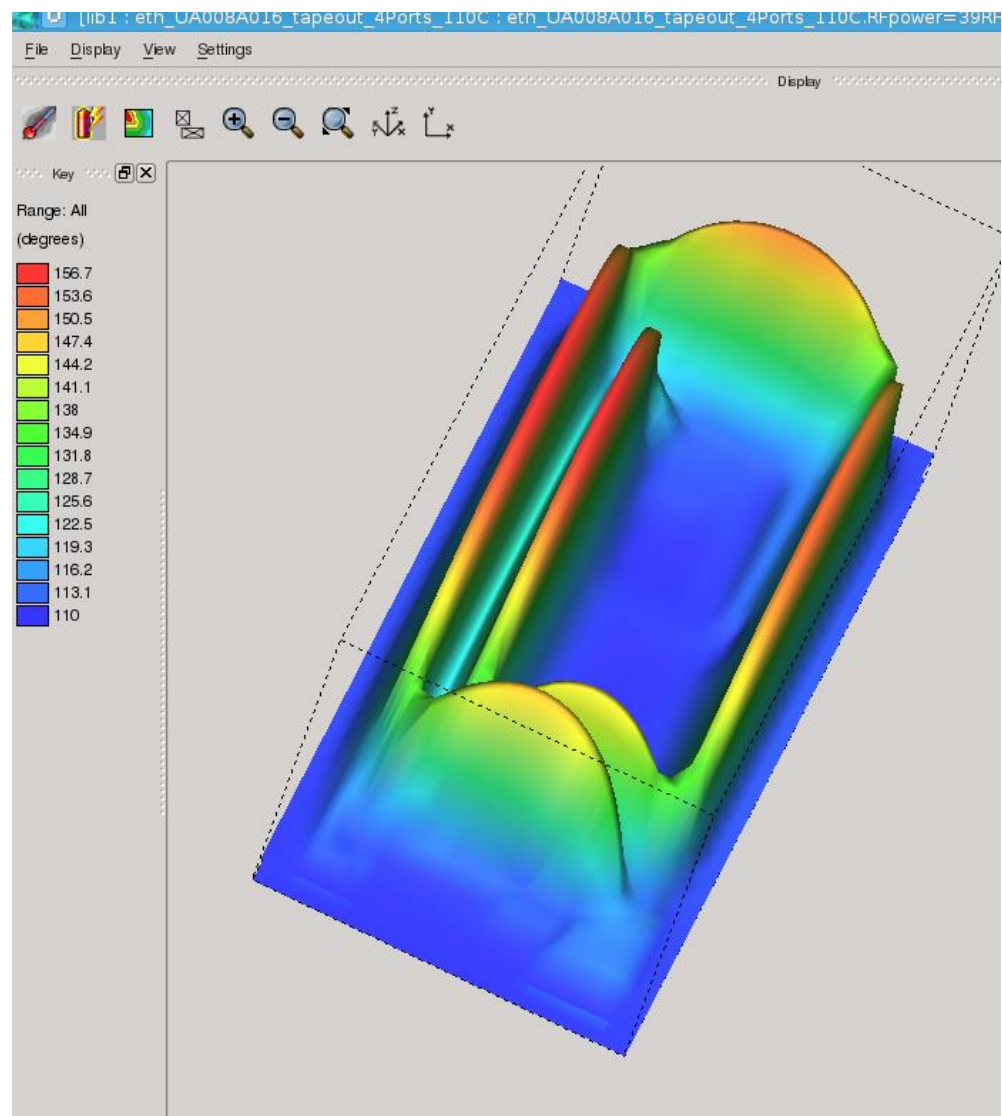
Thermal Analysis Using ETH in Passive Device

Ambient Temp: 110 C; Pin=39 dBm



Thermal Profile

Ambient Temp: 110 C; Pin=39 dBm

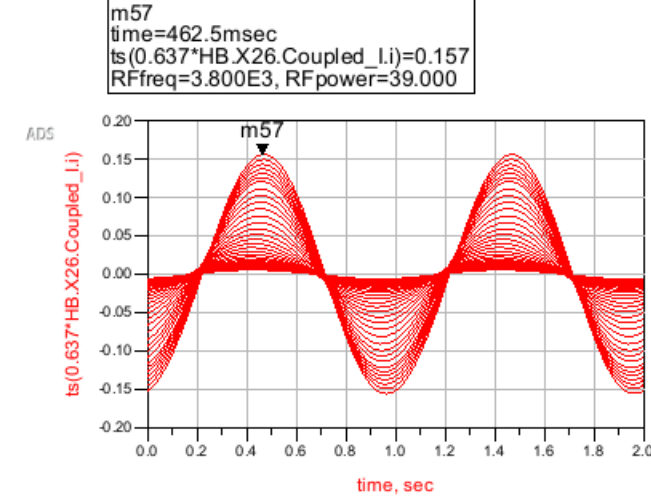
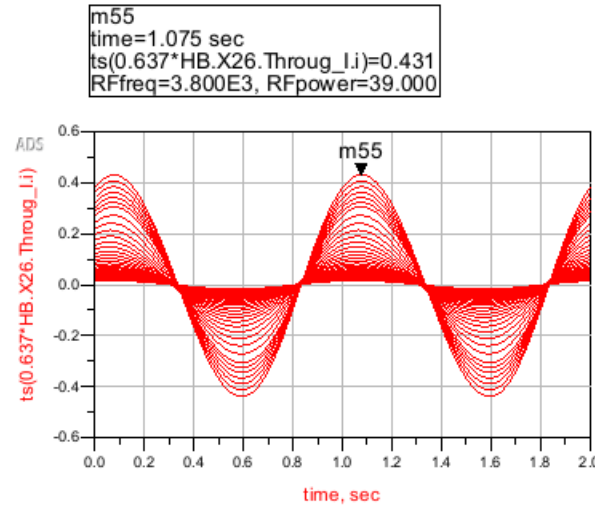
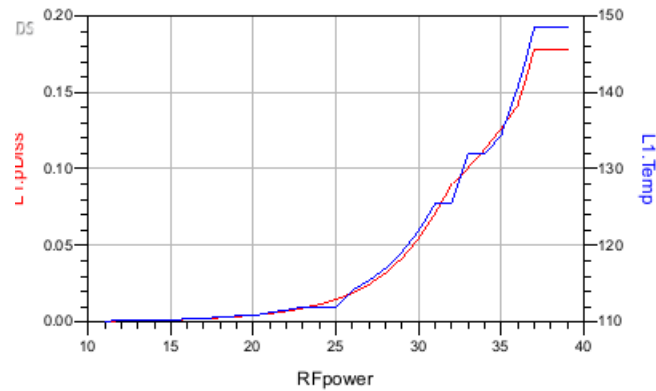
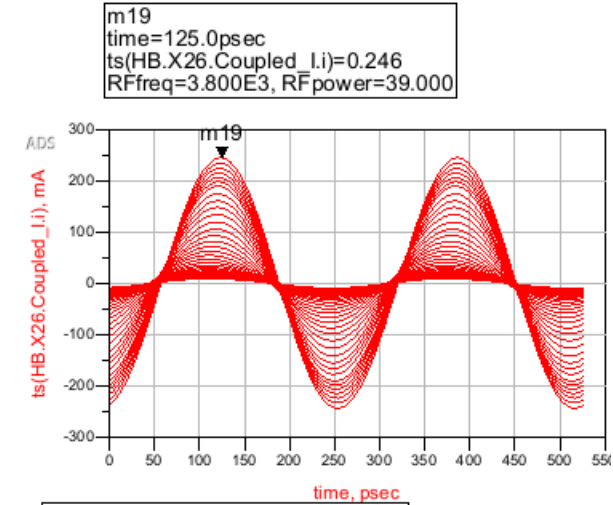
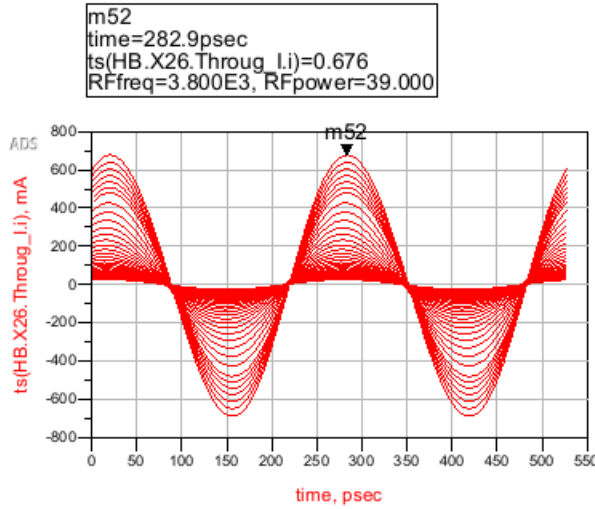
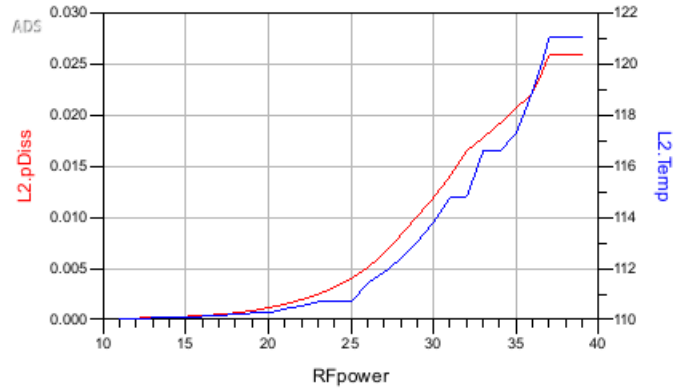


Power distribution in each section

Name	Power L2	Power L2
----	----	----
X 26.L1#6		0.0454429
X 26.L1#1		0.042009
X 26.L1#5		0.0382451
X 26.L1#3		0.020335
X 26.L1#2		0.0198014
X 26.L1#4		0.0124573
X 26.L2#7	0.0075433	
X 26.L2#1	0.00552097	
X 26.L2#6	0.00527516	
X 26.L2#4	0.00218075	
X 26.L2#2	0.0021509	
X 26.L2#3	0.00177965	
X 26.L2#5	0.00152686	
Total Power	0.02597759	0.1782907

Thermal Analysis Using ETH in Passive Device

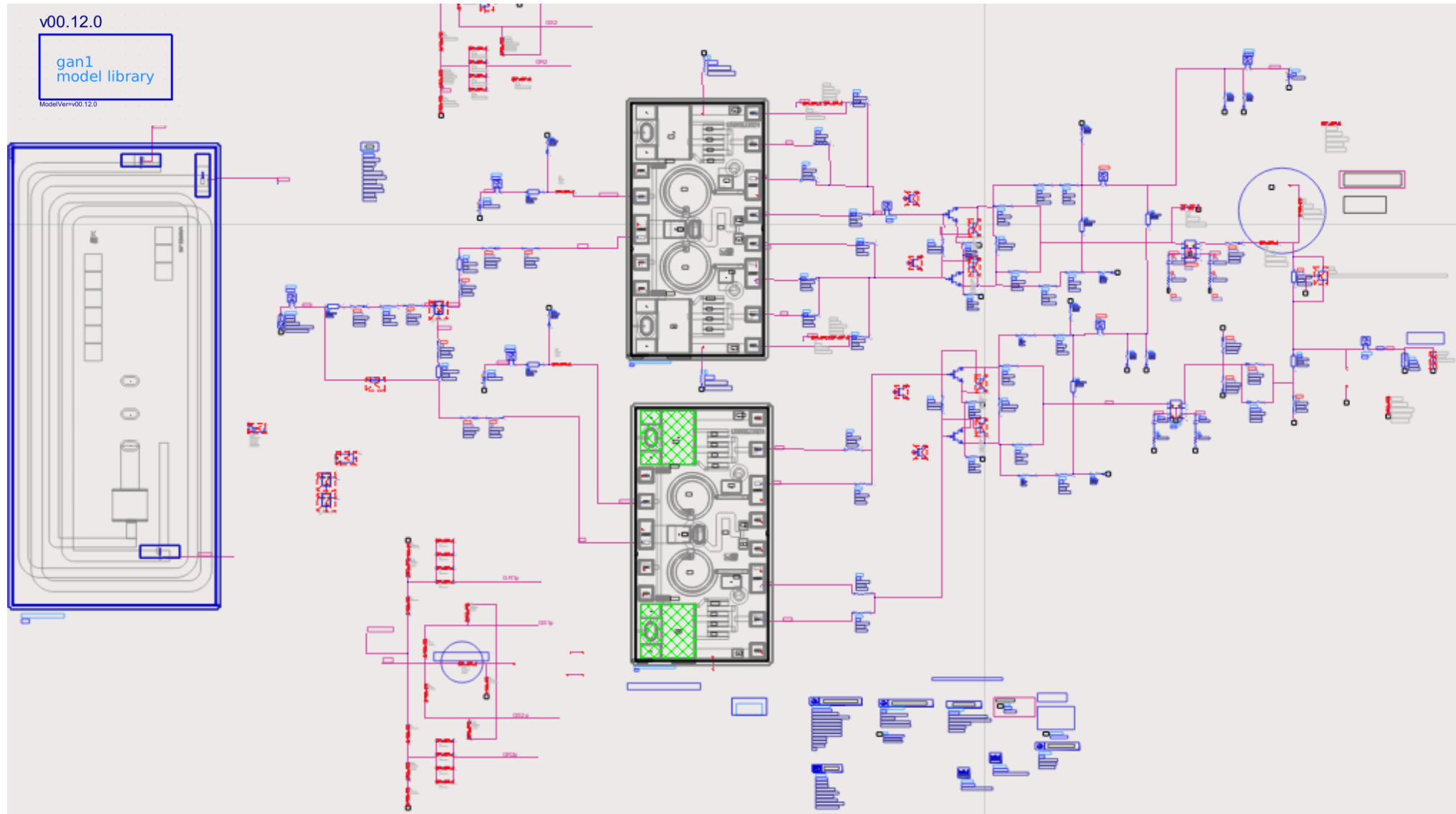
Ambient Temp: 110 C; Pin=39 dBm



Combining Main and Peaking Amplifiers

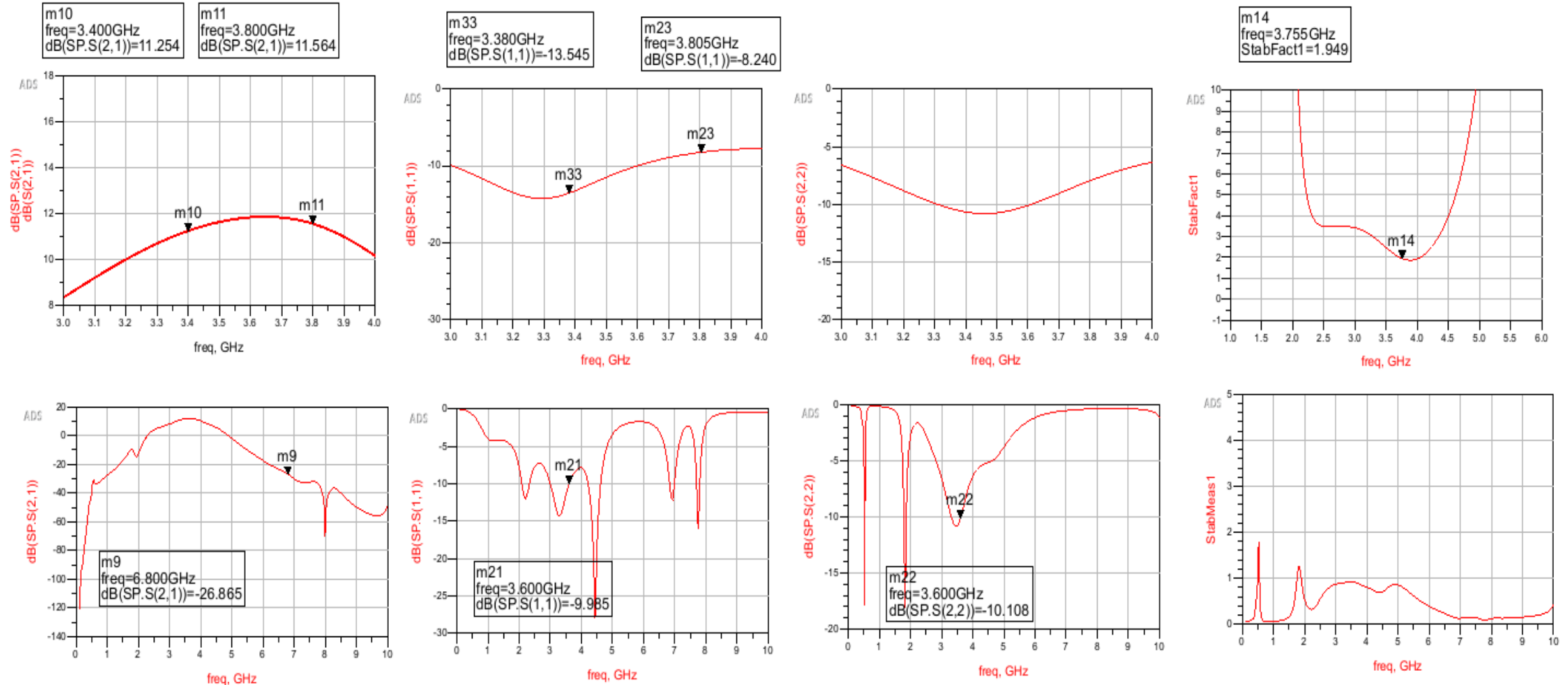
SKYRD033A_WB DPA Lumped model Top Level Sim

Vdd=40V; Vg_main=-2.78V; Vg_peak=-5.6V; Iq=30mA



DPA Small Signal Simulation (Lumped model)

Main IPD = UA006JJ017; Peaking IPD = UA006JJ018

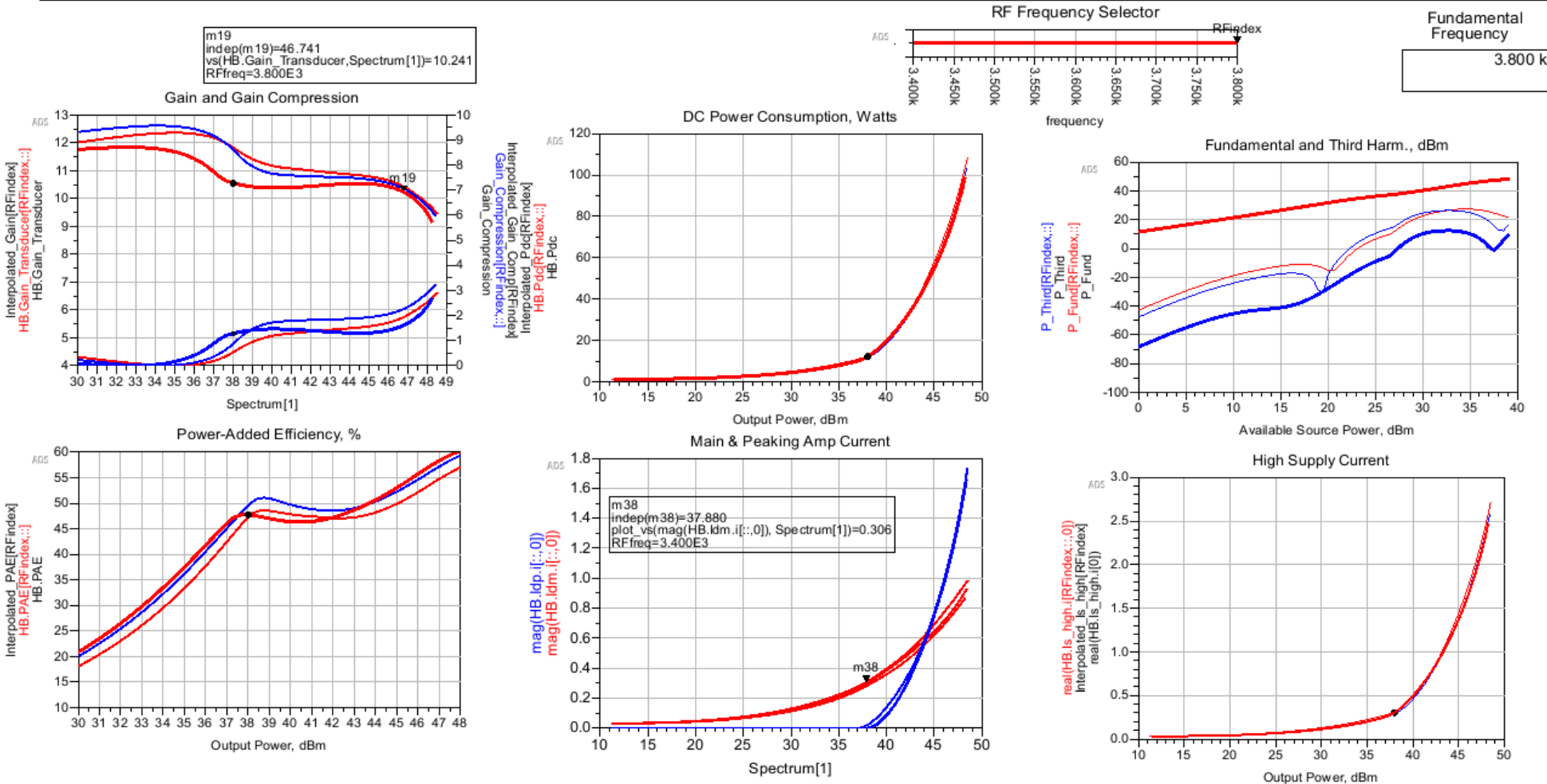


DPA Large Signal Simulation (Lumped model)

Main IPD = UA006JJ017; Peaking IPD = UA006JJ018

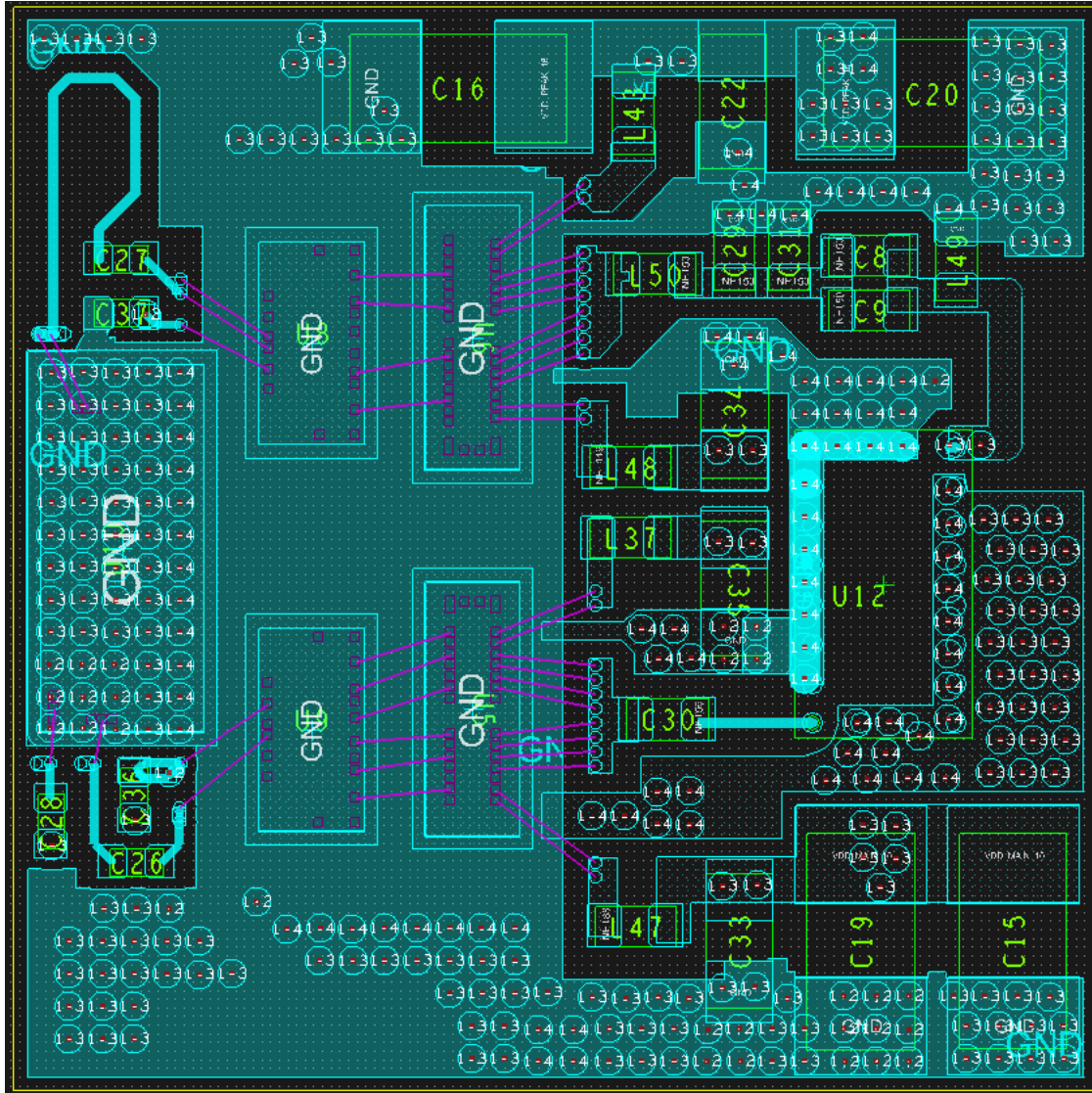
Interpolated Values at 38.000 dBm output power (approximately):

	Fundamental Output Power dBm	Transducer Power Gain	Gain Compression	Power- Added Efficiency, %	DC Power Consumpt. Watts	High Supply Current	Thermal Dissipation Watts	Second Harmonic dBc	Third Harmonic dBc	Fourth Harmonic dBc	Fifth Harmonic dBc	Input Return Loss, dB	Output Return Loss, dB
3.400 k	37.98	11.83	529.7 m	47.27	12.44	311.1 m	6.515	-22.35	-28.77	-56.72	-47.28	-16.08	0.0000 / ...
3.600 k	38.00	11.75	868.1 m	49.54	11.93	298.2 m	5.981	-26.29	-24.40	-53.79	-37.38	-12.18	0.0000 / ...
3.800 k	38.00	10.55	1.290	47.74	12.19	304.8 m	6.358	-26.92	-37.61	-44.53	-38.81	-9.142	0.0000 / ...



Full EM Simulation of DPA

Complete Final Stage DPA Design Layout



Primary die components

- U5 = GE00138
- U6 = GE00138
- U8 = UA006JJ019
- U9 = UA006JJ018
- U10 = UA005-006
- U12 = MK0180_D03

SMDs

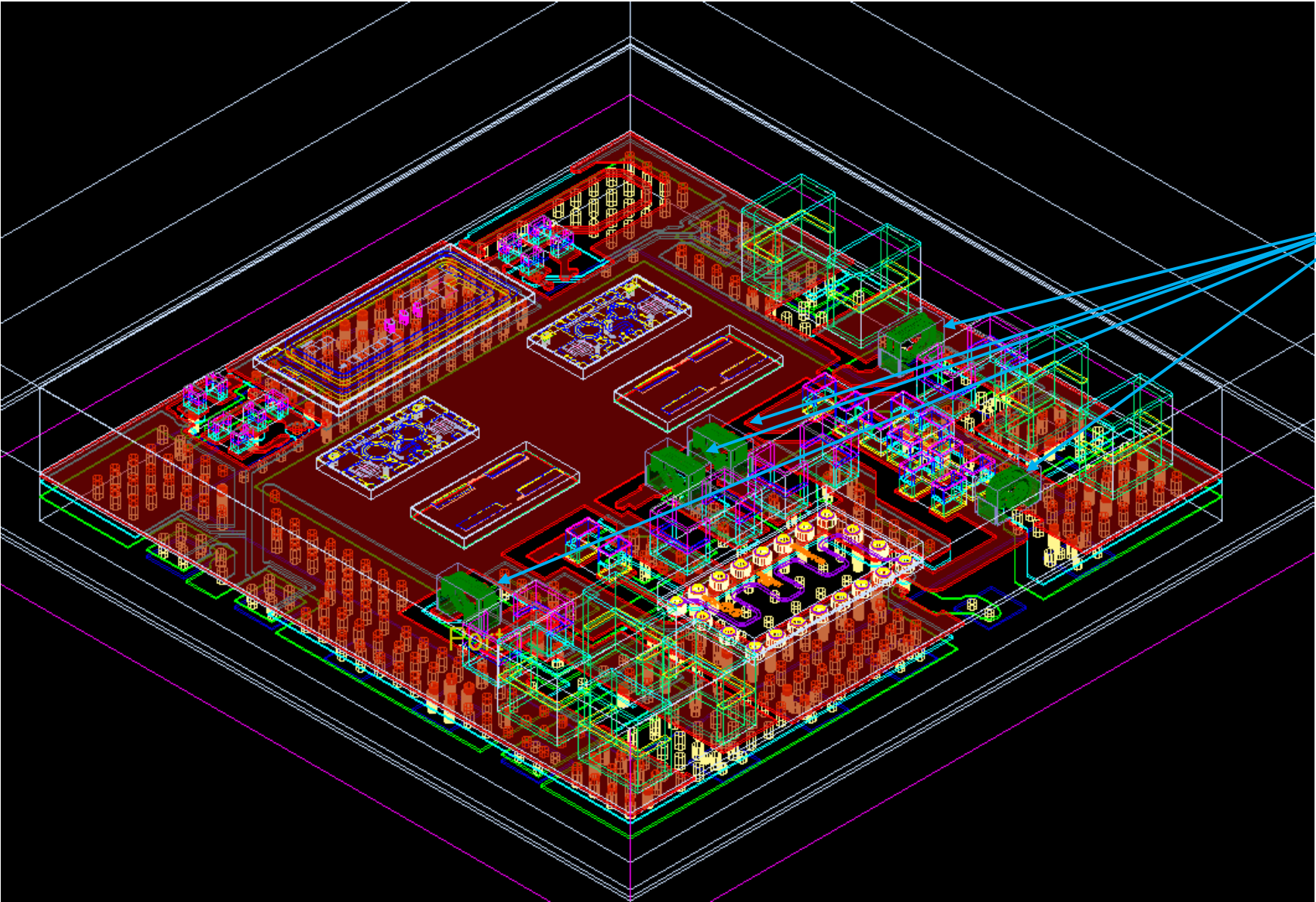
- 0603 = 4
- 0402 = 4
- 0201 = 11
- 01005 = 5

MCM = 4L (Heat Slug)

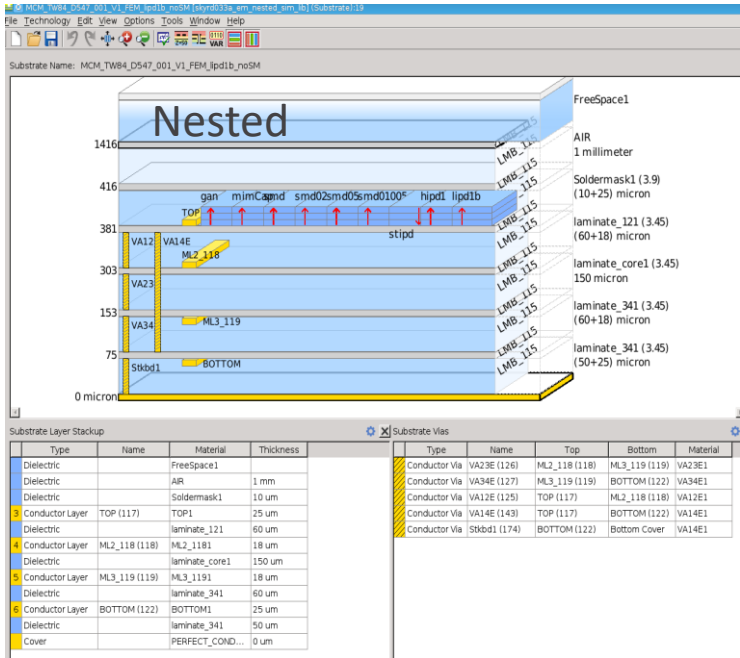
Wirebond = 25um

Epoxy = Atrox 800HT2V

Full MCM EM with 3D SMD Inductors Using FEM

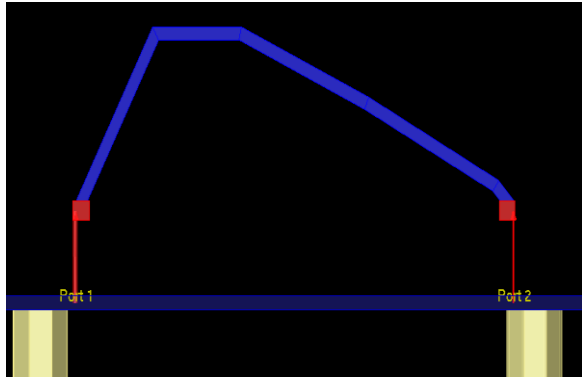


Full 3D SMD inductors

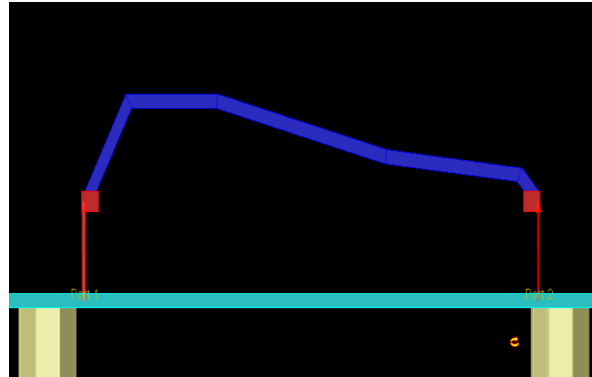


Critical Wire bond Loop Shape

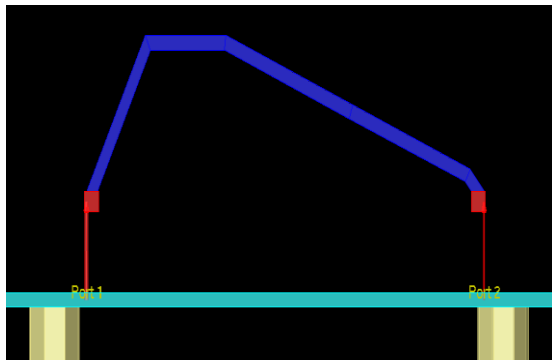
Wire bond E & E2



Wire bond F

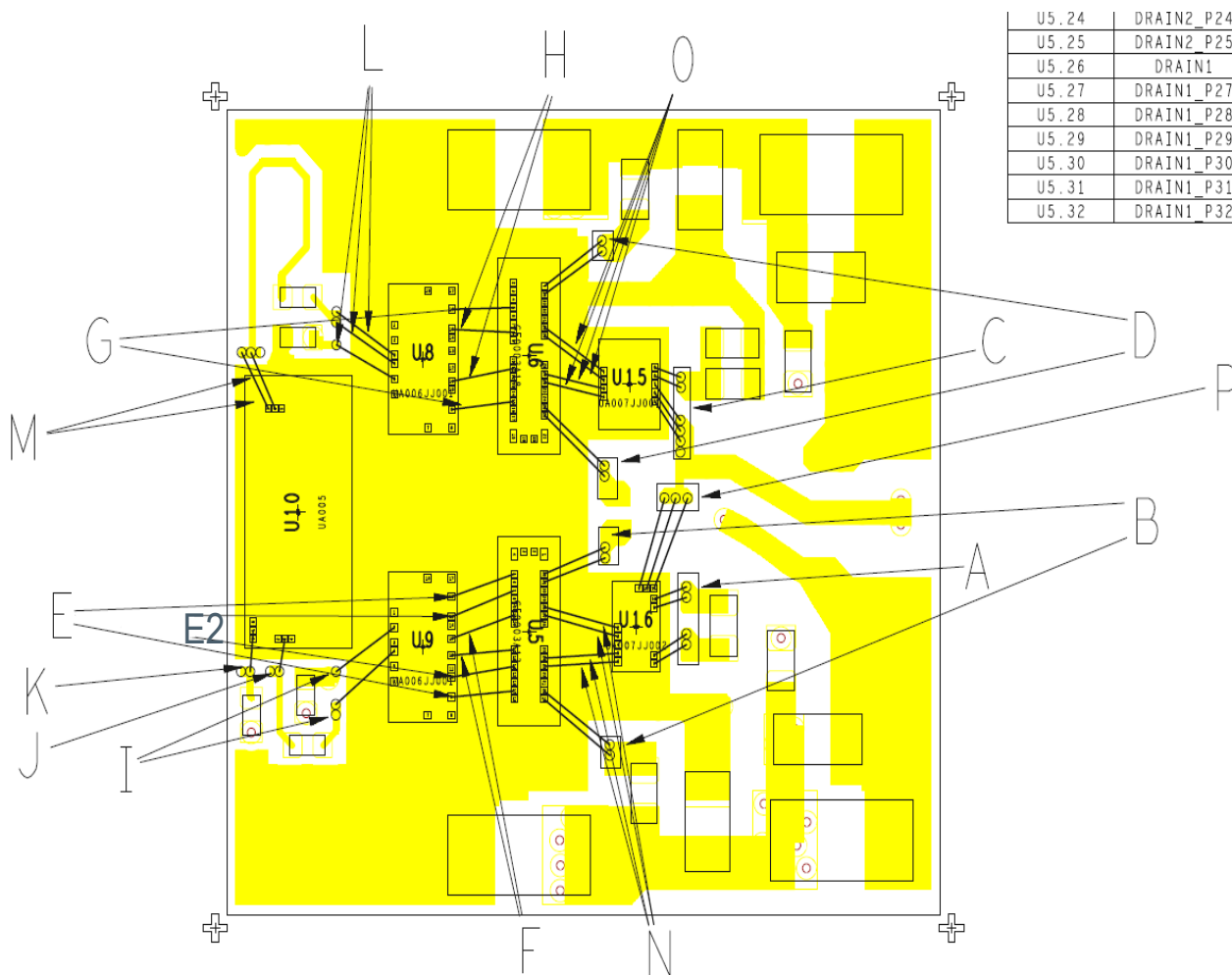


Wire bond G & H



Bonding diagram for Critical Wires for RPL BOM

- Please use 25um diameter gold wire.



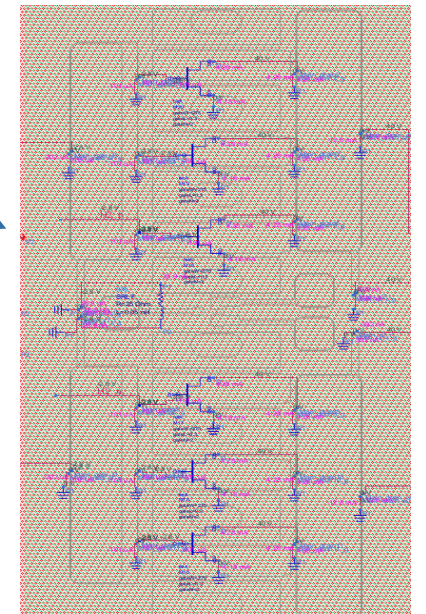
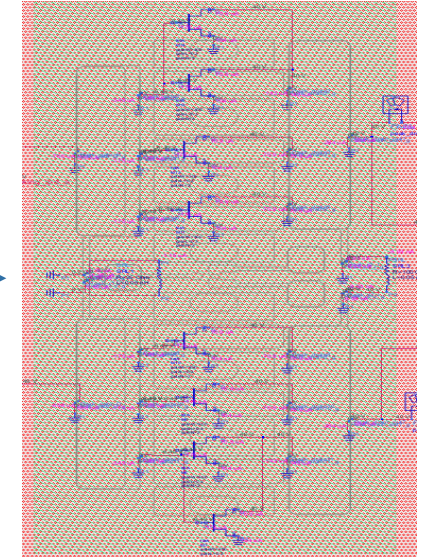
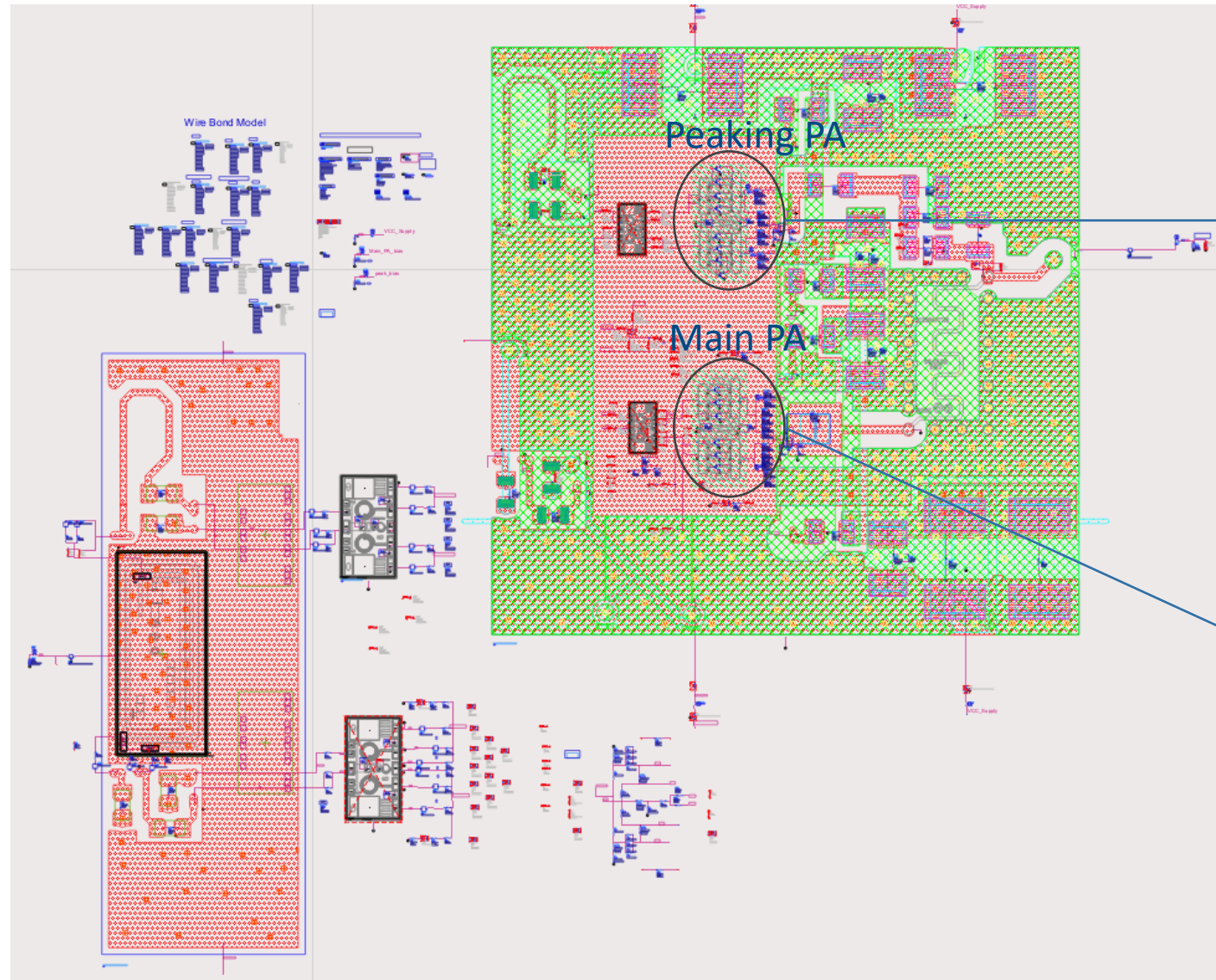
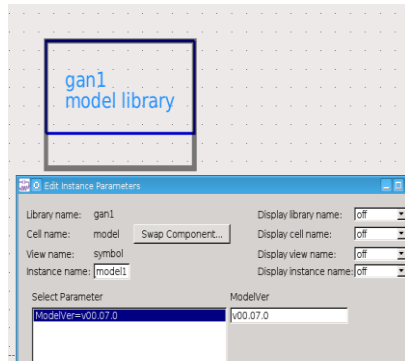
Critical Wire bond Table

Wir e	Loop Height (um)	Loop Shape
A	75 (minimum LH)	Standard shape
B	350	Standard shape
C	75 (minimum LH)	Standard shape
D	350	Standard shape
E	250	Refer to page 24
E2	200	Refer to page 24
F	125	Refer to page 24
G	220	Refer to page 24
H	250	Refer to page 24
I	127	Standard shape
J	127	Standard shape
K	250	Standard shape
L	129	Standard shape
M	127	Standard shape
N	250	Standard shape
O	250	Standard shape
P	100	Standard shape

SKYRD033A WB Simulation bench

4L MCM, GaN die, TPDs, input Splitter, SMDs are full EM using Nested Technology in FEM

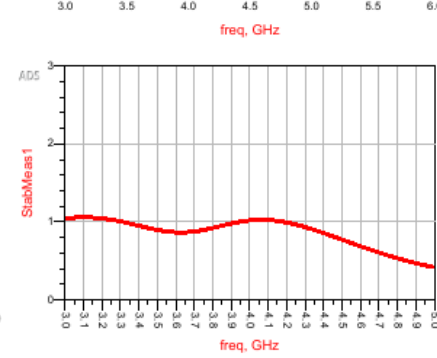
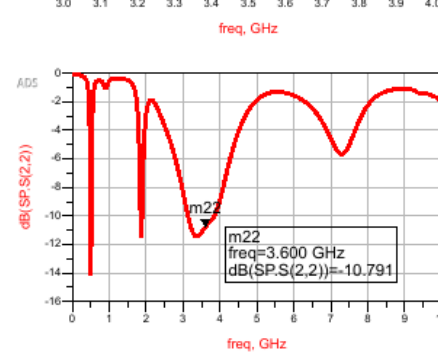
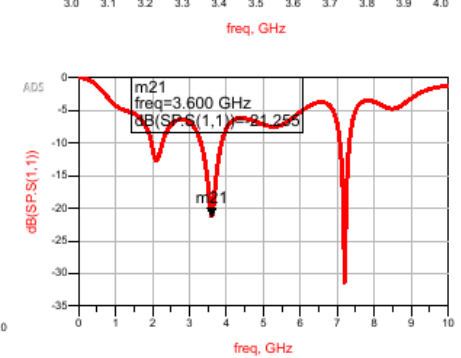
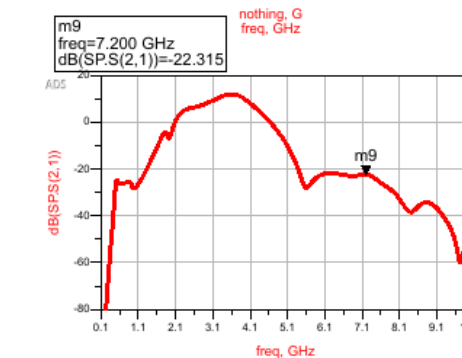
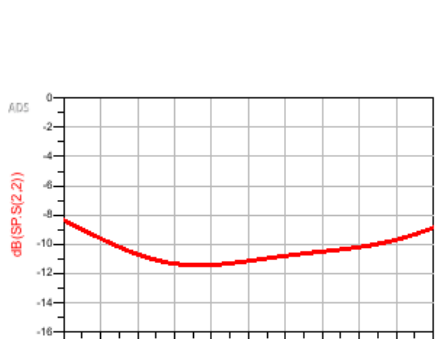
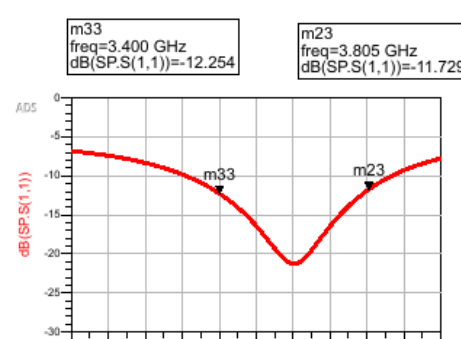
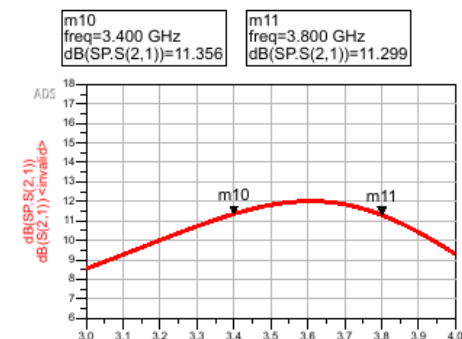
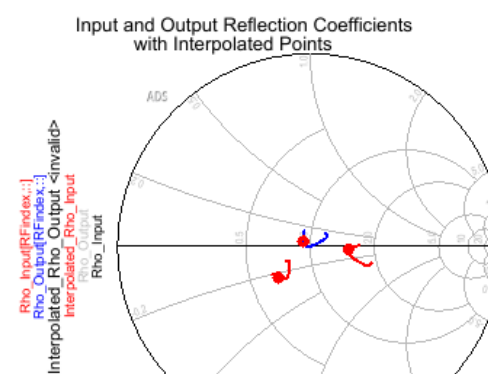
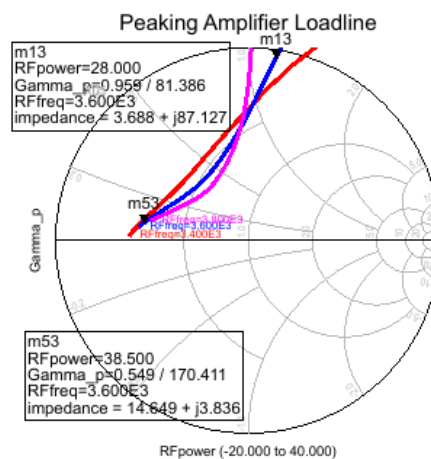
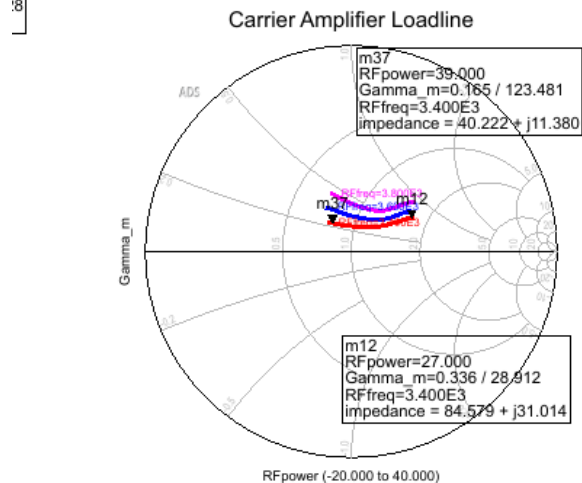
Gan1 model



DPA Small Signal Simulation (EM model)

Vdd=40V; Vg_main=-2.78V; Vg_peak=-5.6V; Iq=30mA

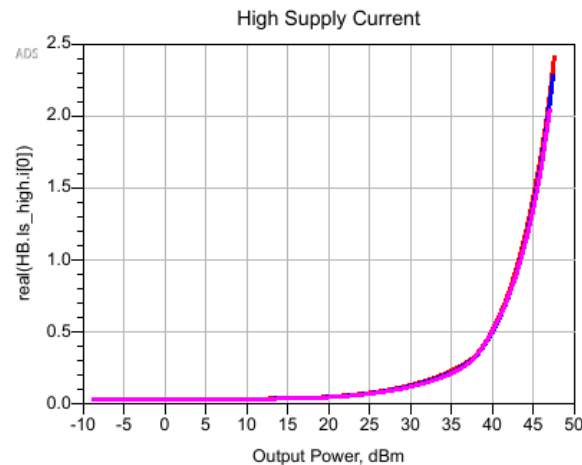
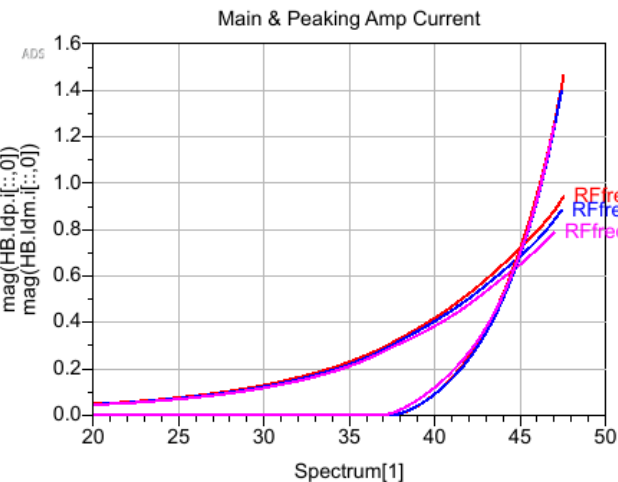
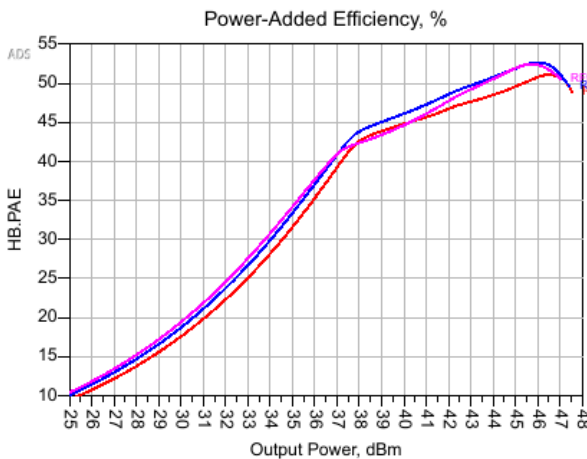
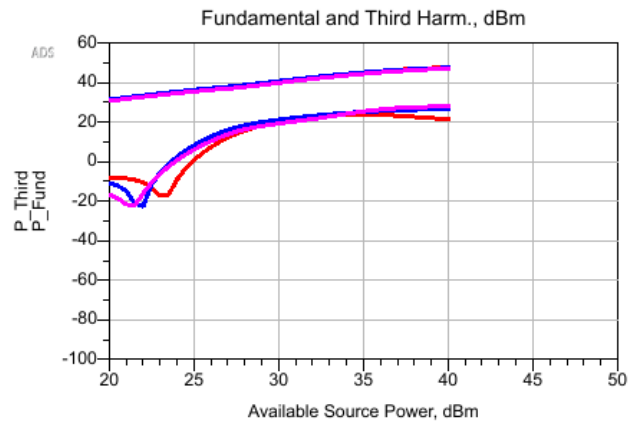
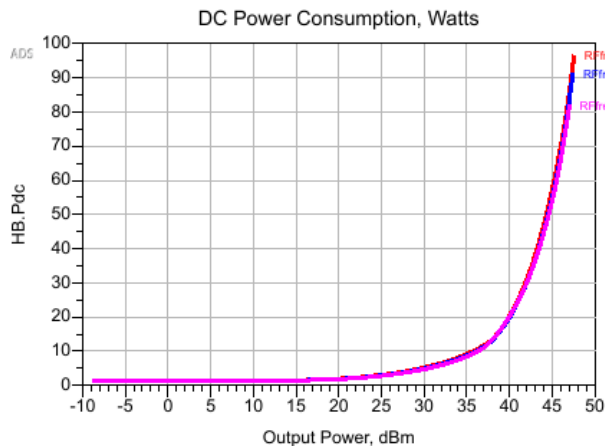
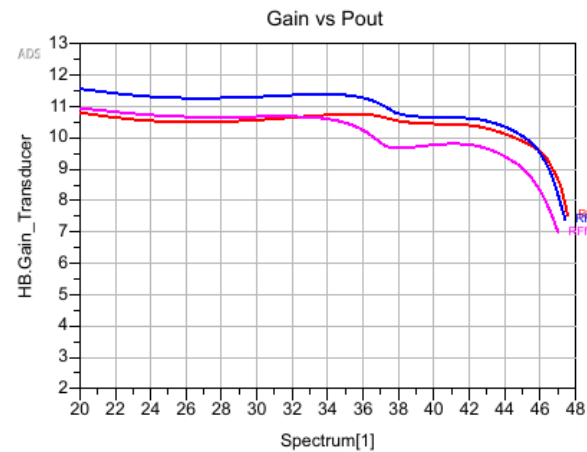
8



DPA Large Signal Simulation (EM model)

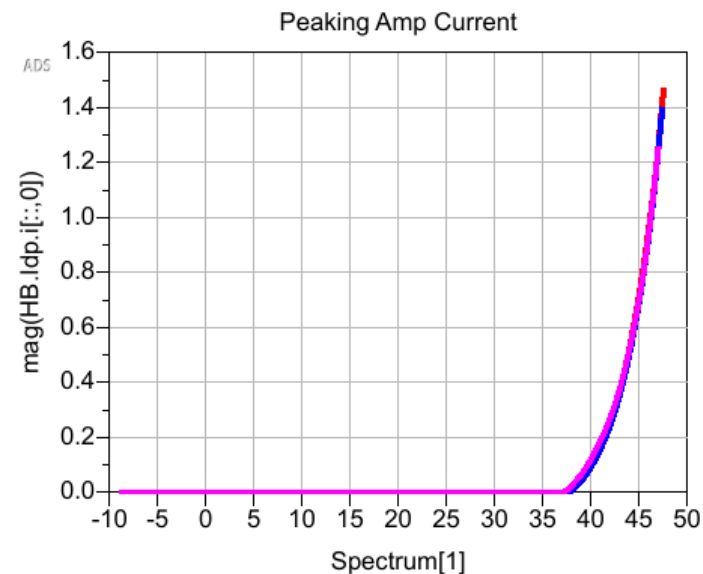
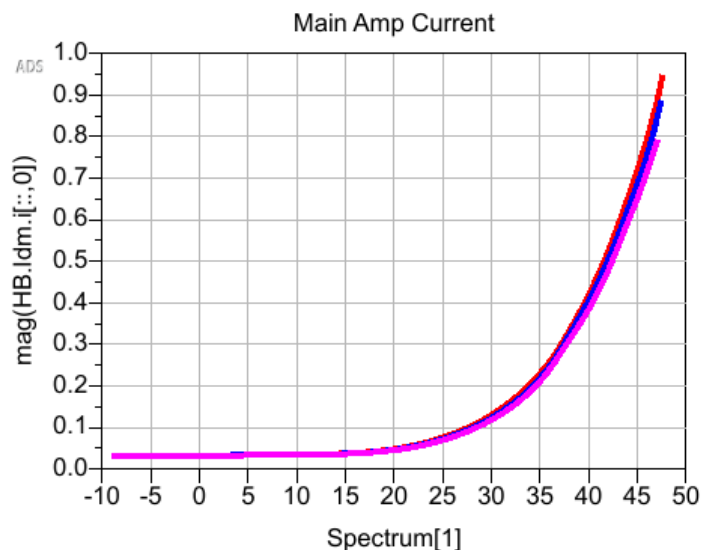
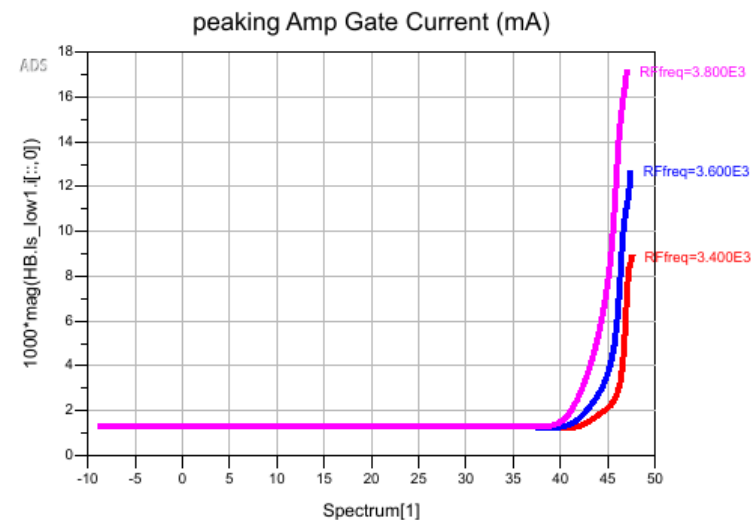
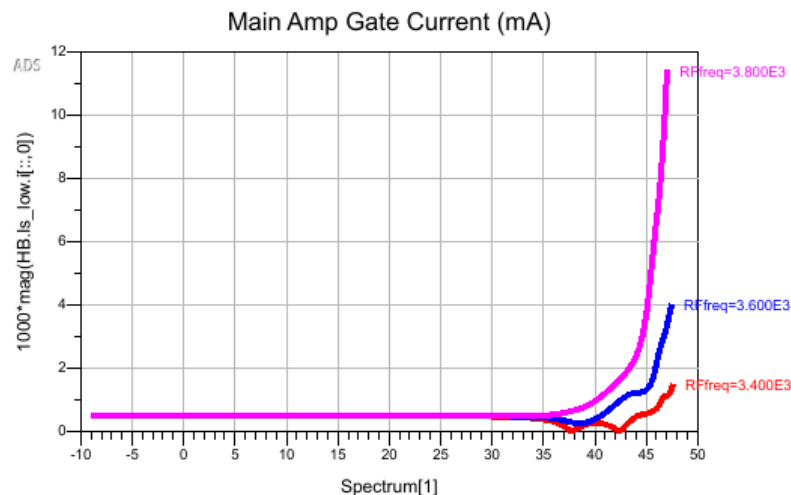
Vdd=40V; Vg_main=-2.78V; Vg_peak=-5.6V; Iq=30mA

	Fundamental Output Power dBm	Transducer Power Gain	Gain Compression	Power- Added Efficiency, %	DC Power Consumpt. Watts	High Supply Current	Thermal Dissipation Watts	Second Harmonic dBc	Third Harmonic dBc	Fourth Harmonic dBc	Fifth Harmonic dBc	Input Return Loss, dB	Output Return Loss, dB
3.400 k	37.99	10.54	818.4 m	42.60	13.53	338.3 m	7.719	-25.26	-24.08	-32.13	-45.90	-12.73	0.0000 / 0...
3.600 k	37.98	10.78	1.235	43.81	13.13	328.2 m	7.311	-27.83	-21.27	-29.87	-46.46	-26.95	0.0000 / 0...
3.800 k	38.02	9.674	1.624	42.40	13.41	335.3 m	7.670	-36.99	-21.09	-30.27	-46.70	-13.97	0.0000 / 0...



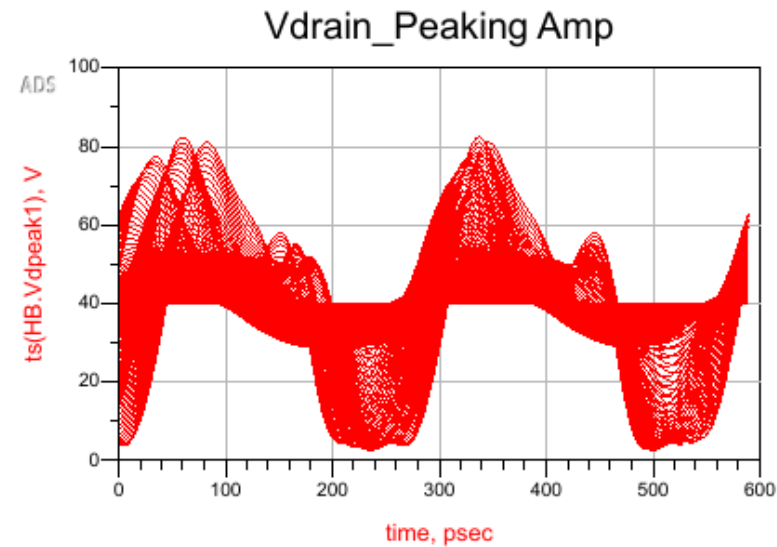
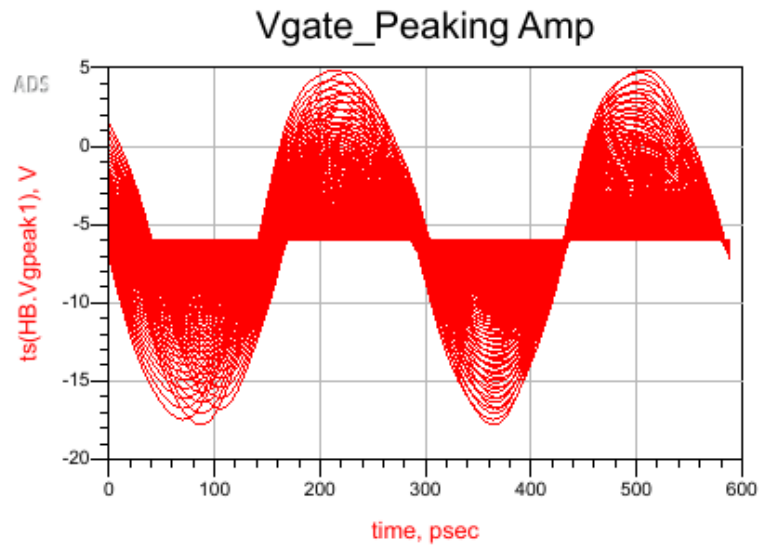
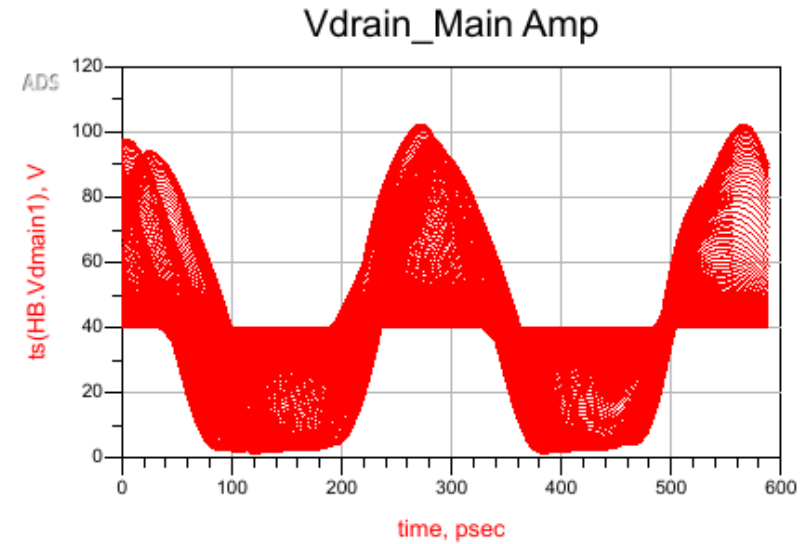
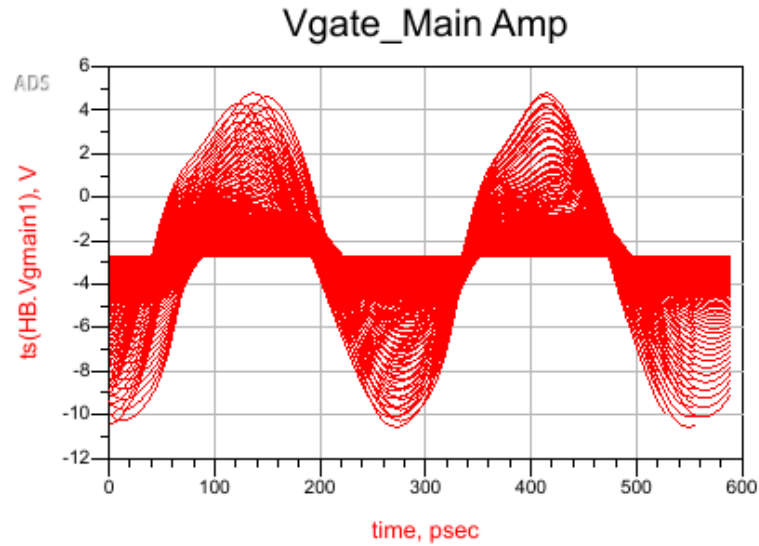
DPA Gate Current Simulation (EM model)

Vdd=40V; Vg_main=-2.78V; Vg_peak=-5.6V; Iq=30mA

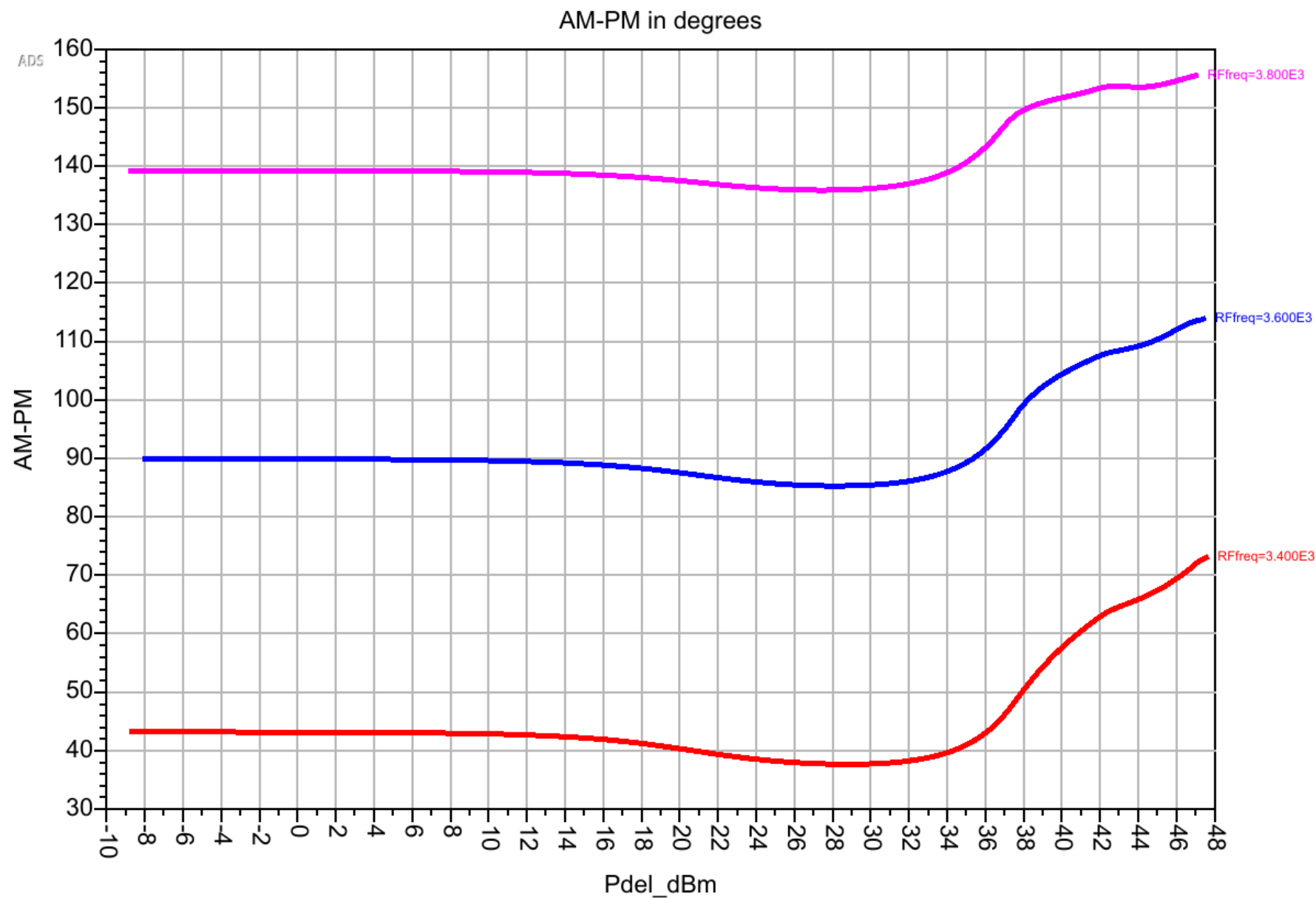


DPA Gate Voltage Swing Simulation (EM model)

Vdd=40V; Vg_main=-2.78V; Vg_peak=-5.6V; Iq=30mA



AM-PM Power Sweep

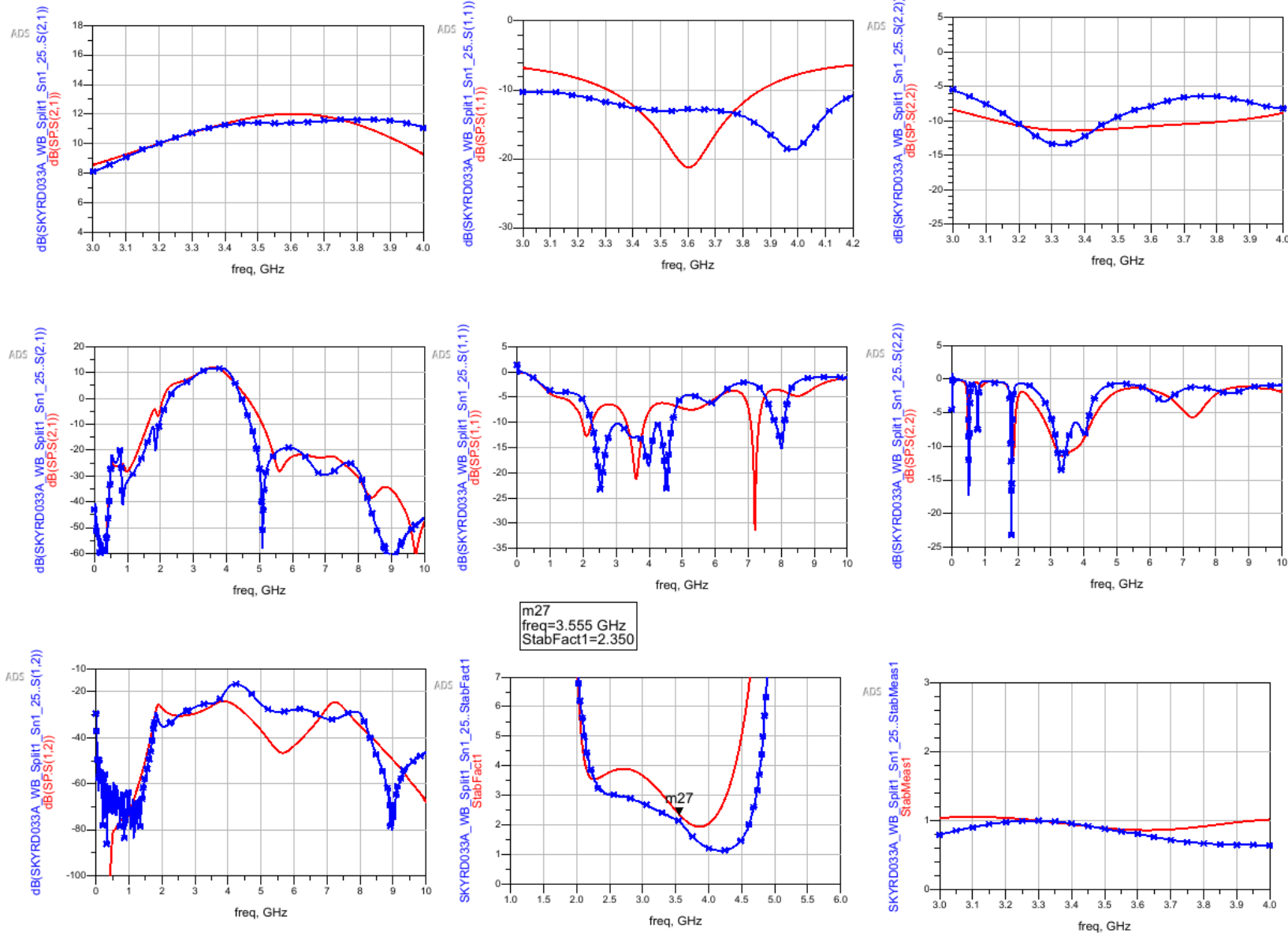


Test Results

(Measurement and Simulation Comparison)

SKYRD033A_WB – Comparison Sim(red) VS Meas(blue)

Vdd=40v; IDDq=30mA; Vg_peak=-6v

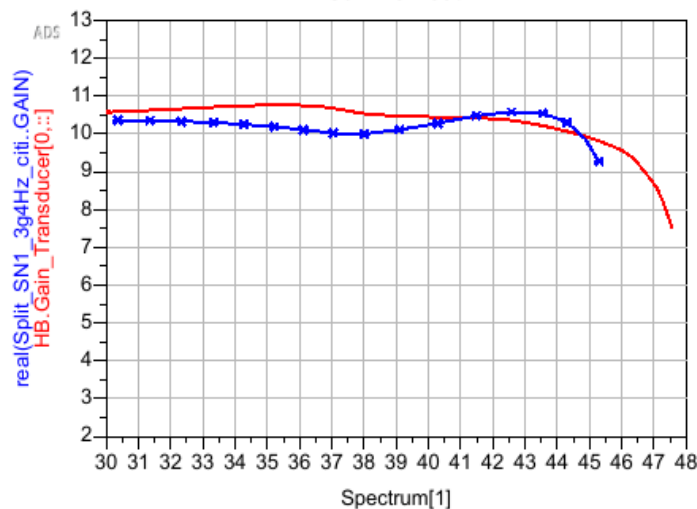


SKYRD033A_WB Comparison Sim(red) VS CW Meas(blue)

Vdd=40v; IDDq=30mA; Vg_peak=-6v

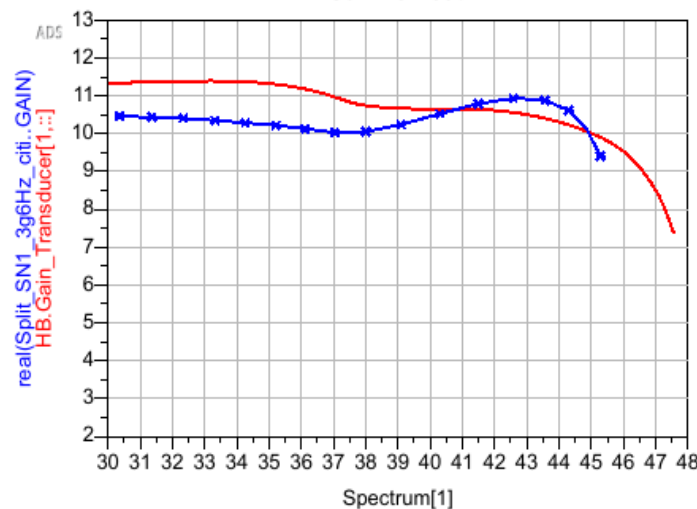
3.4 GHz

Gain vs Pout



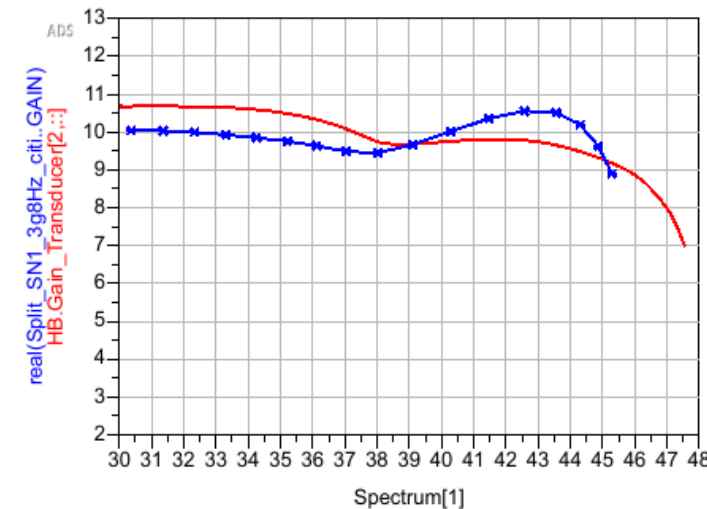
3.6 GHz

Gain vs Pout

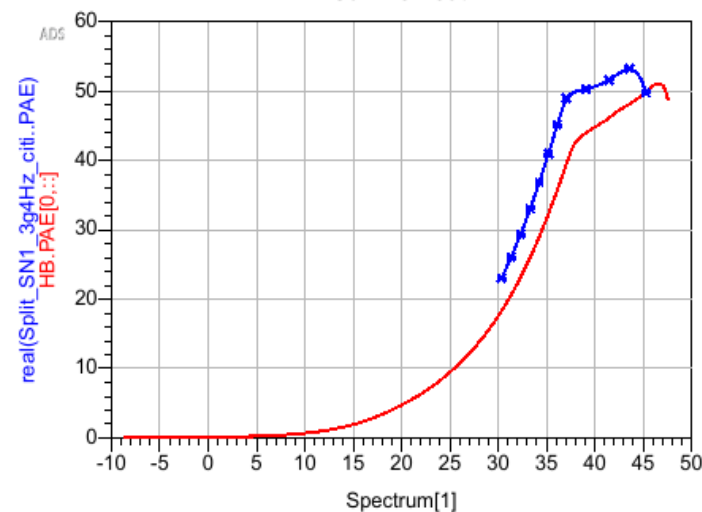


3.8 GHz

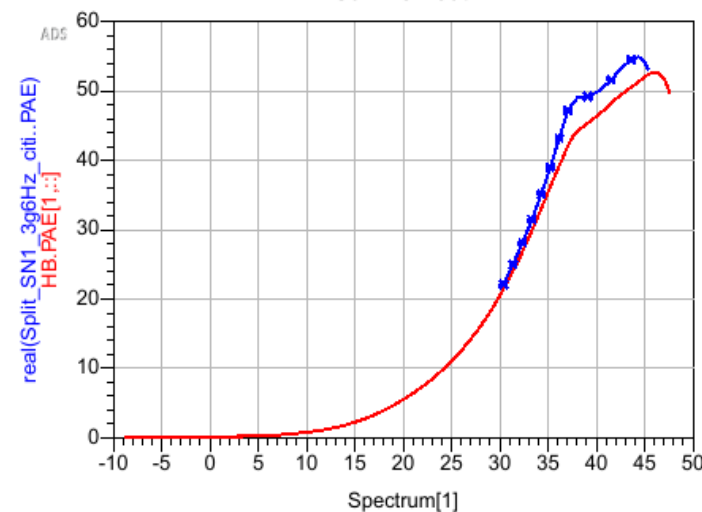
Gain vs Pout



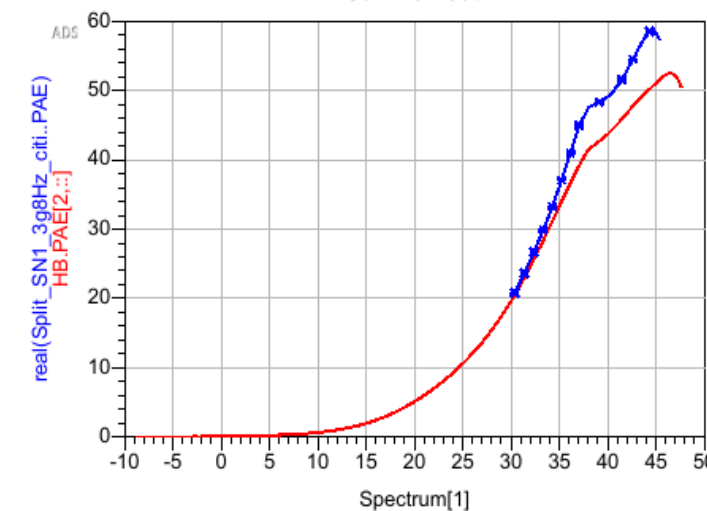
Gain vs Pout



Gain vs Pout



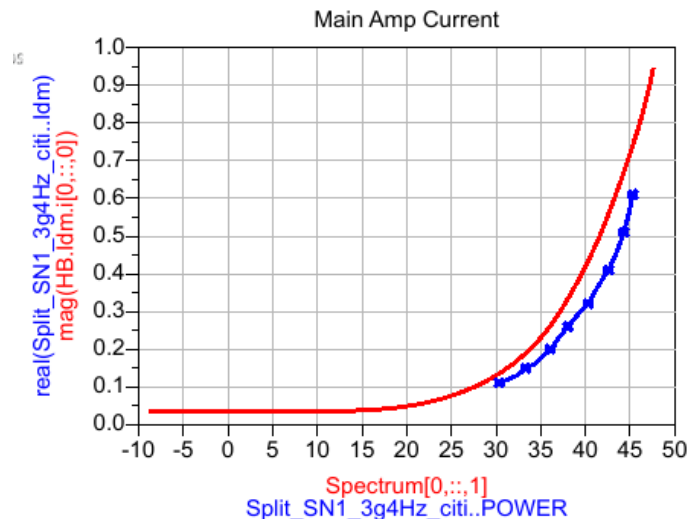
Gain vs Pout



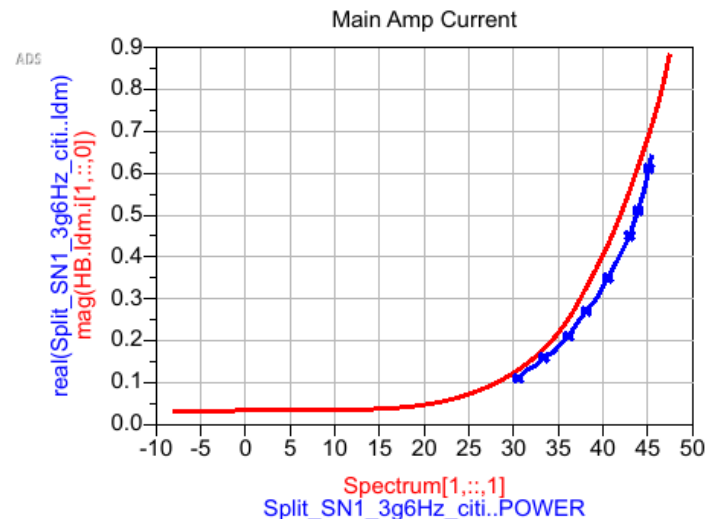
SKYRD033A_WB Comparison Sim(red) VS CW Meas(blue)

Vdd=40v; IDDq=30mA; Vg_peak=-6v

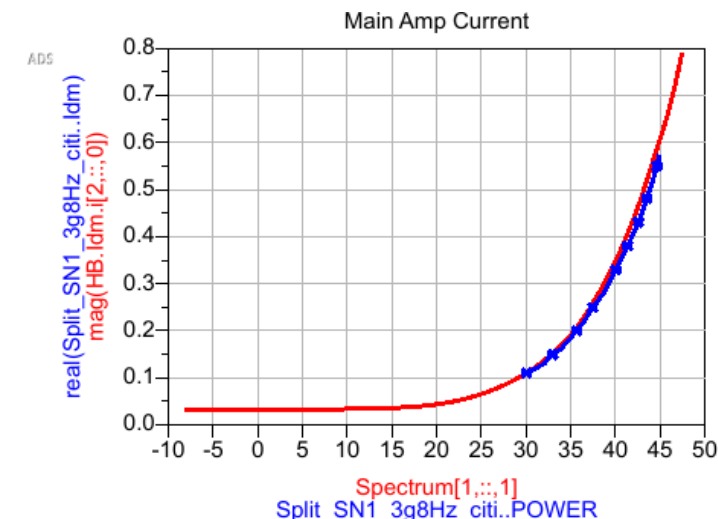
3.4 GHz



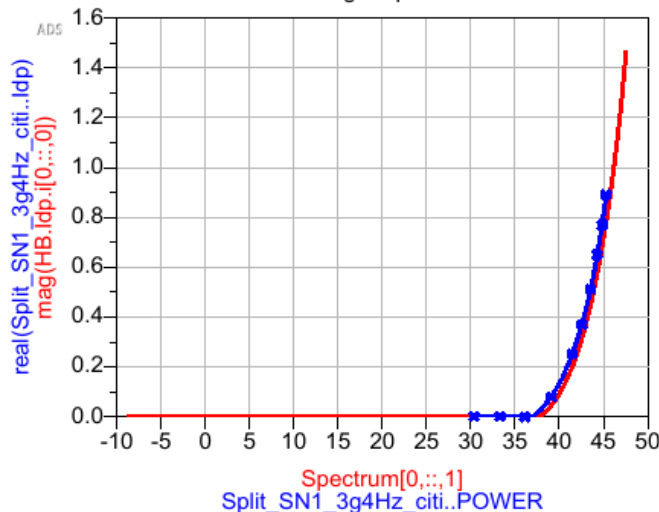
3.6 GHz



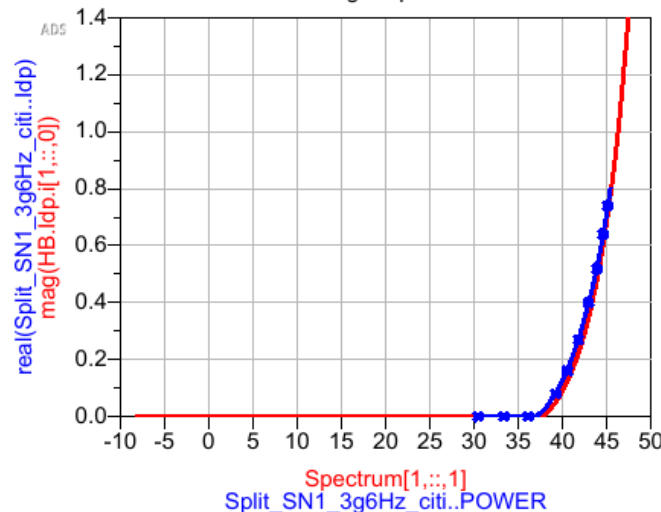
3.8 GHz



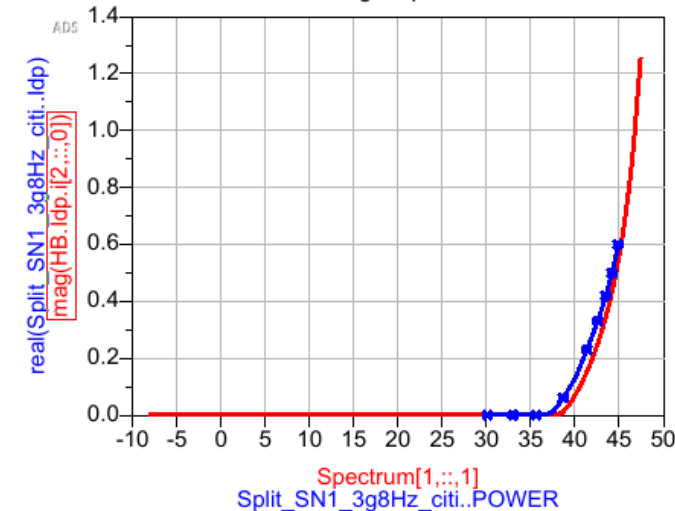
Peaking Amp Current



Peaking Amp Current



Peaking Amp Current

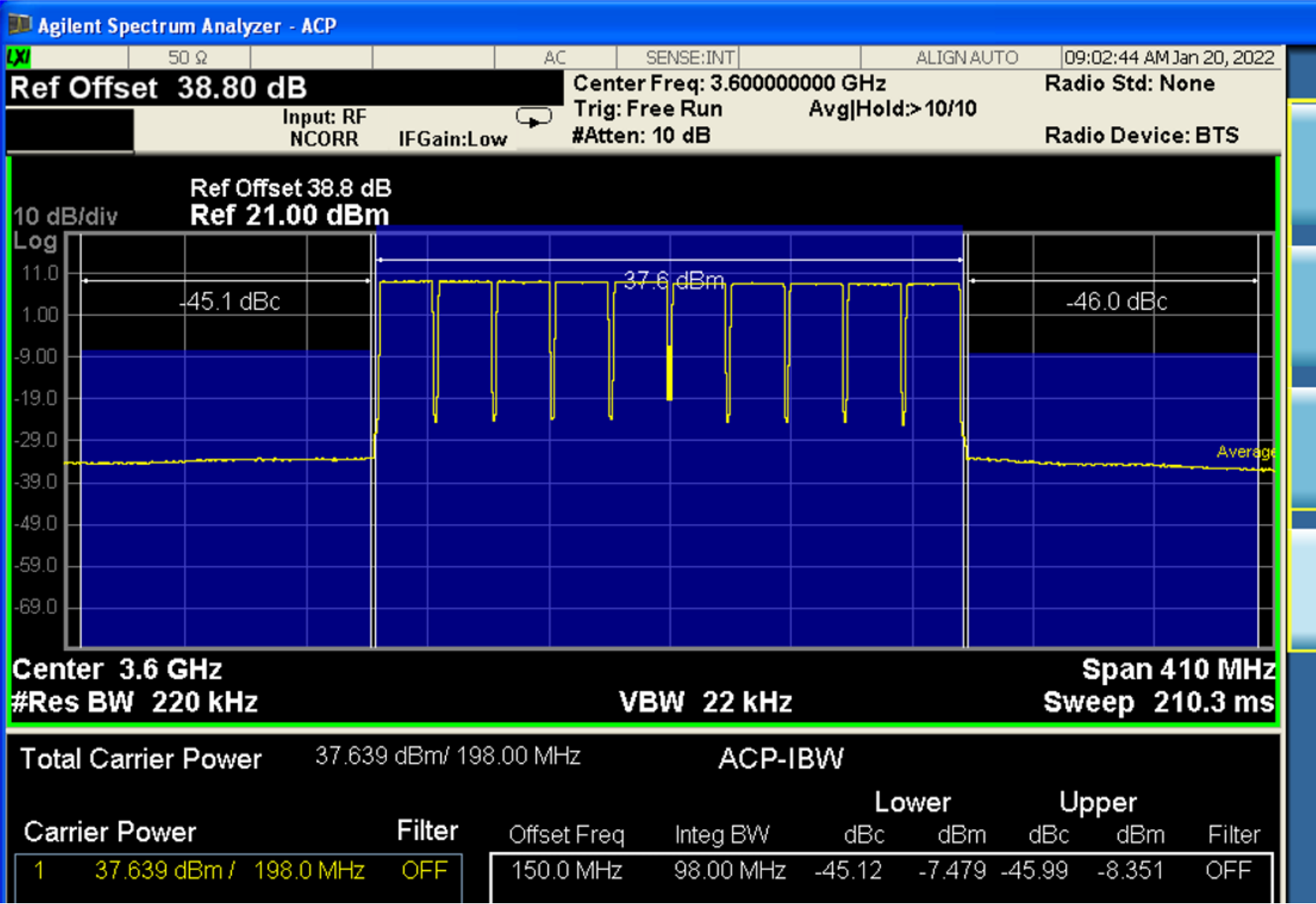


P3dB Measurement Using 10 MHz LTE

VDD=40V; IDq=30mA; Vpk = -6.6V	
Freq	PSAT de-embedded
3.4G	46.85
3.6G	47.07
3.8G	47.03

Split2-SN4 DPD ACLR Measurement

$V_{dd}=40V$; $I_{dq}=30mA$; $V_{g_peak}=-6.0v$; $P_{out} = 38\text{ dBm}$; $F=3.6\text{ GHz}$



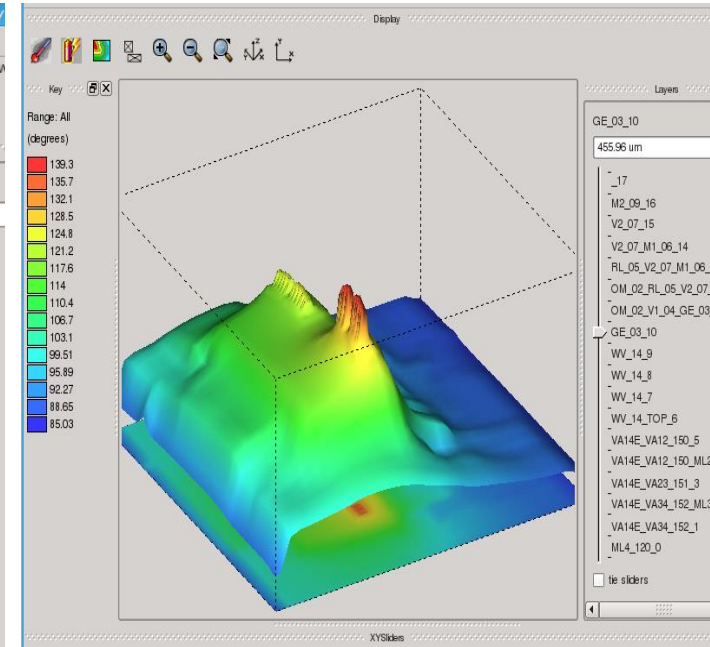
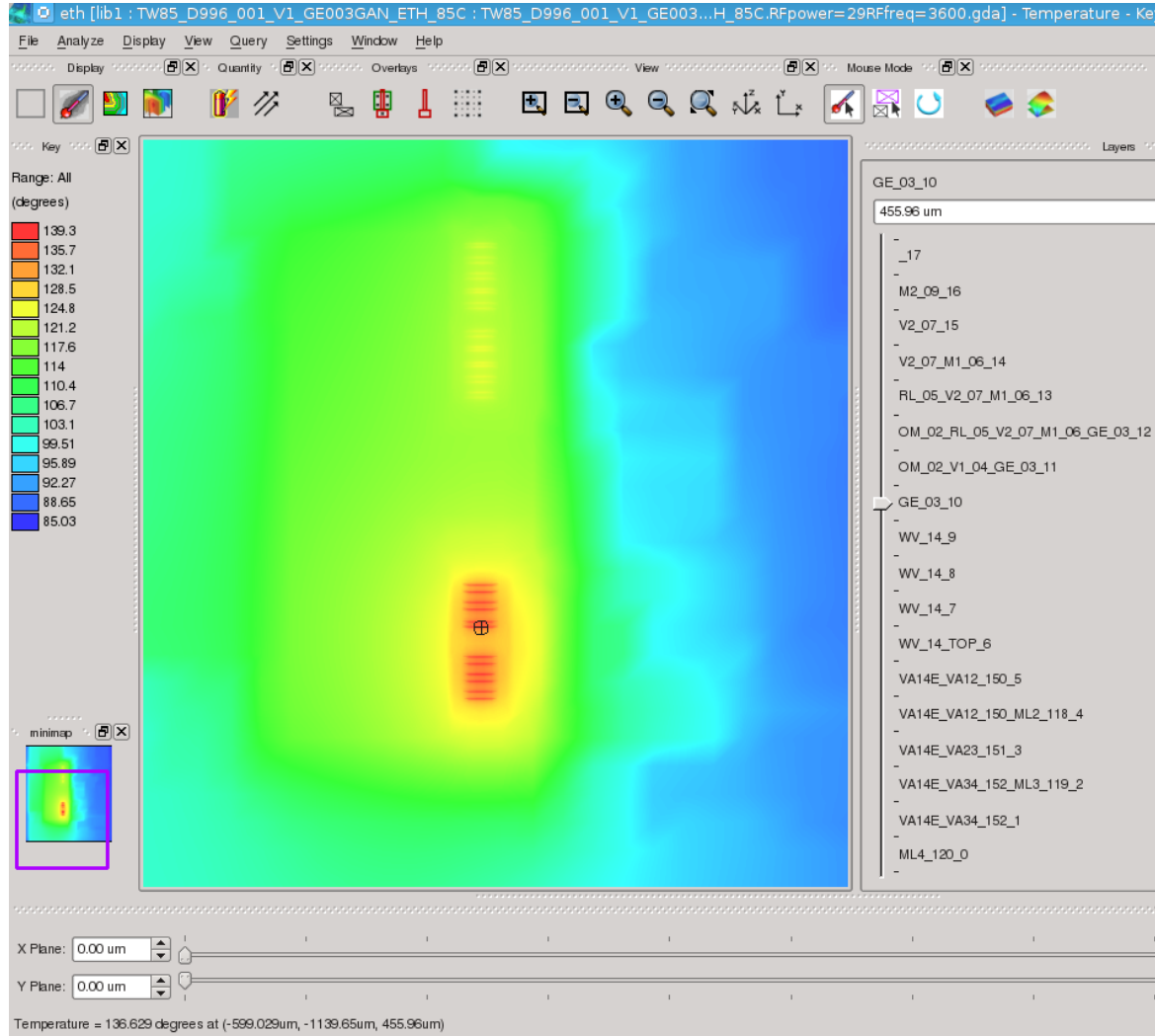


Thermal Considerations & Challenges

Harmonic Balance ETH Simulation for SKY GaN With 4-L MCM

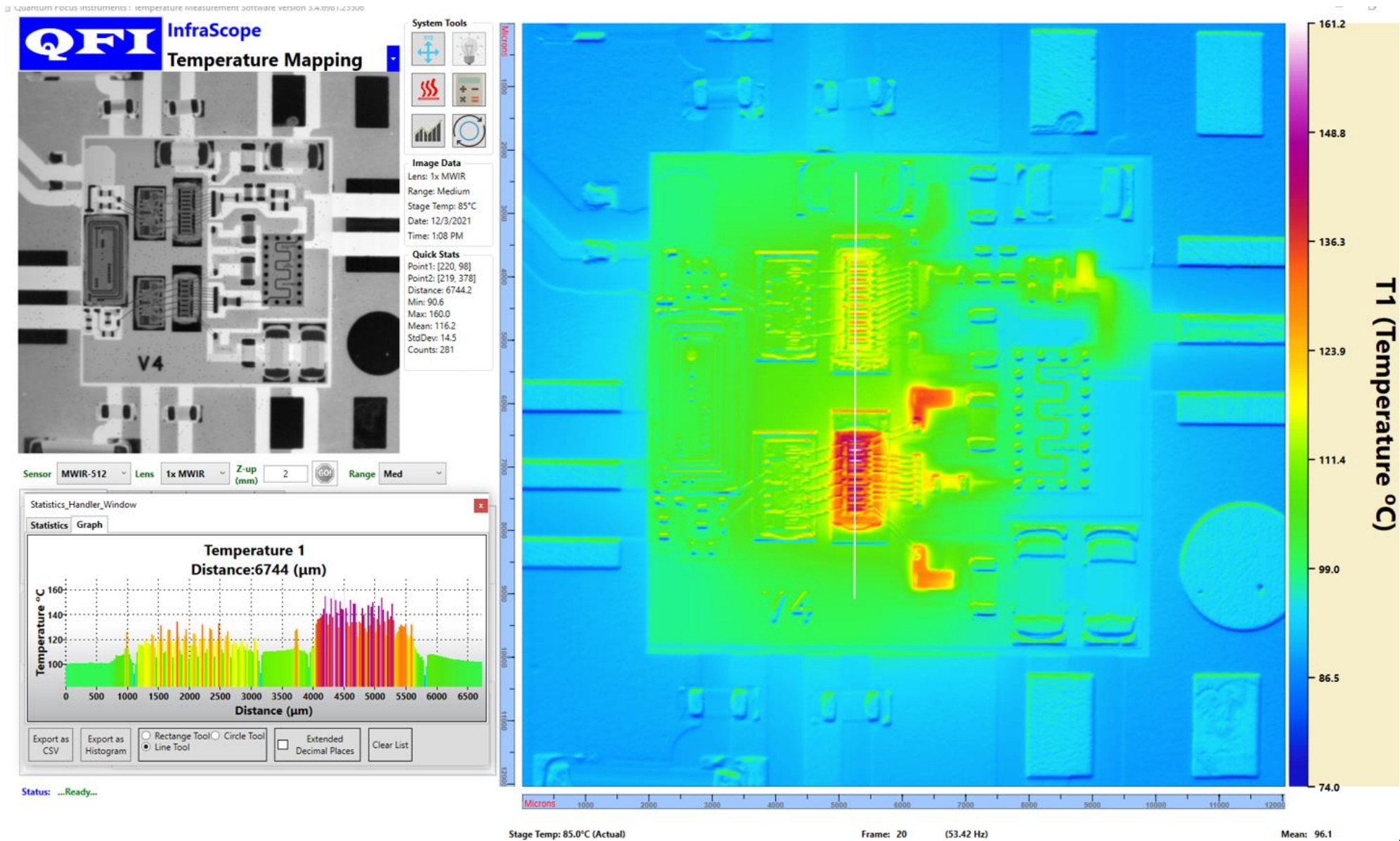
Vdd=40; Vgm= ; Iqm= 27.8 mA; Iqp=0.657 mA; Freq=3.6GHz; Temp=85C

PIN=29 dBm; Pout=38 dBm; CW



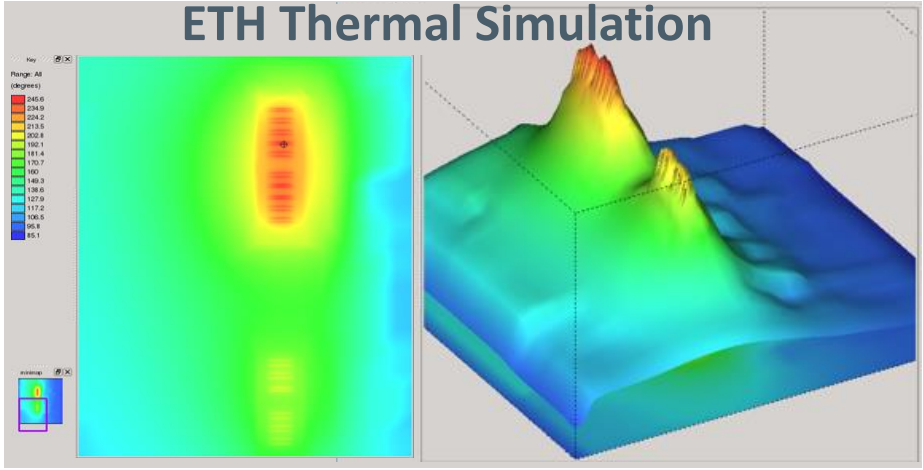
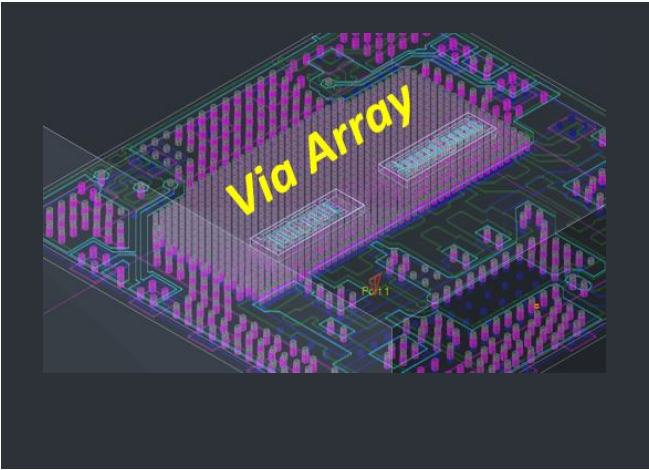
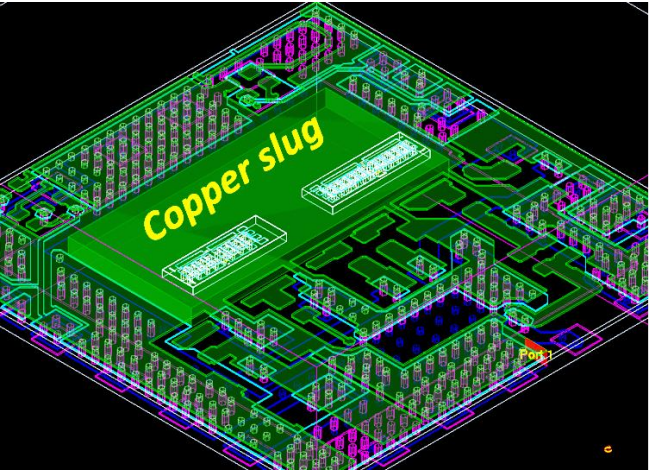
IR SCAN

Vdd=40; Vgm= ; Iqm= 30 mA; Iqp=0.657 mA; Freq=3.6GHz; Pout=38dBm; Temp=85C



Simulation Summary with ETH of Cu-Slug vs Via-Array

SKYRD033: Massive MIMO FEM: Vdd=40; Iqm= 27.8 mA; Iqp=0.657 mA; Freq=3.6GHz



Cu-Slug Temp=85 C

Pout (dBm)	Gain (dB)	PAE (%)
38.0	9.2	36.2
42.0	9.0	41.6

Via-Array Temp=85 C

Pout (dBm)	Gain (dB)	PAE (%)
38.0	8.5	33.3
42.0	8.5	40.5

Performance
Improvements
based on Lower
Die-Temperature



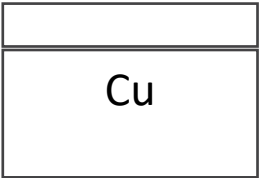
Gain Delta	PAE Delta
Gain (dB)	PAE (%)
0.7 (8.2%)	2.9 %
0.5 (5.8%)	1.1 %

An Increase of 10C in Tj Leads to a 1% Reduction in PAE

Thermal Comparative Summary from ETH simulation

Solid Cu Coin S 4 layers	Tcase= 85C	Tcase=110C	Tcase =125C
Pout (RMS)	6.3W	6.3W	6.3W
Gain (dB)	9.1	8.5	8.1
PAE (%)	37.7	35.6	34.3
Pdiss (W)	9	9.7	10.1
Tj (degC)	137	167	184
Rth (degC/W)	5.7	5.8	5.8
Bottom Slug (degC)	127	155	171

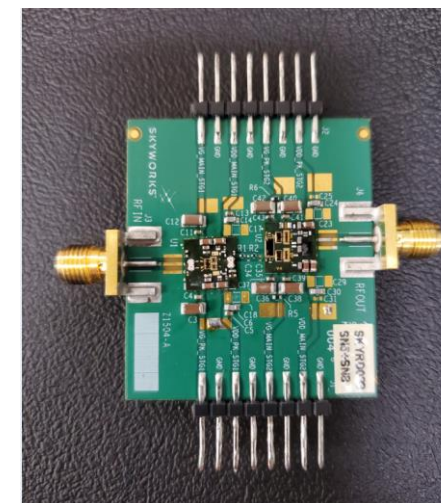
Vdd=40; Iqm= 27.8 mA; Iqp=0.657 mA; Freq=3.6GHz
Doherty PA Output stage
Atrox die attach (800HT2V): 130W/mK
4ML MCM





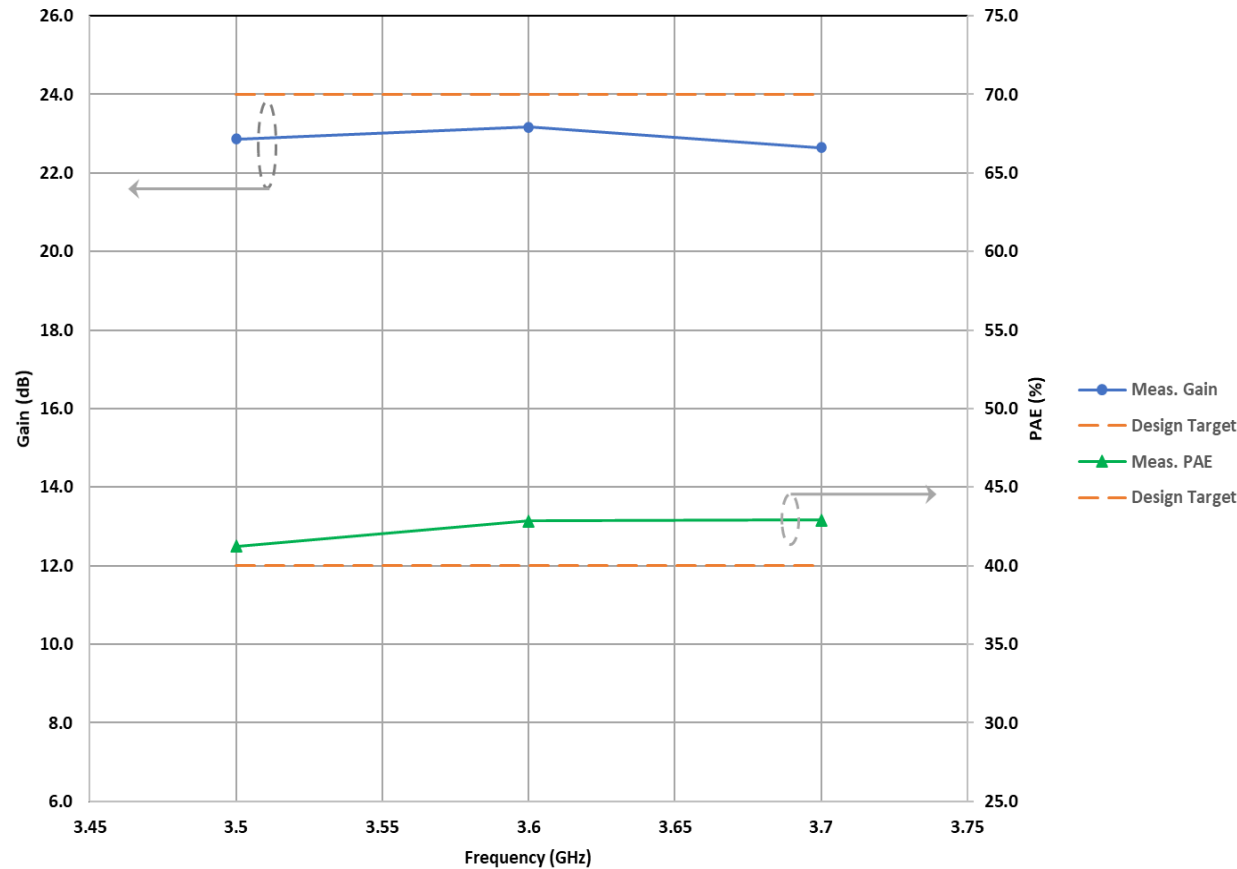
Cascaded Performance

Doherty Lite + Final



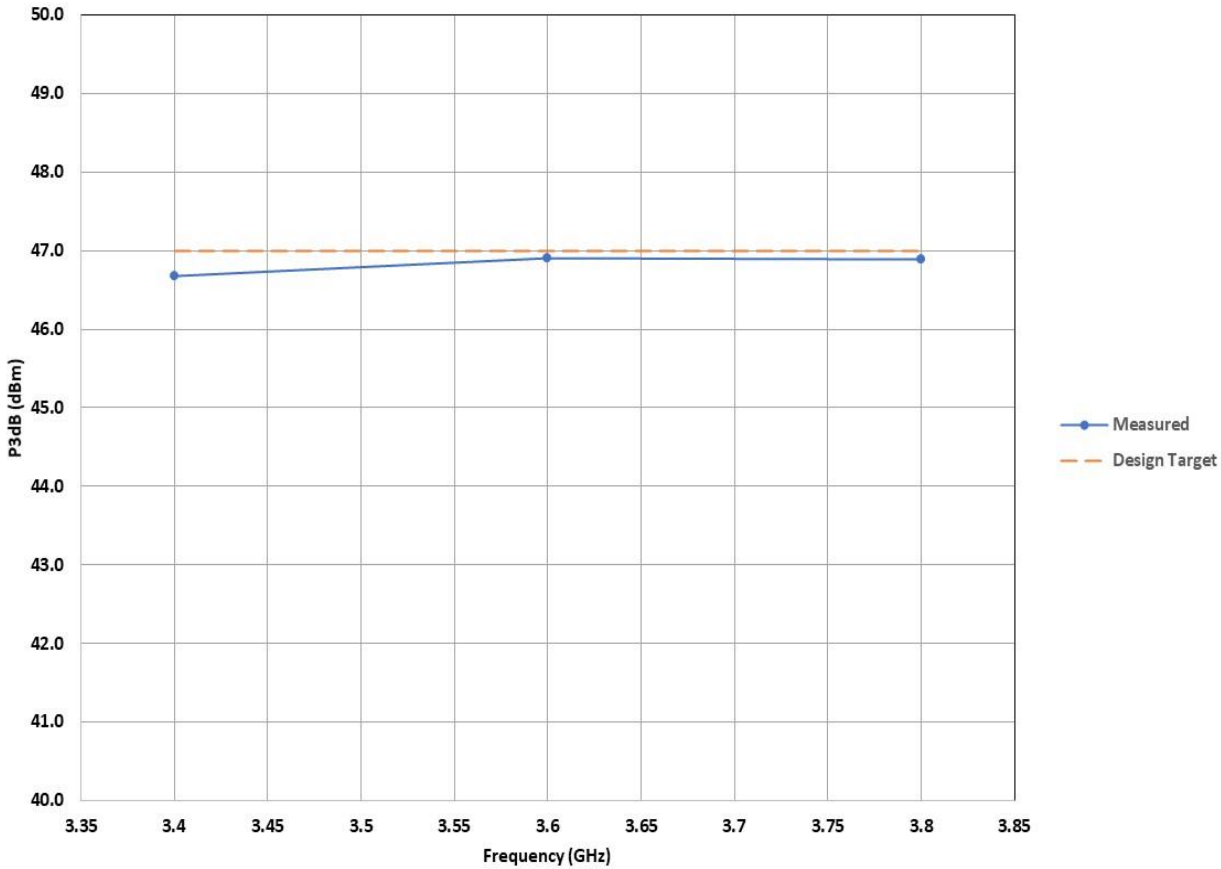
Cascode Performance

SKYRD033 Cascaded Performance
Gain & PAE vs. Frequency & Temperature



Measured using 200MHz LTE
8.5dB PAPR

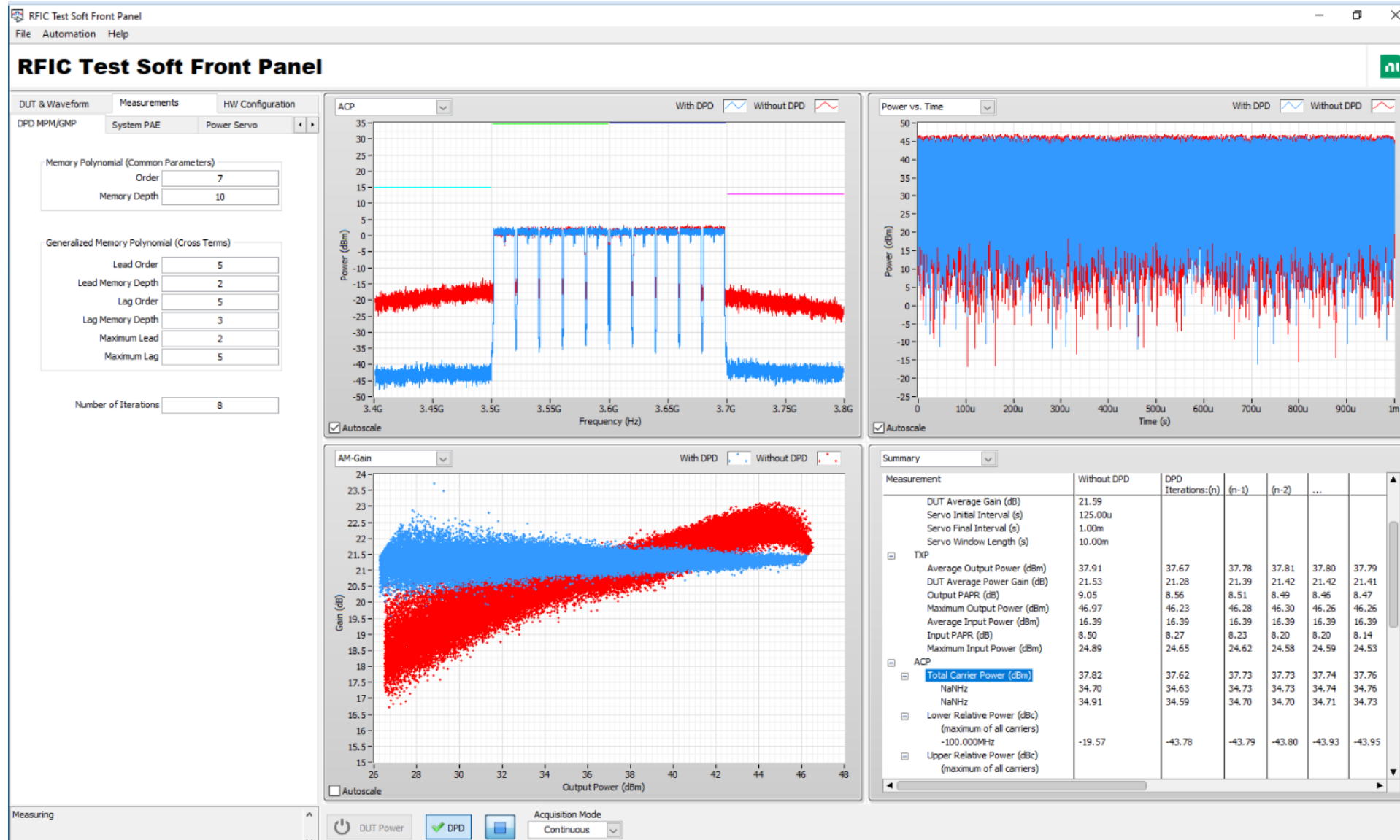
SKYRD033 Cascaded Performance
P3dB vs. Frequency & Temperature



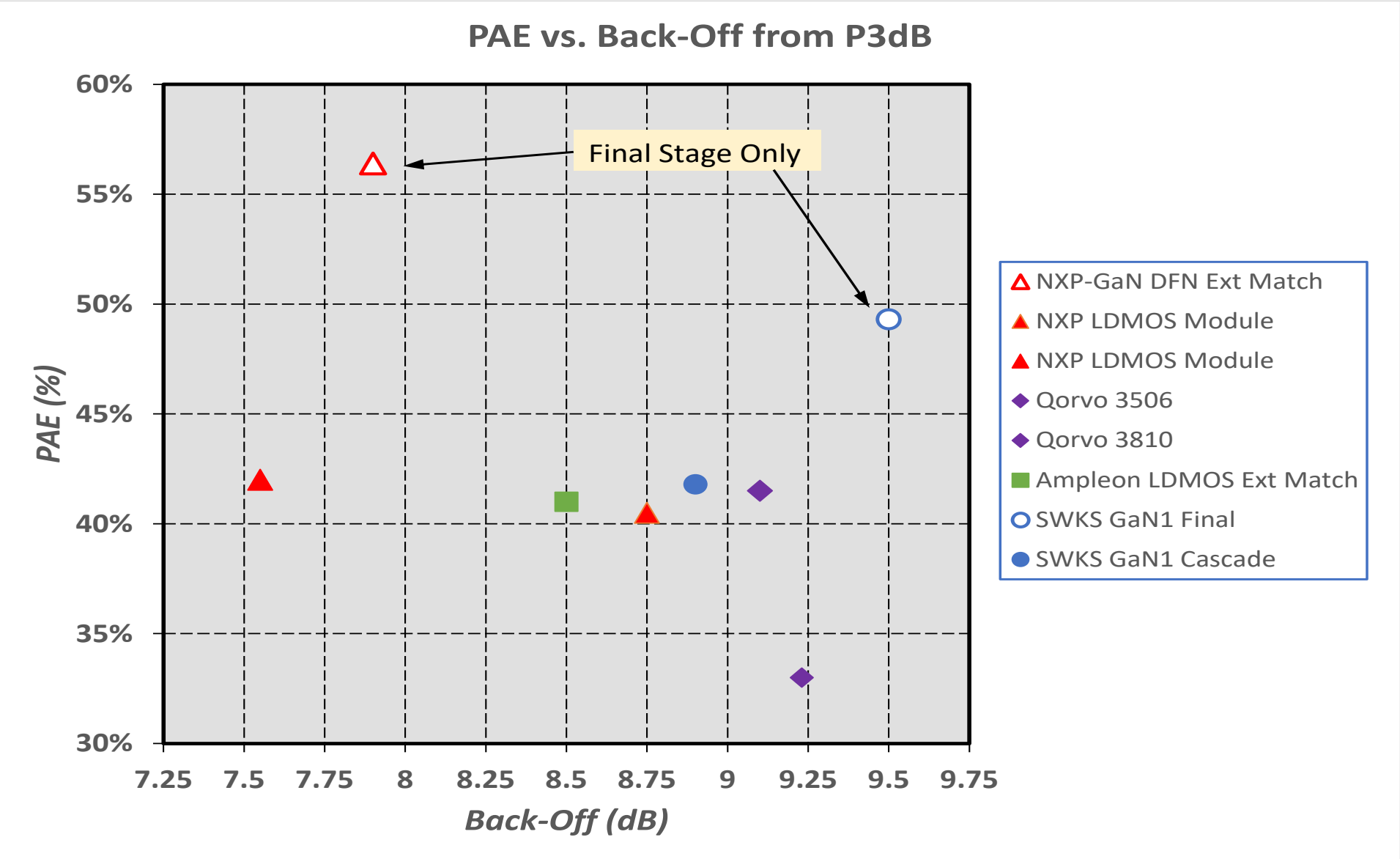
Measured using unclipped 10MHz LTE

Cascaded ACLR – ($f = 3.6\text{GHz}$, $i\text{BW} = 200\text{MHz}$, $\text{PAPR} = 8.5\text{dB}$ @ 25C)

Stage 1 $\text{ICQ} = 18.5\text{mA}$, Stg2 $\text{ICQ} = 30\text{mA}$, $\text{P}_{\text{OUT}} = +38\text{dBm}$



Performance Comparison of 3.5 GHz PAM Solutions



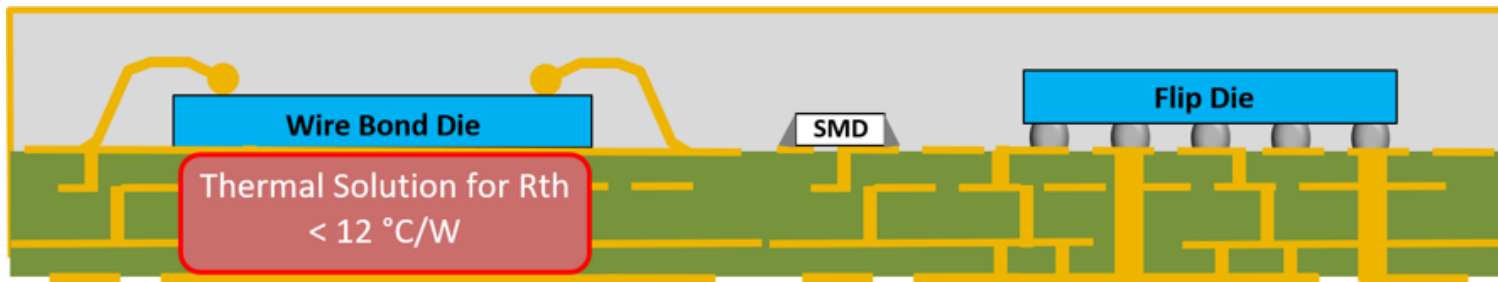


Appendix / Back up Slides

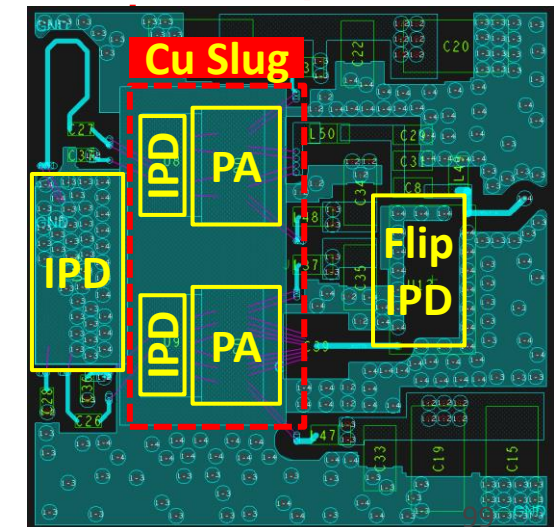
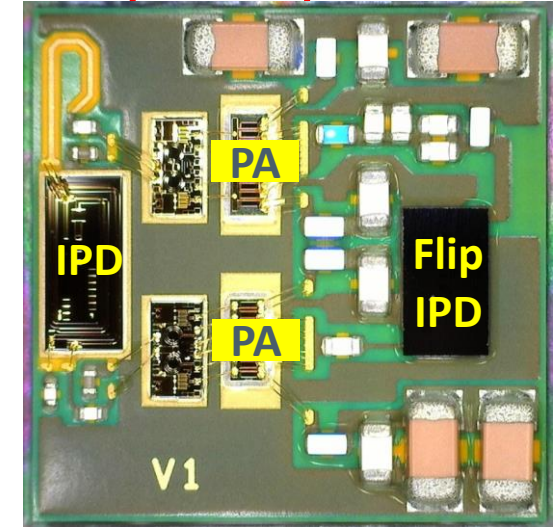
Overview

High Power MCM Package NTI P0

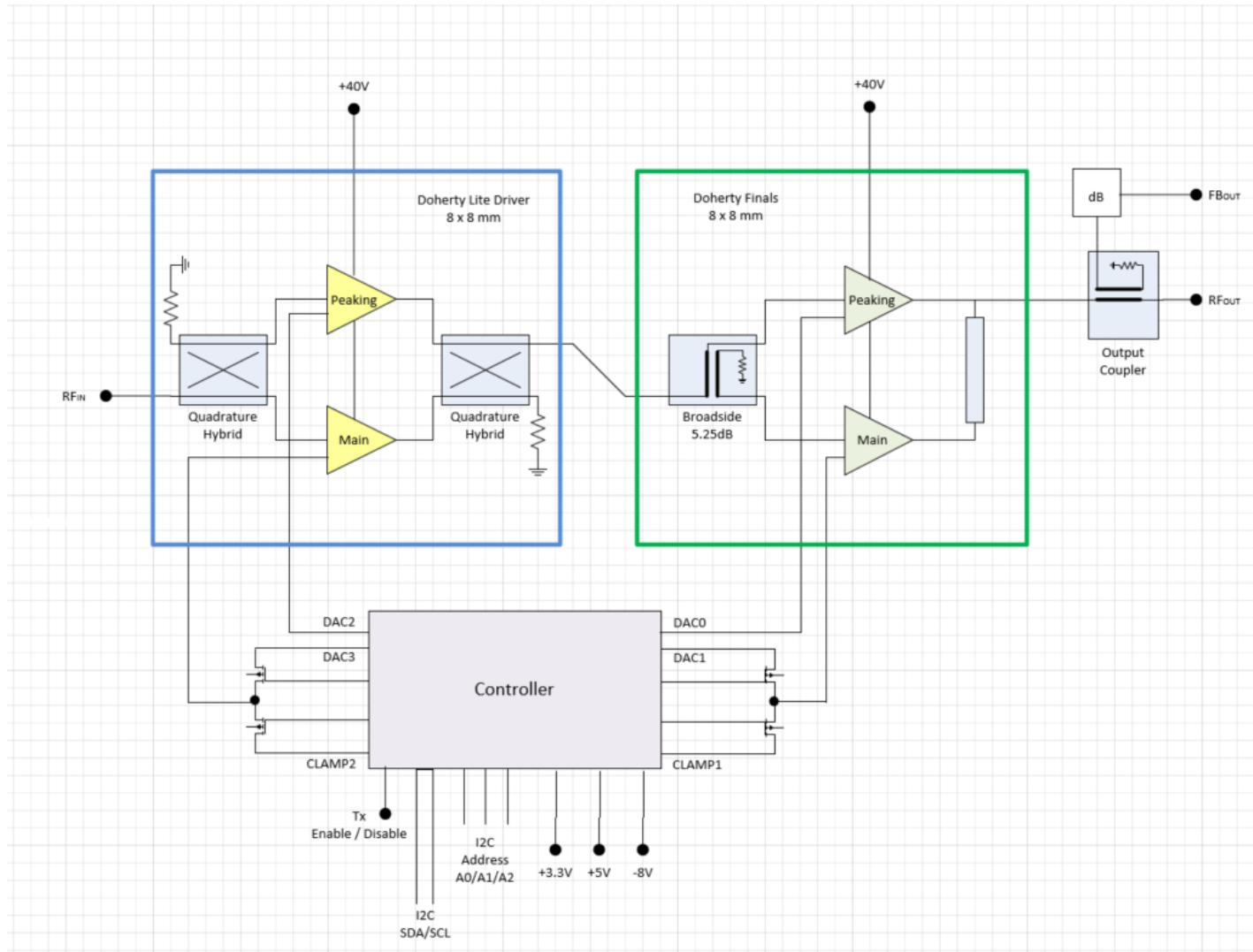
- Qualify a Substrate and associated Assembly solution for a high-power MCM platform with the following characteristics
 - Large Module Size ~ 8x12mm (<=1mm height)
 - 4-6 Layers for Routing
 - Top-Metal Finish for Wire-bond+SMD+Flip-chip
 - Low loss full-stack via-connections
 - Capability for Conformal Shielding
 - Thermal Solution for < 12°C/W (comparable to QFN)
 - Wirebond SiC Die ~1x2mm
 - 16W power dissipation
 - Capability for multiple-die in different locations
 - Baseplate ambient to 115°C



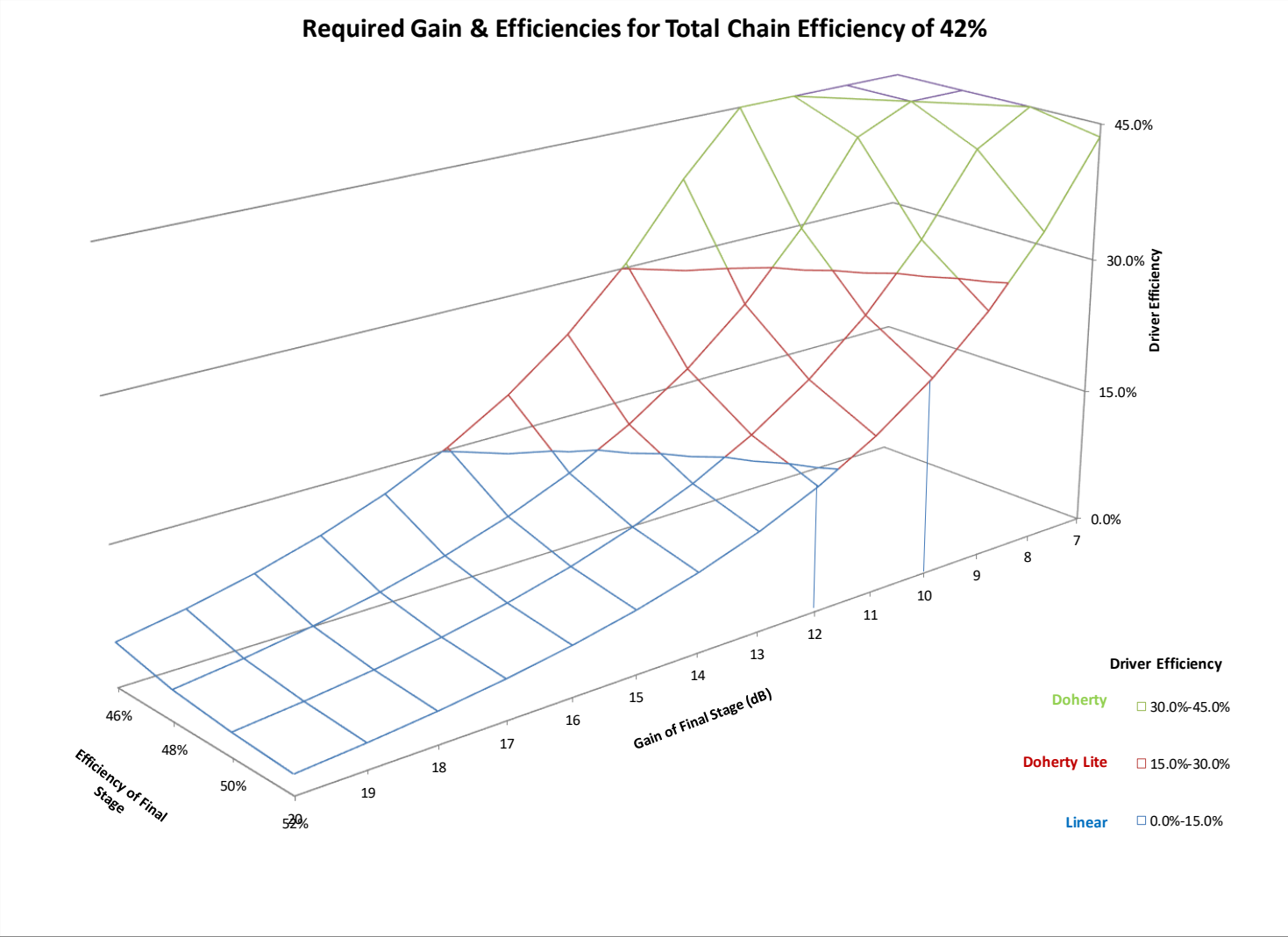
Example Solution:
Crosssection



Doherty Lite Driving Doherty Final

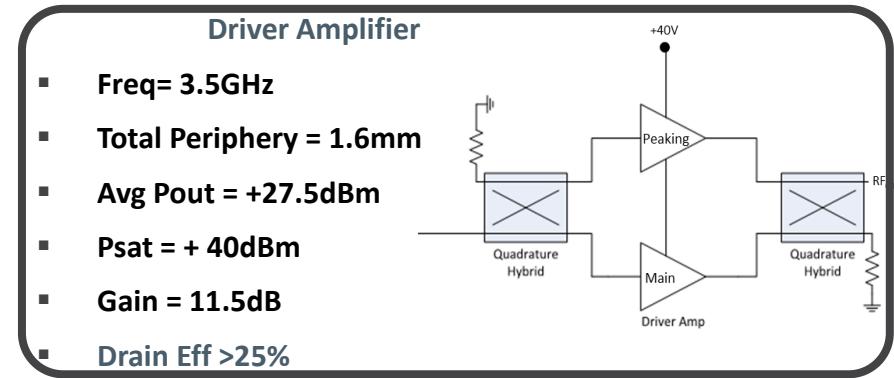


Surface Plot of Gains & Efficiency Required for Total Chain PAE of 42%

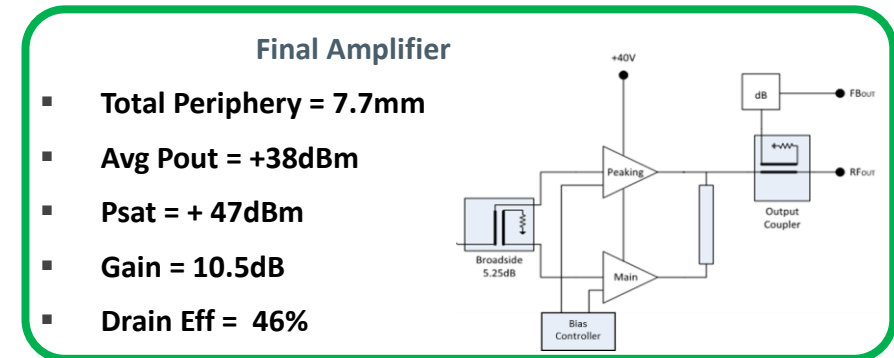


Target Tx Chain Requirements

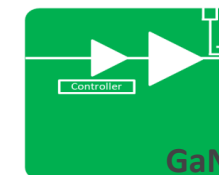
- **Freq. Range: 3.4 – 3.8 GHz**
- **PAVG : 38dBm / 6.3W @ PA output (LTE)**
 - *Post PA loss ~ 0.3 to 0.5 dB (~6W required at input to ANT port)*
- **PSAT/3dB : 47dBm (PA output)**
- **PAE: >40% @ 9dB OBO**
- **Total Gain: > 20dB typ @ 25C**
- **IBW: 200MHz**
- **PAR: 8.5dB typ.**
- **40V rail**



+



Integrated PAM+



Integrated PAM+

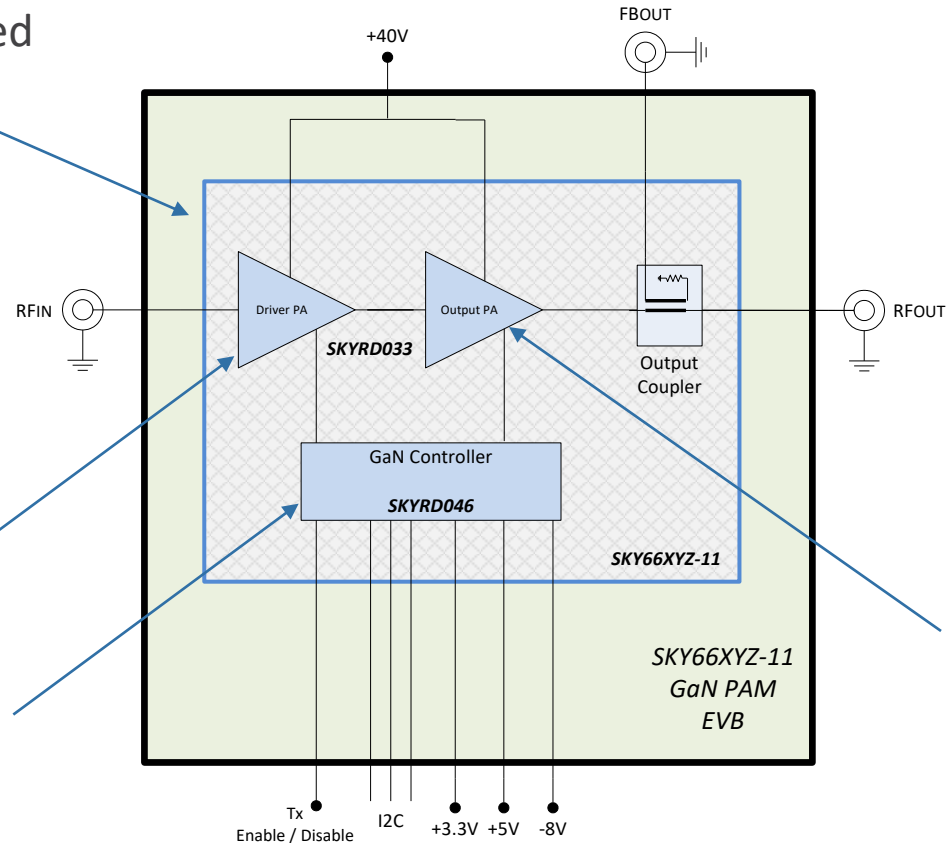
38 dBm Pout Avg
47 dBm Psat (**50W**)
Integrated Controller
PAE > 40%

Integrated GaN PAM with Controller

2 Stage GaN PA with Integrated
Controller and Coupler
Assembled at OSAT

Doherty Lite Driver Stage: > 1W

Si Bias Controller
Sets & Controls Bias
of GaN PA Stages



Final Output Doherty Stage: > 6W

Summary / Overview of SKYRD033

Objective:

To have necessary IP developed & in place when Internal GaN is available.

- GaN PA for use in Tx Chain.
- High Power MCM / Packaging
- Control Circuitry

Deliverables:

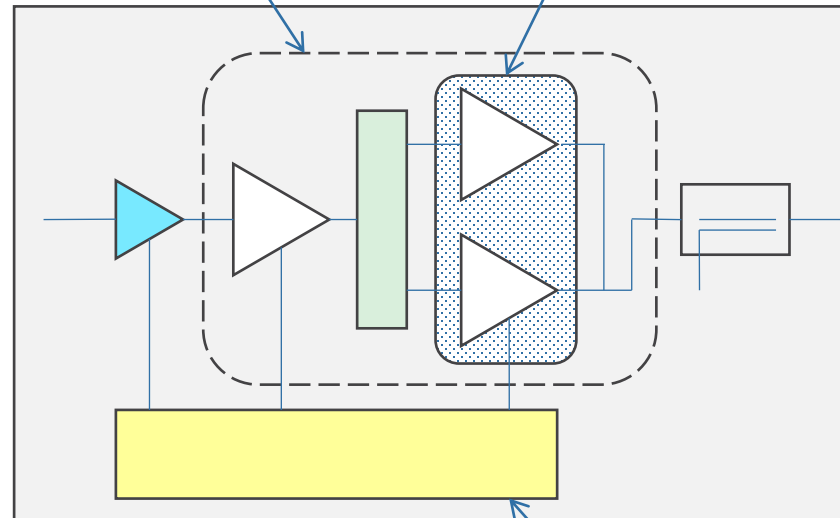
**An integrated chip and wire MCM GaN Tx Chain.
Defined Tx architecture**

GaN Tx Chain

- Doherty Final
 - Single stg or 2 stg core
- IPD matching & splitter
- Driver

MCM

- Chip and Wire Assembly
- Thermal Heat Slug
- Embedded Coupler



This is from the SKYRD033 P0. Over time, portions of other IP development needs were split off as separate SKYRD programs.

The Analog Controller became SKYRD046

The MCM & Assembly become SKYRD056

Internal HVIPD process became SKYRD520

Analog Controller

- Gate Bias Control (negative)
- Temperature Compensation
- PA enable / disable
- Supply sequencing

Final Stage Device Size Calculation

r^*	PAPR (dB)
1	-6
1.5	-8
2	-9.5
2.5	-10.9
3	-12

* $r = P_p/P_c$ (ratio of peaking to carrier power)

Peaking Amp Device Size Estimation:

Total power requirement: @ 47dBm = 50.1 W

Peak power is assumed 47dBm-2dB = 45 dBm = 31.62W

Peak Power needed (P_{t_p}) = 45dBm+0.5dB (IL)
= 45.5dB = 35.48 W

Calculate Device size (WT) based on GaN1

Power density ~ 6.5W/mm, $V_d=40V$ @ 4.6 GHz

$$WT = P_{out}/P_j = 35.48W/(6.5W/mm) \\ = 5.45 \text{ mm}$$

Main Amp Device Size Estimation:

Power needed by Main Amp = 50.1W-31.62

$$= 18.48W = 42.67\text{dBm}$$

Assume IL = 0.5dB

Total Main Amp power(P_{t_m})= 43.17 dBm
= 20.75 W

Calculate Device size (WT) based on GaN1

$$WT = 20.75W/(6.5W/mm) = 3.19 \text{ mm}$$

Load-line Estimation Using Class-B Amplifier; @ Peak Both PAs are ON

Peaking Amp Load Estimate:

Peaking Amp Peak Power = 35.48 W

$$R_{\text{peak_p}} = \frac{(V_p^2)}{(2 * \text{Power})}$$

$$V_p = (V_{\text{dd}} - V_k) = 40 - 5 = 35\text{V}$$

$$\begin{aligned} R_{\text{peak}} &= \frac{(35^2)}{(2 * 35.48)} \\ &= 17.26 \, \Omega \end{aligned}$$

Main Amp Load Estimate:

Main Amp Peak Power = 20.75 W

$$R_{\text{main_p}} = \frac{(V_p^2)}{(2 * \text{Power})}$$

$$V_p = (V_{\text{dd}} - V_k) = 40 - 5 = 35\text{V}$$

$$\begin{aligned} R_{\text{main_p}} &= \frac{(35^2)}{(2 * 20.75)} \\ &= 29.5 \, \Omega \end{aligned}$$

Main Amp Ave Power = 6.8 W @ Peak=OFF

$$\begin{aligned} R_{\text{main_ave}} &= \frac{(35^2)}{(2 * 6.8)} \\ &= 90 \, \Omega \end{aligned}$$



SKYWORKS®

Connecting Everyone and Everything, All the Time.

skyworksinc.com